

DISLOCATIONS

- ❑ Edge dislocation
- ❑ Screw dislocation
- ❑ Dislocations in crystals

Further reading

Introduction to Dislocations

D. Hull and D.J. Bacon

Pergamon Press, Oxford (1984)

Advanced reading (comprehensive)

Theory of Dislocations

J. P. Hirth and J. Lothe

McGraw-Hill, New York (1968)

Recommended website

http://www.tf.uni-kiel.de/matwis/amat/def_en/

Part of

**MATERIALS SCIENCE
& *A Learner's Guide*
ENGINEERING**

AN INTRODUCTORY E-BOOK

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<http://home.iitk.ac.in/~anandh/E-book.htm>

- ❑ As seen before dislocations are 1D (line) defects.
- ❑ The role of dislocations goes *far beyond* just ‘plasticity by slip’.
- ❑ They play an important role in a variety of deformation processes like
 - creep, fatigue and fracture.
- ❑ They can play a ‘constructive role’ in crystal growth.
- ❑ They can provide short circuit paths for diffusion (pipe diffusion).
- ❑ Understanding the importance of dislocations in material behaviour cannot be overstated
 - hence it is very important to thoroughly understand the structure and behaviour of dislocations.

Caution Note: In any chapter, amongst the first few pages (say 5 pages) there will be some ‘big picture’ overview information. This may lead to ‘overloading’ and readers who find this ‘uncomfortable’ may skip particular slides in the first reading and come back to them later.

Dislocations

- ❑ The importance of understanding dislocations and their effect on material behaviour cannot be overstated.
 - ❑ Though often the importance of dislocations in the context of **plastic deformation*** by **slip**** is highlighted; its role in materials science is far greater.
(The next slide shows some of these roles).
 - ❑ In this context it is important to note that even in crystalline materials there are alternate mechanisms of plastic deformation (as shown in an upcoming slide)
→ **Twinning***** also being an important one.
 - ❑ The important thing to be kept in mind is the role of dislocations in *weakening* crystals (taken up after the above mentioned slides).
-
- ❑ The overall processes leading to shape change occurring during plastic deformation are very complicated (and is well beyond the scope of this book!). However, the first step involves the motion of dislocations leaving the crystal surface (or internal interfaces).

* Plastic deformation

→ permanent deformation that remains when all external loading/constraints are removed

** **Slip** → is a technical term referring to plastic deformation caused by dislocations

■ the 'first step' of the process is the small surface step which is created when a dislocation leaves a crystal

*** **Twinning**

→ process by which one part of the crystal gets related to another part, by a symmetry operator (usually a mirror); which is not a symmetry operator of the crystal.

Path to understanding the role of Dislocations in material behaviour

Though these points are written as a sequence many of these have to be considered in parallel

Consider a dislocation in an infinite crystal

Stress fields, strain fields, energy etc.

$E_d, \sigma_{xx}, \sigma_{yy}, \tau_{xy}, \epsilon_{xx}, \epsilon_{xy} \dots$

Take into account finite crystal effects

Free surfaces, grain boundaries etc.

Consider interaction of dislocations with other defects

Interactions with other dislocations, interstitials, precipitates etc.

Collective behaviour and effects of external constraints

*Long range interactions & collective behaviour & external constraints***

■ Static and dynamic effects and interactions should be included*

* Dynamic effects include:

- (Altered) Stress field of a moving dislocation
- Interactions evolving in time

Note: the above step by step method may often not be the most practical one and there are techniques which take up collective behaviour directly

Role of Dislocations

Deformation Processes

Kinetics

**Diffusion
(Pipe)**

**Crystal Growth
(Screw dislocation)**

Creep

Fatigue

Fracture

Slip

Structural

Incoherent Twin

Grain boundary
(low angle)

Semicoherent Interfaces

Disc of vacancies
~ edge dislocation

and more...!!

Creep
mechanisms in
crystalline
materials

Cross-slip

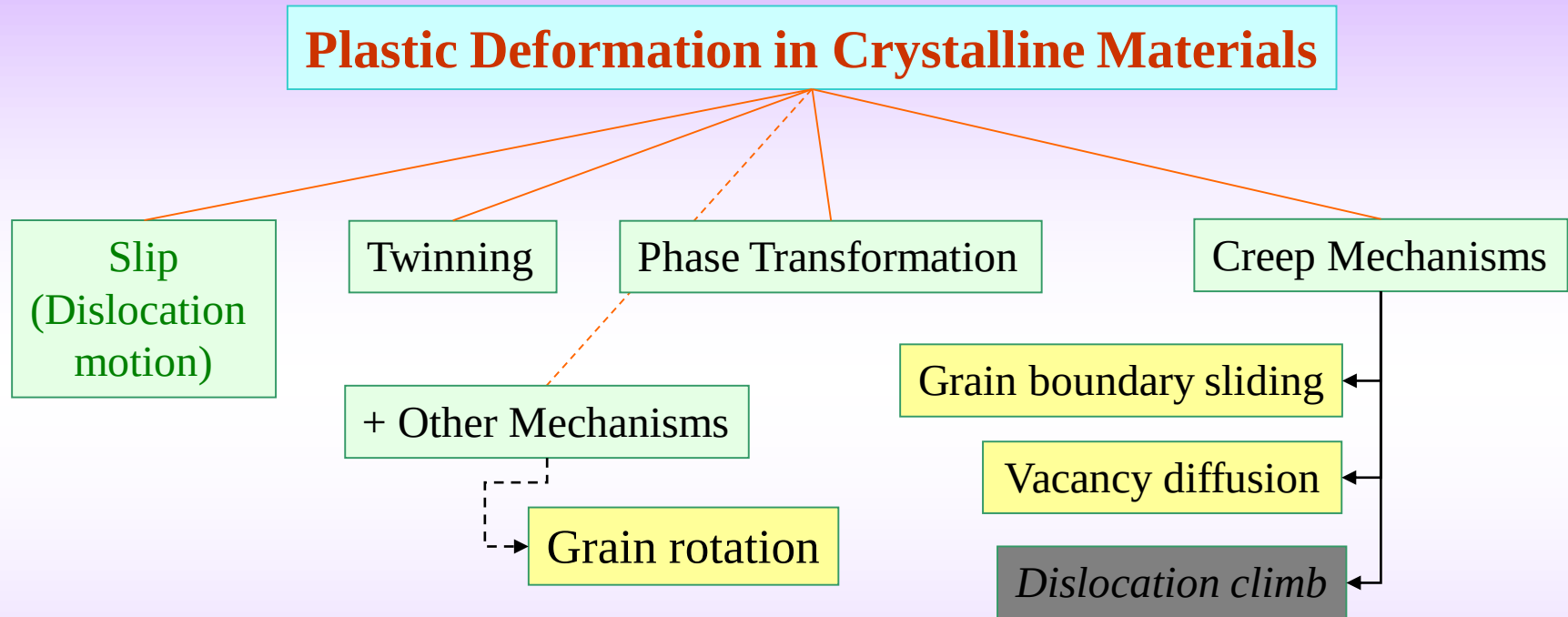
Dislocation climb

Vacancy diffusion

Grain boundary sliding

Note: Structural dislocations can also play a role in deformation and kinetic processes

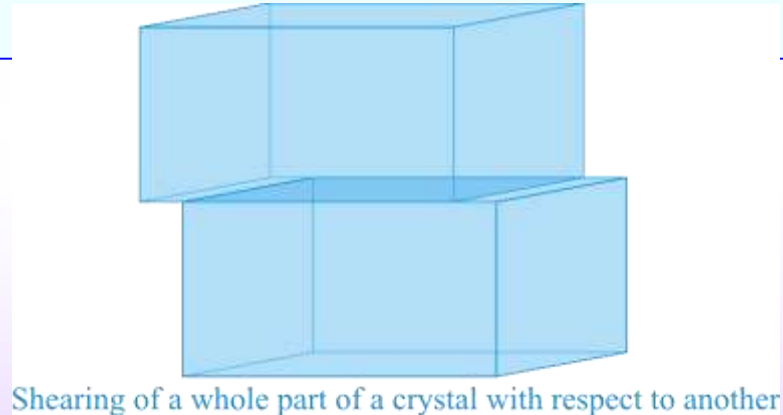
Though plasticity by slip is the most important mechanism of plastic deformation, there are other mechanisms as well (*plastic deformation here means permanent deformation in the absence of external constraints*):



Note: Plastic deformation in amorphous materials occur by other mechanisms including flow (~viscous fluid) and shear banding

Weakening of a crystal by the presence of dislocations

- ❑ To cause plastic deformation by shear (*all of plastic deformation by slip require shear stresses at the microscopic scale**) one can visualize a plane of atoms sliding past another (fig below**).
- ❑ This requires stresses of the order of GPa (calculation in the next slide).
- ❑ But typically crystals yield at stresses ~MPa.
- ❑ $\Rightarrow$ This implies that ‘something’ must be weakening them drastically.
- ❑ It was postulated in 1930s[#] and confirmed by TEM observations in 1950s, that the agent responsible for this weakening are dislocations.



- * Even if one does a pure uniaxial tension test with the tension axis along the z- axis, except for the horizontal and the vertical planes all other planes ‘feel’ shear stresses on them.
- ** As to how this atomic slip is connected to macroscopic permanent shape changes will be considered later.

[#] By Taylor, Orowan and Polanyi

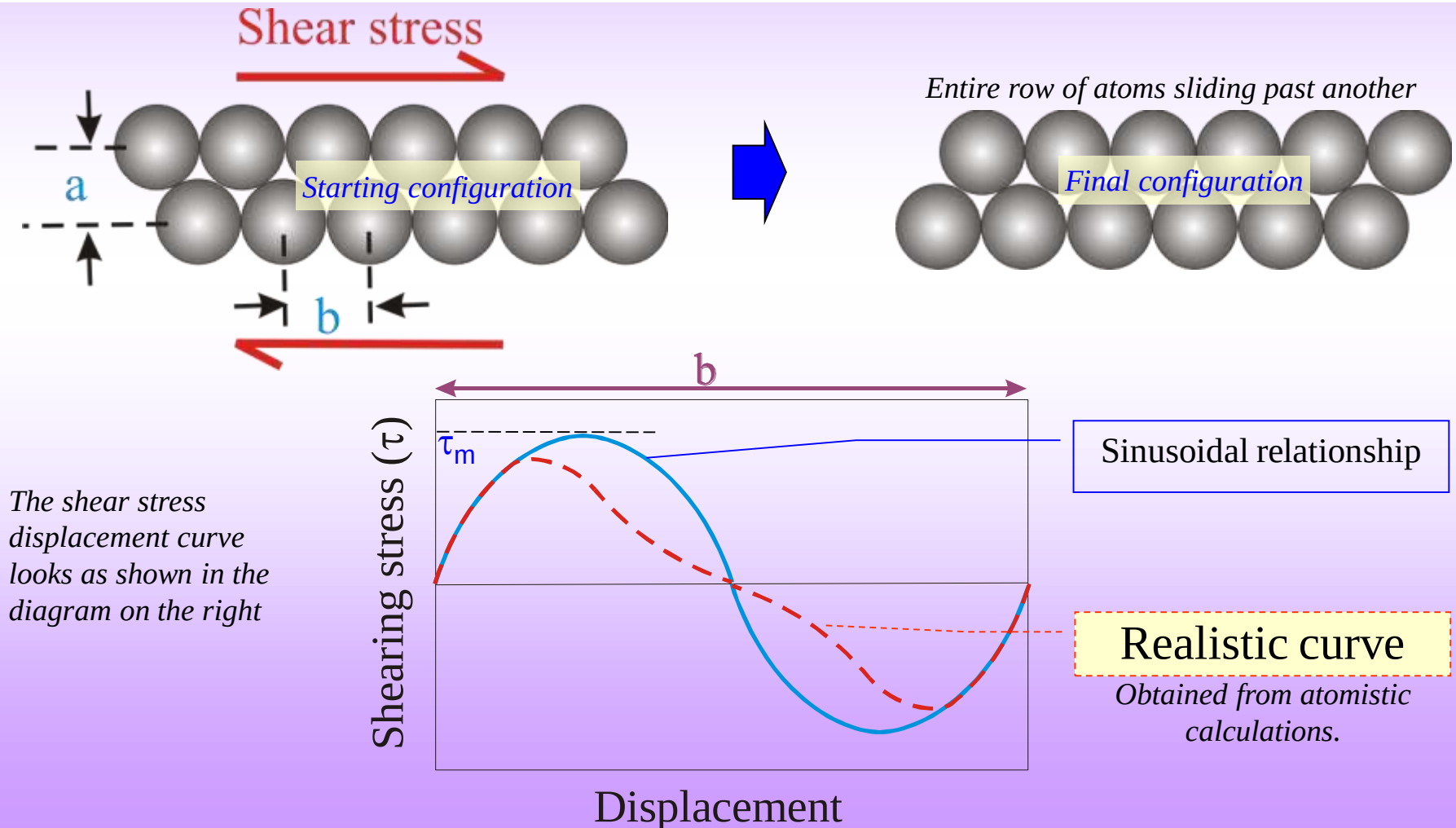
E. Orowan, Z. Phys. 89 (1934) 605, 614, 634.

M. Polanyi, Z. Phys. 89 (1934) 660.

G.I. Taylor, Proc. Roy. Soc. London, A145 (1934) 362.

Plastic deformation of a crystal by shear

- Let us consider the shearing of an entire plane of atoms over one another
→ causing plastic deformation by shear.
- If the atoms 'climb' only to the top of the peak (of another atom, corresponding to the peak in shear stress in the plot below), then it can 'slide' back to its original position and the deformation will be elastic.



As a first approximation the stress-displacement curve can be written as

$$\tau = \tau_m \sin\left(\frac{2\pi x}{b}\right)$$

At small values of displacement Hooke's law should apply

$$\tau = G\gamma = G \frac{x}{a}$$

$$G \frac{x}{a} = \tau_m \left(\frac{2\pi x}{b}\right)$$

$\Rightarrow$ For small values of x/b

$$\tau = \tau_m \left(\frac{2\pi x}{b}\right)$$

Hence the maximum shear stress at which slip should occur

$$\tau_m = \frac{G}{2\pi} \frac{b}{a}$$

If $b \sim a$

$$\tau_m \sim \frac{G}{2\pi}$$

$$\tau_m \sim \frac{G}{2\pi}$$

- ❑ The **Theoretical Shear Stress (TSS)** will be in the range **3-30 GPa**.
- ❑ TSS is also called the *Theoretical Shear Strength*.

❑ The shear modulus of metals is in the range 20 – 150 GPa.

$$\tau_m \approx \frac{G}{2\pi}$$



❑ The theoretical shear stress will be in the range 3-30 GPa

❑ Actual shear stress is 0.5 – 10 MPa (*experimentally determined*).

❑ I.e. (Shear stress)_{theoretical} > 100 × (Shear stress)_{experimental} !!!!



DISLOCATIONS

Dislocations severely weaken the crystal

Material	Strength	
	Theoretical [MPa] G/2π	Experimental [MPa]
Al	4200	0.7-0.8
Cu	7700	0.5-3
Fe	13000	25-30
Al ₂ O ₃	20000	19000
Diamond	46000	21000

Note that the discrepancy in the values for non-metals (listed in the last two rows) is not 'that bad'. This is because at 'low' temperatures they do not deform plastically by the motion of dislocations.

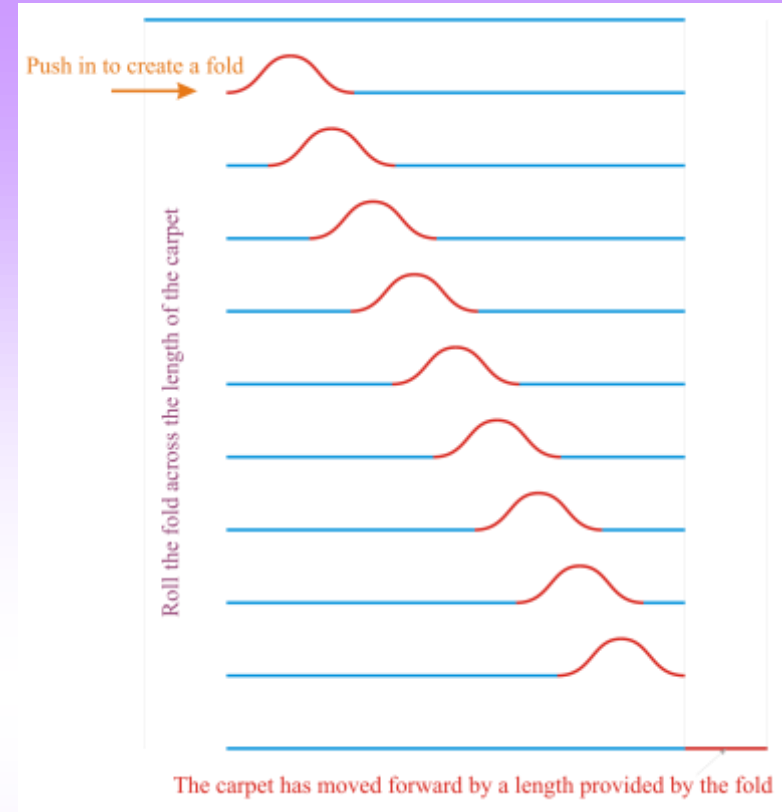
▪ Whiskers of metals (*single crystal free of dislocations, Radius ~ 10⁻⁶m*) can approach theoretical shear strengths. Whiskers of Sn can have a yield strength in shear ~10⁻² G (10³ times bulk Sn).

- ❑ As we have seen before dislocations can play diverse kinds of role in materials structure and its behaviour
- ❑ Perhaps the most important of these is the weakening of the crystal in the presence of dislocations
- ❑ From a slide before we know the path to understanding the role of dislocations in materials involves their interactions with other dislocations and defects present in the material (and the evolution of the system with time/deformation)
 - *This path will include the 'hardening' of the crystal, i.e strengthening of the weakened crystal*
 - *In this context it will be noted that many dislocations will interact with each other and there will be a strengthening effect*

Dislocations: path breaking ideas

- As late as 1930 the reason behind this weakening of the crystal was not clear (*to imagine that this was the post Relativity, post Quantum Mechanics era, wherein deep questions regarding the larger scale of the universe and the sub-atomic realms were being conquered → but why a rod of copper can be bent easily was not known!*)
 - Taylor, Orowan and Polanyi (independently) postulated the presence of dislocations as a mechanism leading to the weakening of the crystal
 - The continuum construction of a dislocation (and other defects) was proposed by Volterra in 1905.
 - The presence of dislocations was Electron microscopically confirmed in the 1950s
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- ■ G.I. Taylor, Proceedings of the Royal Society A, 145 (1934) 362. ■ E. Orowan, Zeit. Physics, 89 (1934) 605. ■ N. Polanyi, Zeit. Phys. 89 (1934) 660.
 - Vito Volterra, 1905.

- ❑ The analogy usually given to understand the role of dislocations in weakening a crystal is the one of ‘pulling a carpet’.
- ❑ If one tries to pull an entire carpet (a long and wide one), by sliding it against the floor, the effort required is large.
- ❑ However, if a ‘bump’ is made in the carpet (as in the figure in the following slide) and this bump is moved across the length of the carpet, then the carpet moves forward by a small distance (as provided by the bump).
- ❑ The force required to move the bump will be considerably small as compared to the force required to pull the entire carpet.
- ❑ By creating and moving a series of bumps successively the carpet can be moved forward ‘bit by bit’. (*Graphic on next slide*).



The ‘carpet-pulling’ analogy

The Volterra Dislocation

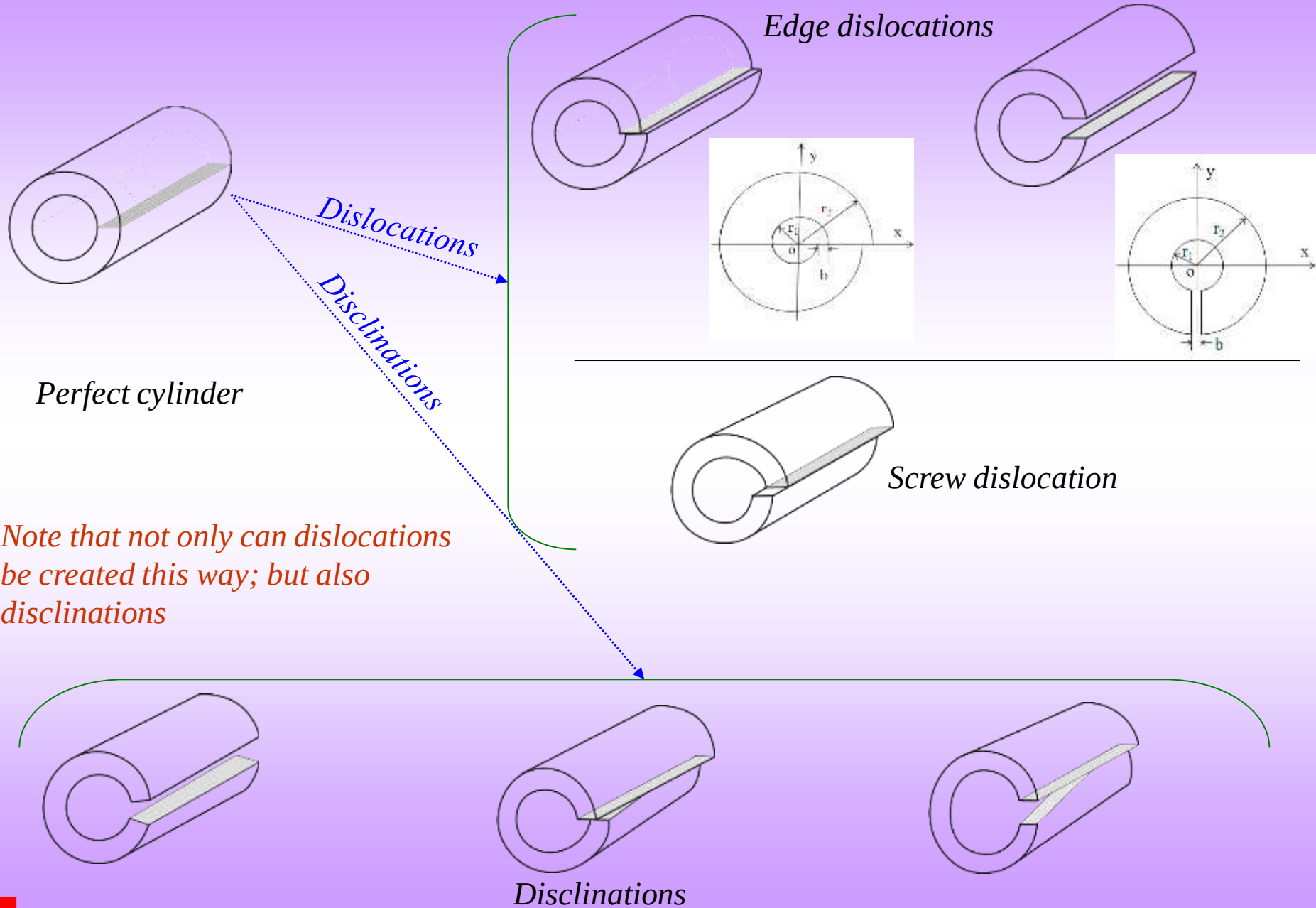
Continuum description of a dislocation

- ❑ The continuum* concept of a dislocation (and other defects) was proposed by **Vito Volterra** in 1905.
- ❑ His ideas and calculations based on his ideas predate their application to crystals. However, continuum calculations based on Volterra's idea are used even today to understand the behaviour of dislocations in crystals.
- ❑ Continuum calculations of stress fields, displacement fields etc. related to dislocations are found to be valid to within a few atomic spacing (i.e. the continuum description fails only within about 5 atomic diameters/Burgers vector).
- ❑ In this chapter the stress fields of dislocations shown are based on **elastic continuum theories**.**

* Continuum implies that we are not 'worried' about atoms! (*Antonym* → *Discretum*).

** This is a 'strange' aspect: we have used elasticity theory to calculate the stress fields of the very (most important) agent responsible for plastic deformation

Deformations of a hollow cylinder showing the formation of various defects (Volterra constructions)



DISLOCATIONS

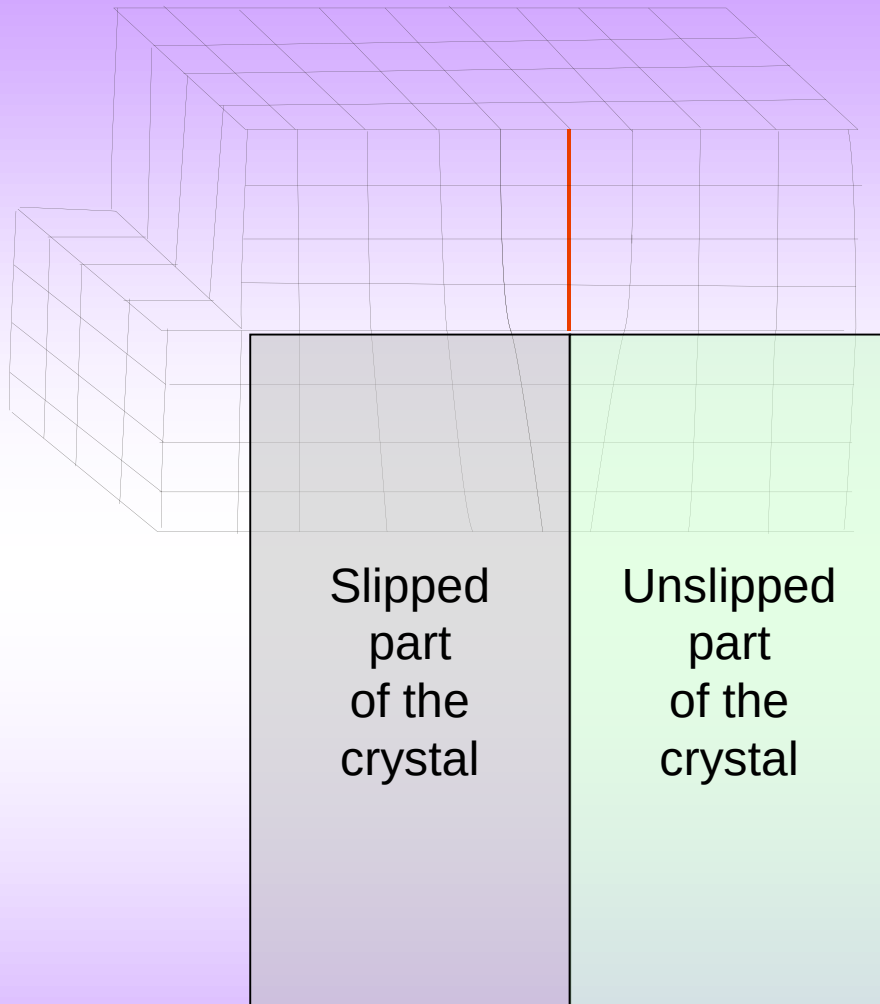


- ❑ If one looks at a sample of Aluminum under a TEM, one usually finds curved dislocation lines $\Rightarrow$ Usually dislocations have a **mixed** character and *Edge* and *Screw* dislocations are the *ideal extremes*.
- ❑ Not only this, the character of the dislocation (i.e the percentage of screw and percentage of edge character) will change from position to position along the dislocation line.
- ❑ However, under special circumstances **Pure Edge**, **Pure Screw** or **a Mixed Dislocation with a fixed percentage of edge character** can form.
(e.g. in GeSi epitaxial films on Si substrate 60° misfit dislocations form- i.e. the dislocation lines are straight with the angle between $\mathbf{b}$ and $\mathbf{t}$ being 60°)
 $\rightarrow$ more about these aspects will be considered later
- ❑ The edge dislocation is easier to visualize and hence many of the concepts regarding dislocations will be illustrated using the example of the pure edge dislocation.

TEM micrograph
showing dislocation lines



Dislocation lines



Dislocation can be considered as a boundary between the slipped and the unslipped parts of the crystal lying over a slip plane*

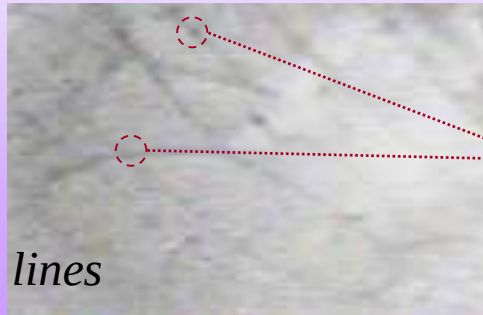
** this is just a way of visualization and often the slipped and unslipped regions may not be distinguished*

A dislocation has associated with it two vectors:

$\vec{t} \rightarrow$ A unit tangent vector along the dislocation line

$\vec{b} \rightarrow$ The Burgers vector

- ❑ The Burgers vector is like the 'SOUL of a dislocation'. It 'can be defined' even if there is no dislocation in the crystal (it is the shortest lattice translation vector for a full/perfect dislocation), it defines every aspect of the dislocation (its stress fields, energy, etc.) and expresses itself even in the 'death' of the dislocation (i.e. when the dislocation leaves the crystal and creates a step of height 'b'). It is a parameter which characterizes a dislocation and is measure of the strength of the dislocation.
- ❑ Burgers vector of a perfect dislocation is the shortest translation vector (for a full/perfect dislocation) and can be determined by the Burgers circuit (coming up).
- ❑ Hence, Burgers vector is an invariant for a given crystal, while the line vector is not. If one looks at a transmission electron micrograph showing a dislocation line in Aluminium, it will not be straight– i.e. the line vector is not fixed (usually*).



Dislocation lines

TEM micrograph
showing dislocation lines

* In special cases we might have straight dislocation lines– this implies that $\vec{t}$ is constant.

Note: the Burgers circuit does not define the Burgers vector. If you have a dislocation in a crystal, this is a method to determine it.

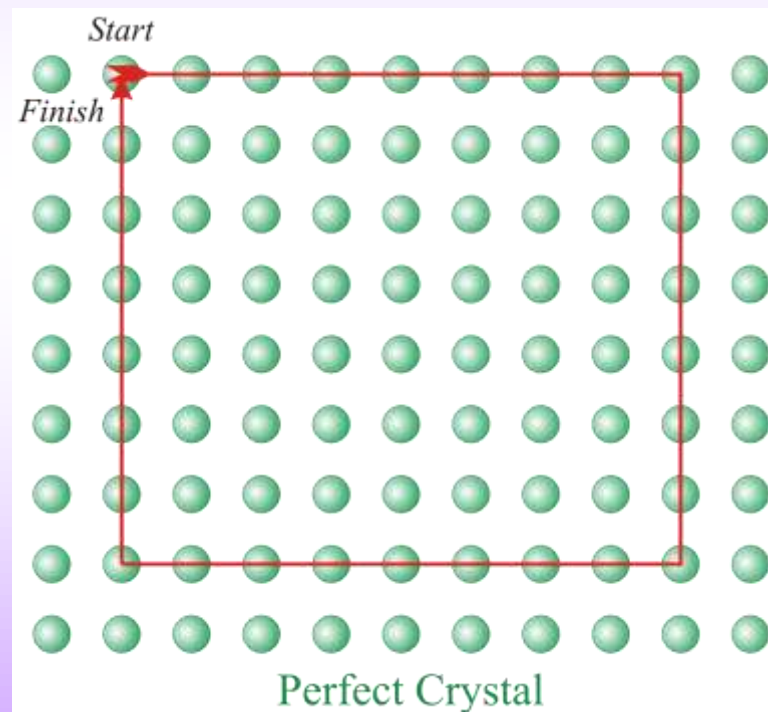
Burgers Vector

$\vec{b}$

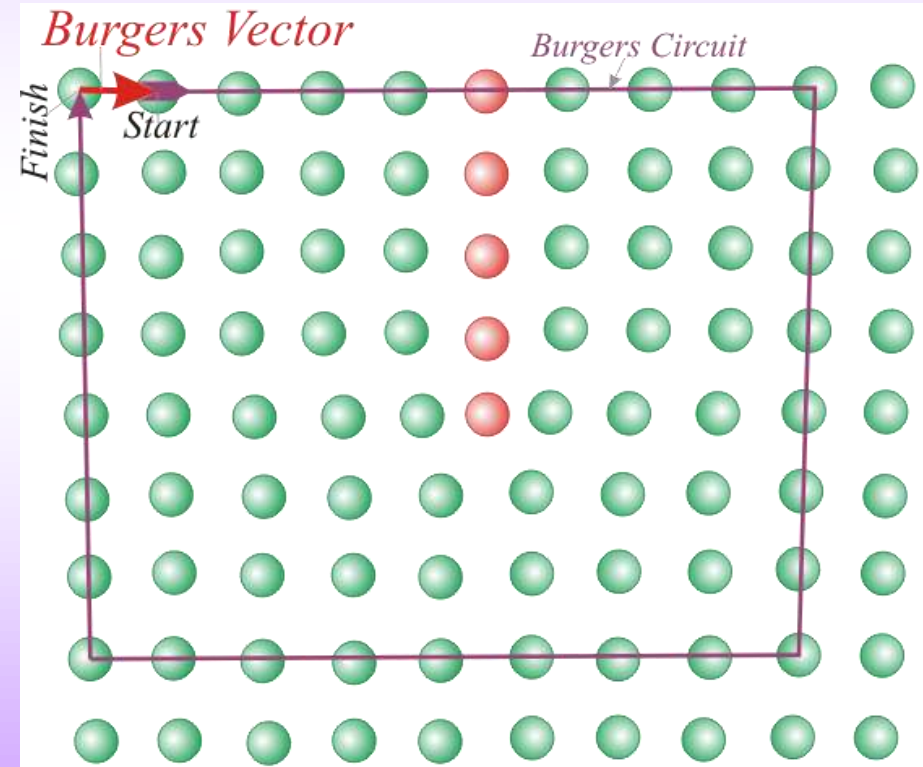
Edge dislocation

Determination of Burgers vector in a dislocated crystal using Right Hand Finish to Start Rule (RHFS)

- In a perfect crystal make a circuit (e.g. as in the figure shown: 8 atomic steps to right, 7 down, 8 left & 7 up). The circuit is *Right Handed*.
- Follow the same steps in the dislocated crystal. The 'missing link' (using some convention like RHFS) is the Burgers vector.
- 'Burgers circuit' and RHFS convention can be applied to both edge and screw dislocations.

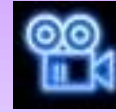


RHFS:
Right Hand Finish to Start convention

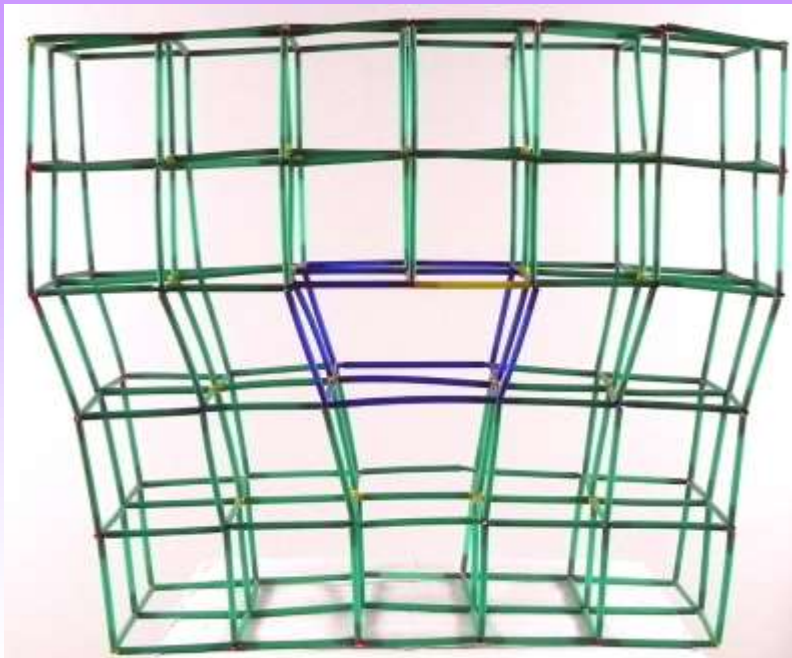


Note: the circuit has to be drawn 'far' away from the dislocation line !
(but for convenience this has been shown close to the dislocation line)

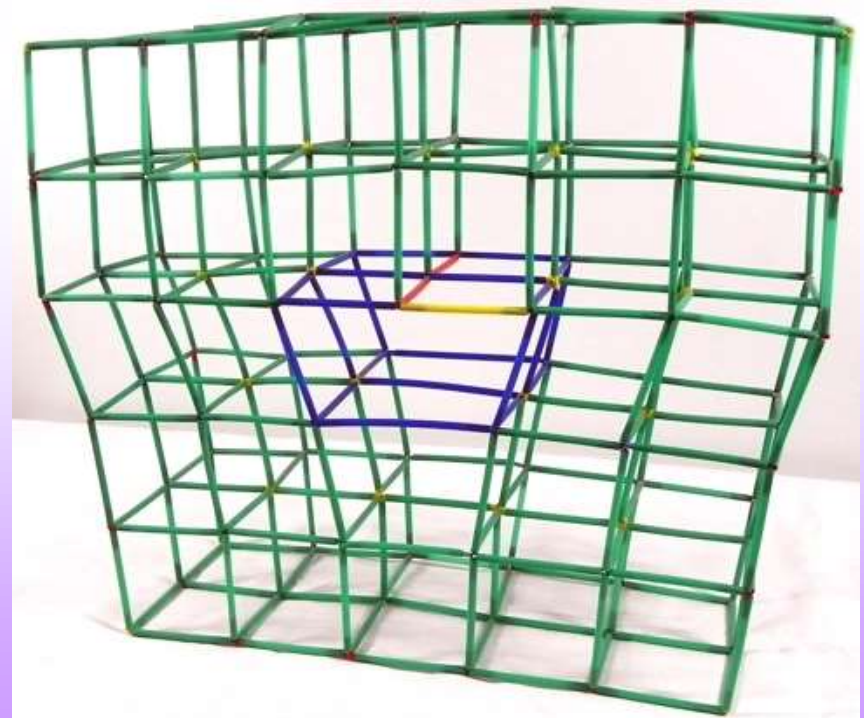
Some models of Edge Dislocation



Video: Edge Dislocation

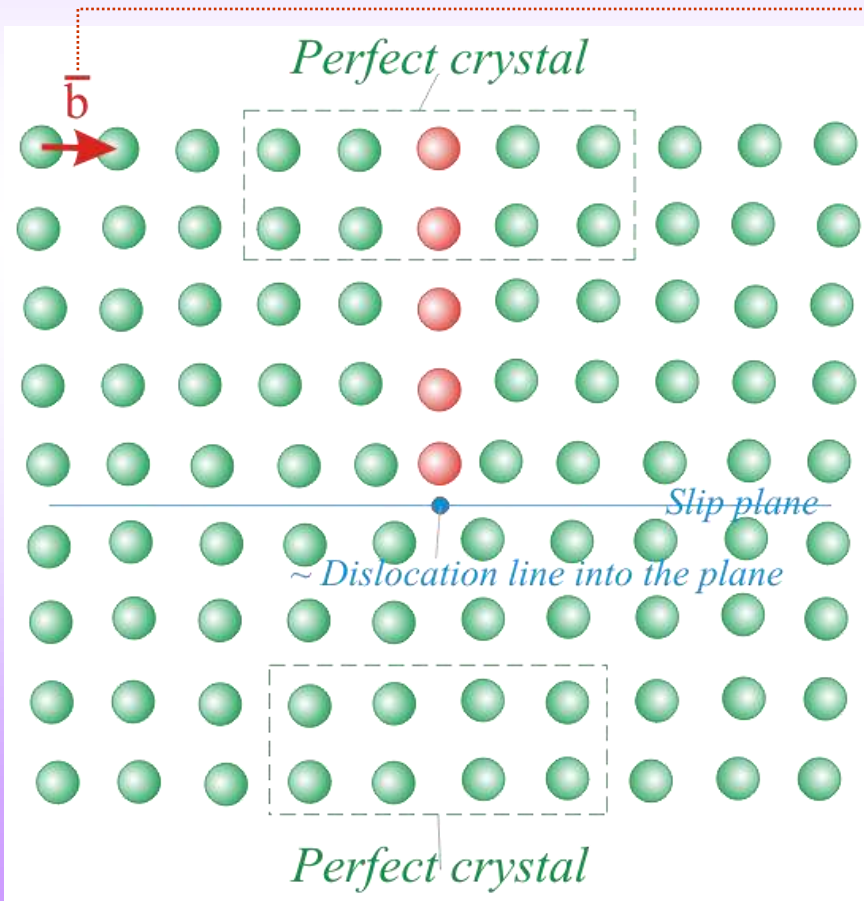


Model using magnetic balls
(not that accurate!)



Understanding the Edge dislocation

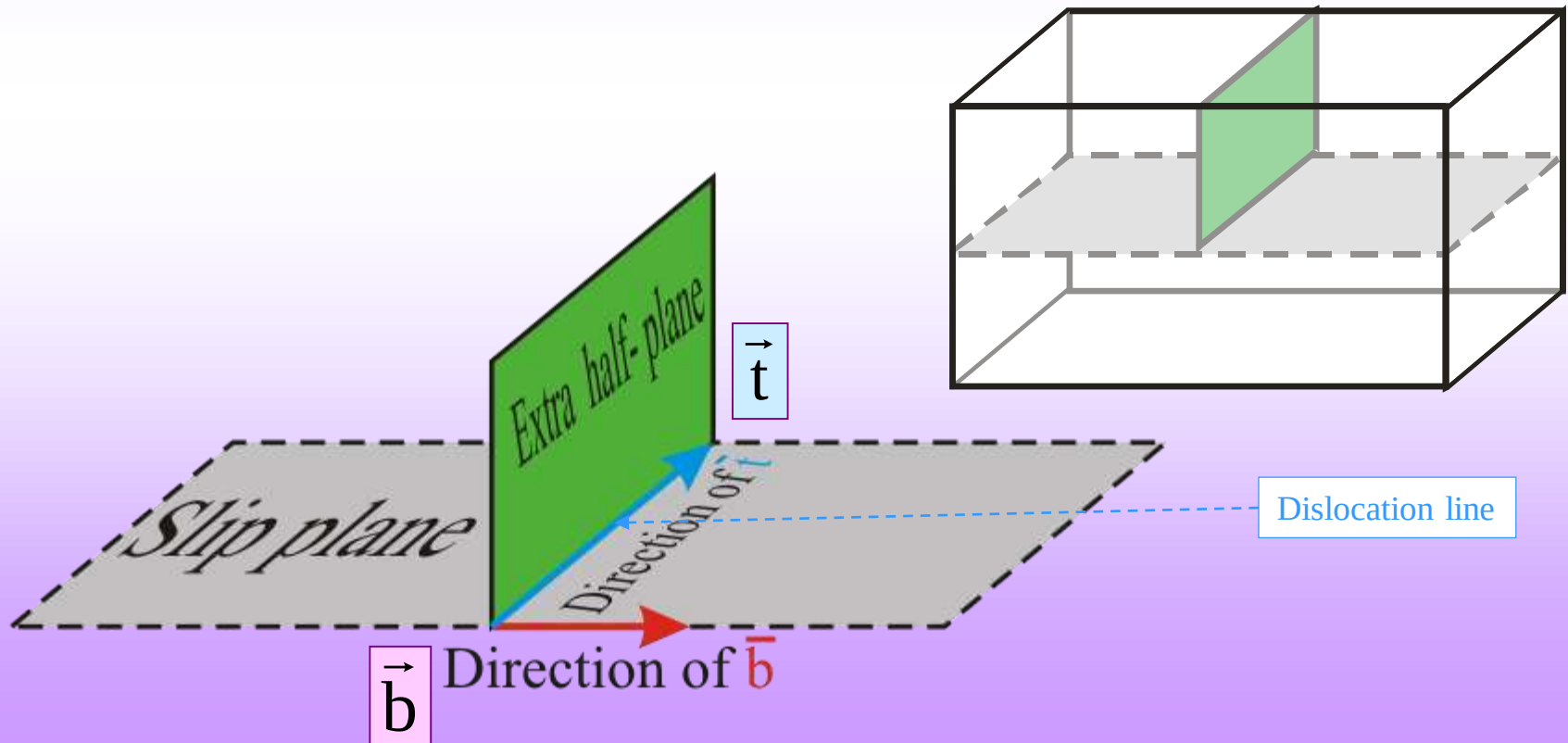
- ❑ The edge dislocation is NOT the 'extra half-plane', it is *neither* the 'missing half-plane' → it is the line between the 'extra' and the 'missing' half-planes.
- ❑ The regions far away from the dislocation line are perfect → all the 'deformation' is concentrated around the dislocation line.
- ❑ However, the stress field of the dislocation has a 'long range'.



Note:

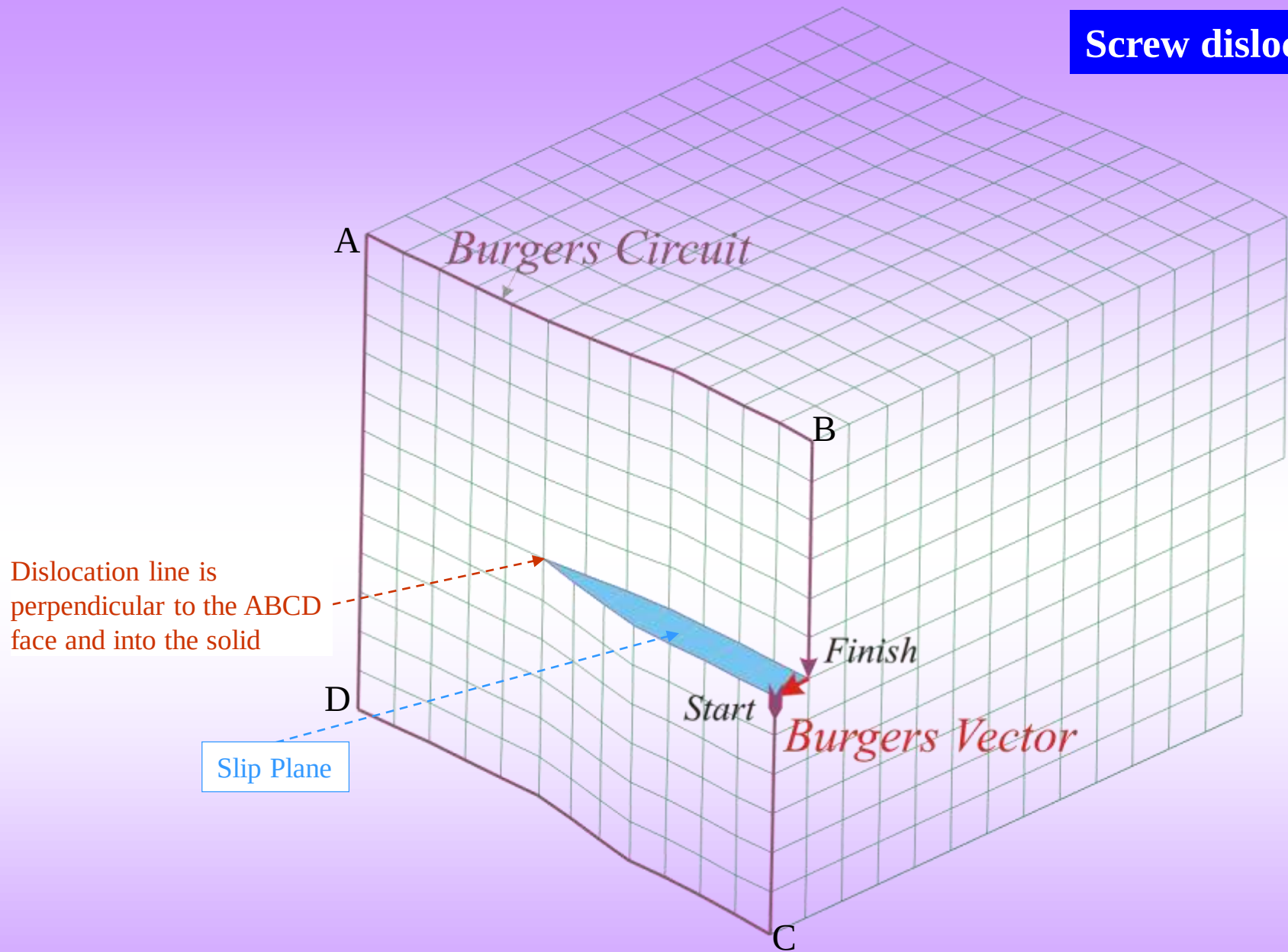
- The Burgers vector has to be drawn '*far away*' from the dislocation line (sometimes it may be drawn close to dislocation line for convenience).
- The edge dislocation line is between the 'missing' and 'extra' half-planes.
- I.e. it is the '*fictitious/imaginary*' line between the missing and extra half-planes.

- ❑ Often to visualize the edge dislocation, only the extra 'half'-plane and slip plane are shown. The remaining crystal is hidden away.
- ❑ The intersection of the extra half-plane and slip plane can be visualized as the dislocation line (one of the two possible directions is represents the line vector- shown in blue colour).
- ❑ Since the $\vec{b}$ and $\vec{t}$ vectors are not collinear, they will define a plane– the *slip plane*.
- ❑ Under the action of shear stress, the dislocation may move on the slip plane.



- ❑ Dislocation can be considered as the boundary between the slipped and the unslipped parts of the crystal lying over a **slip plane**.
The slip plane is a close packed plane in close packed structures and a high density plane otherwise.
- ❑ For an edge dislocation, the intersection of the extra half-plane of atoms with the slip plane defines the **dislocation line**.
- ❑ Direction and magnitude of slip is characterized by the Burgers vector of the dislocation
(A dislocation is born with a Burgers vector and expresses it even in its death!).
- ❑ The Burgers vector can be determined by the **Burgers Circuit**.
- ❑ Right hand screw (finish to start) convention is usually used for determining the direction of the Burgers vector.
- ❑ As the periodic force field of a crystal requires that atoms must move from one equilibrium position to another $\Rightarrow \mathbf{b}$ must connect one lattice position to another *(for a full dislocation)*.
- ❑ **Dislocations tend to have as small a Burgers vector as possible.**
- ❑ Dislocations are **non-equilibrium** defects and would leave the crystal if given an opportunity. *(The presence of dislocations increases the configurational entropy and hence the $-T\Delta S$ term would be negative. However, the 'T' at which they are stabilized is beyond the melting point of all crystals).*

Screw dislocation



Dislocation line is perpendicular to the ABCD face and into the solid

Slip Plane

Notes:

The figure shows a Right Handed Screw (RHS) dislocation (RHS is structurally distinct from LHS). As for the edge dislocation the Burgers circuit has to be drawn far away from the dislocation line.

Geometric properties of dislocations

- In a edge dislocation : **b** is perpendicular to **t**.
- In a screw dislocation : **b** is parallel to **t**. Since the vectors (**b** & **t**) are collinear, they do not uniquely define a plane. In the case of screw dislocations, the slip plane (which contains these vectors) is a high density plane and multiple such planes (belonging to the same family or having a high density) can act as slip planes.
- Other properties are as in the table below.

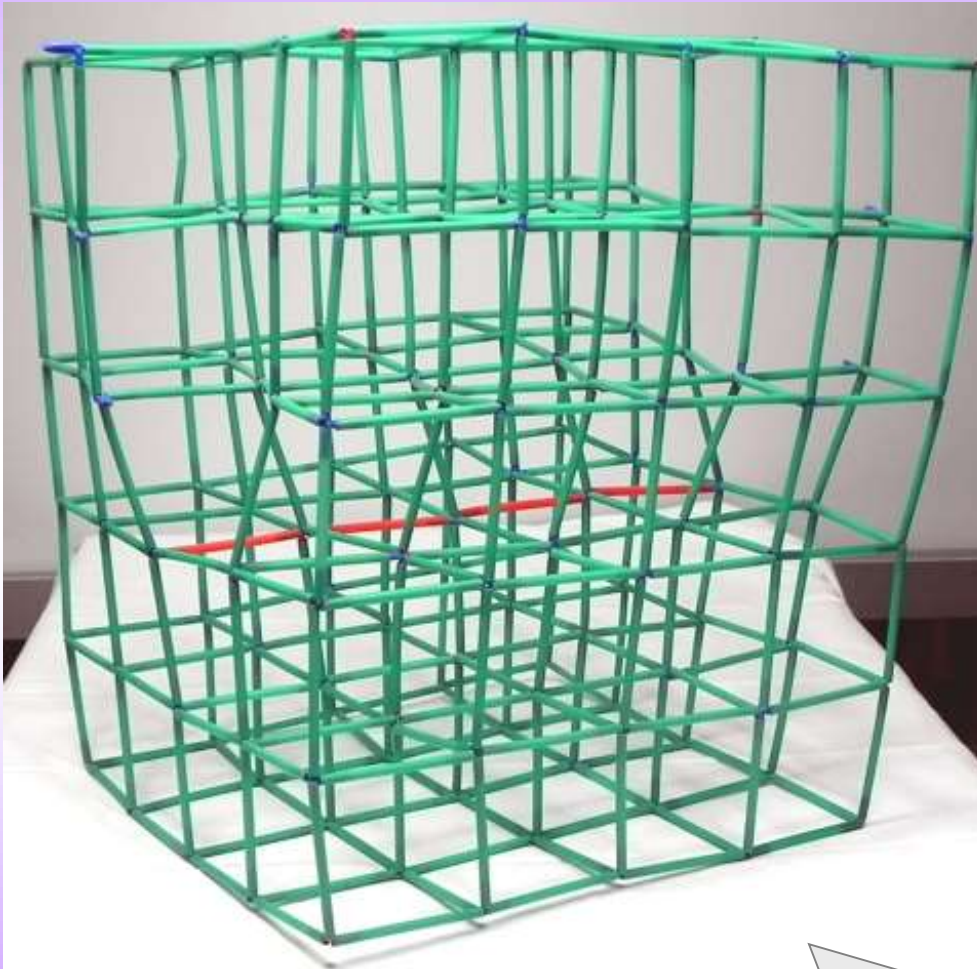


Dislocation Property	Type of dislocation	
	Edge	Screw
Relation between dislocation line (t) and b	$\perp$	$\parallel$
Slip direction** (& the ‘direction’ of step created when dislocation leaves the crystal)	$\parallel$ to b	$\parallel$ to b
Direction of dislocation line movement relative to b	$\parallel$	$\perp$
Process by which dislocation may leave slip plane*	climb	Cross-slip

* Note: edge dislocations cannot cross slip & screw dislocations cannot climb.

** Slip is the end result when dislocation leaves the crystal to create a step. Slip is hence always parallel to **b**. Motion of a dislocation line is... well... “motion of a dislocation line”.

Model of Screw Dislocation

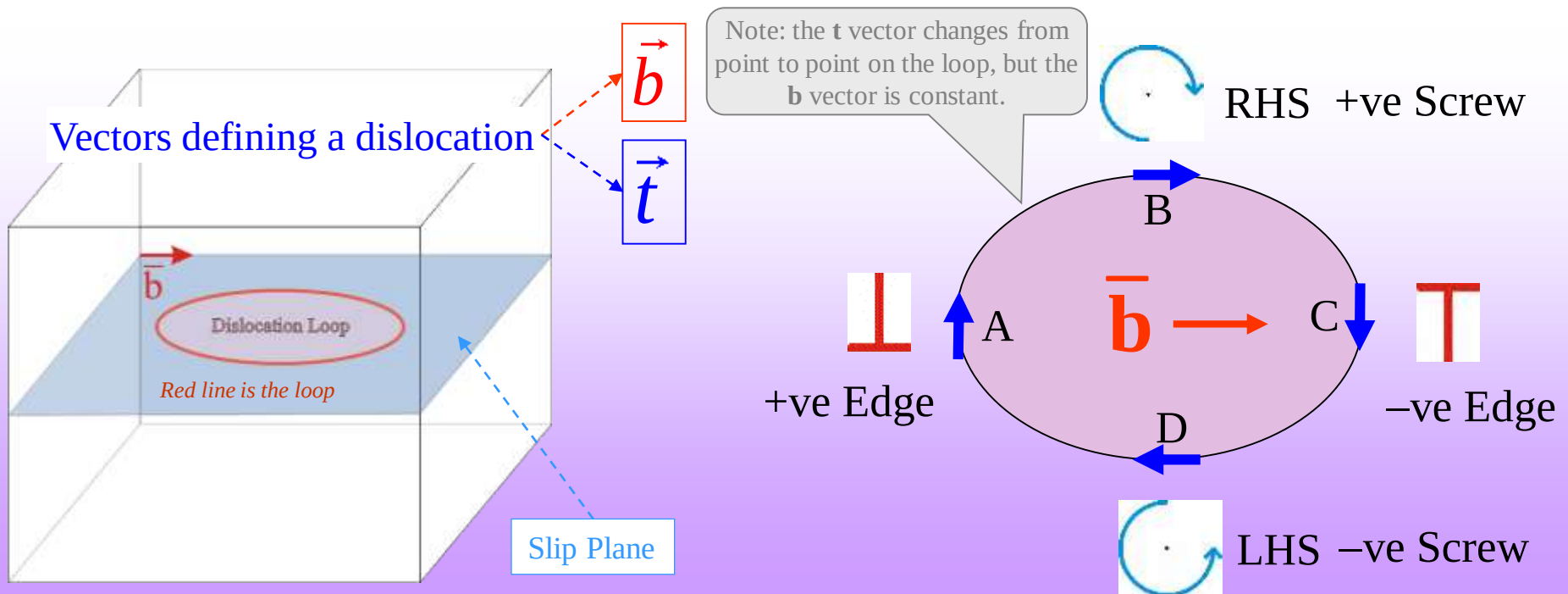


Though it is difficult to understand anything from the photo of the model!

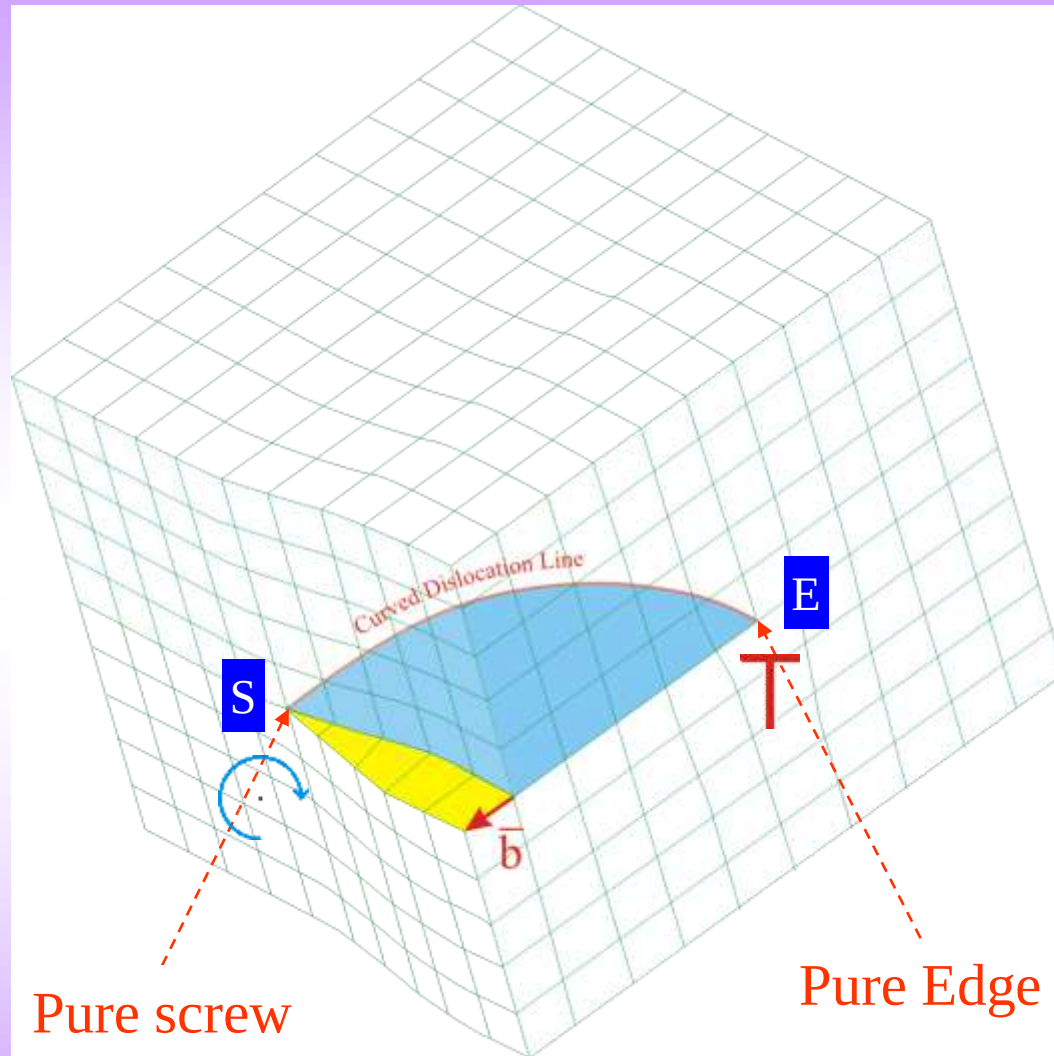
Mixed dislocations

Dislocations with mixed edge and screw character

- ❑ As we had noted, except in special circumstances, dislocations have mixed edge and screw character.
- ❑ In a curved dislocation the edge and screw character change from point to point.
- ❑ Typically in a dislocation loop only 'points' have pure edge or pure screw character
Edge: $\mathbf{b} \perp \mathbf{t}$ → two points A,C
Screw: $\mathbf{b} \parallel \mathbf{t}$ → two points B,D.
- ❑ The region enclosed by the loop can be considered as the 'slipped region'. And when 'sufficient' shear stress exists on the slip plane the loop will expand or contract.



One good way of understanding the dislocation loop is by considering a 'quarter' of a loop. One end of this loop is of edge character and the other has a screw character.



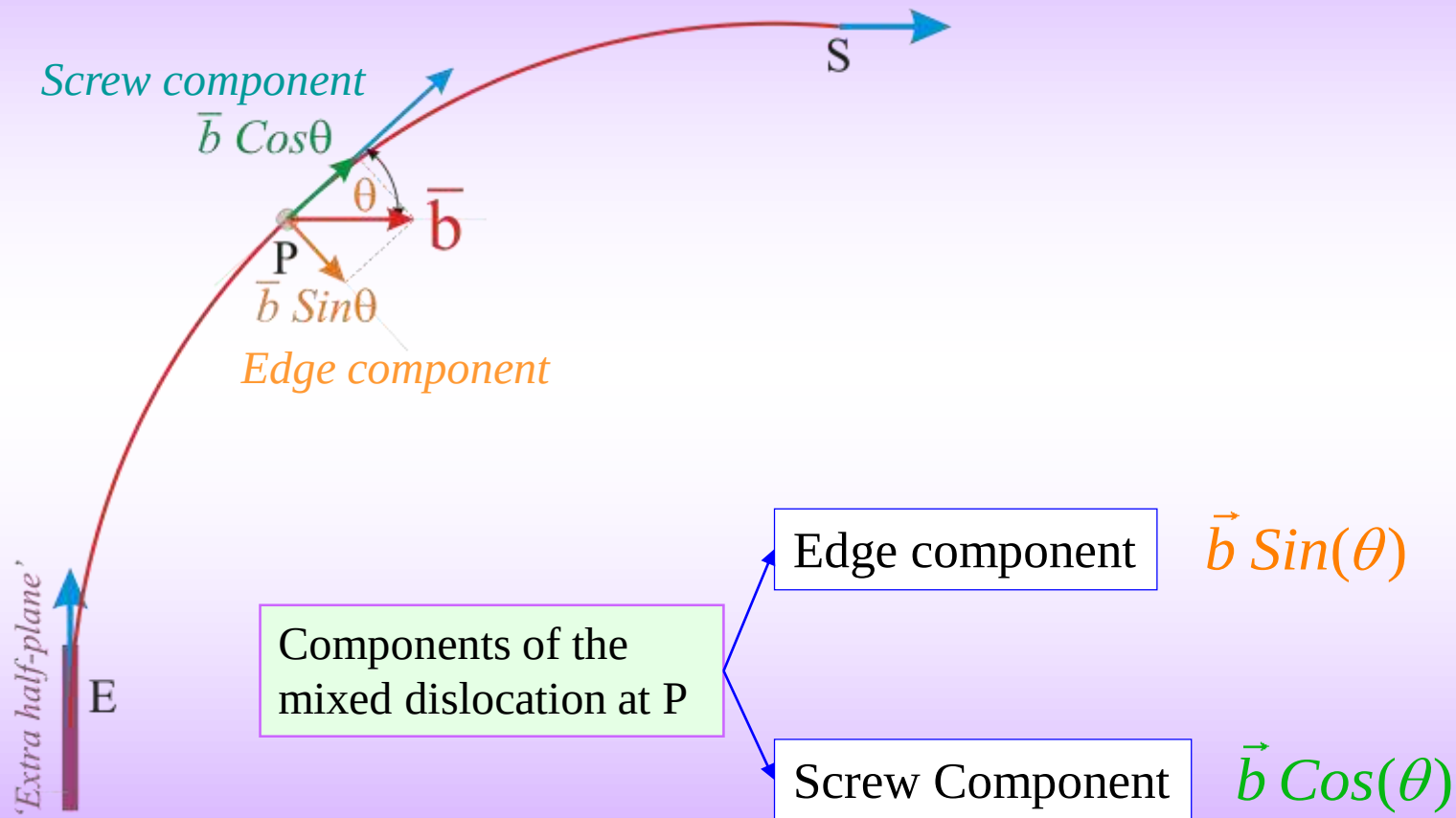
Except for points S and E the remaining portion of the dislocation line has a mixed character

Edge and Screw components: the 'usual' way to get the effective Burgers vector

The $\mathbf{b}$ vector is resolved into components:

'parallel to $\mathbf{t}$ ' $\rightarrow$ screw component and

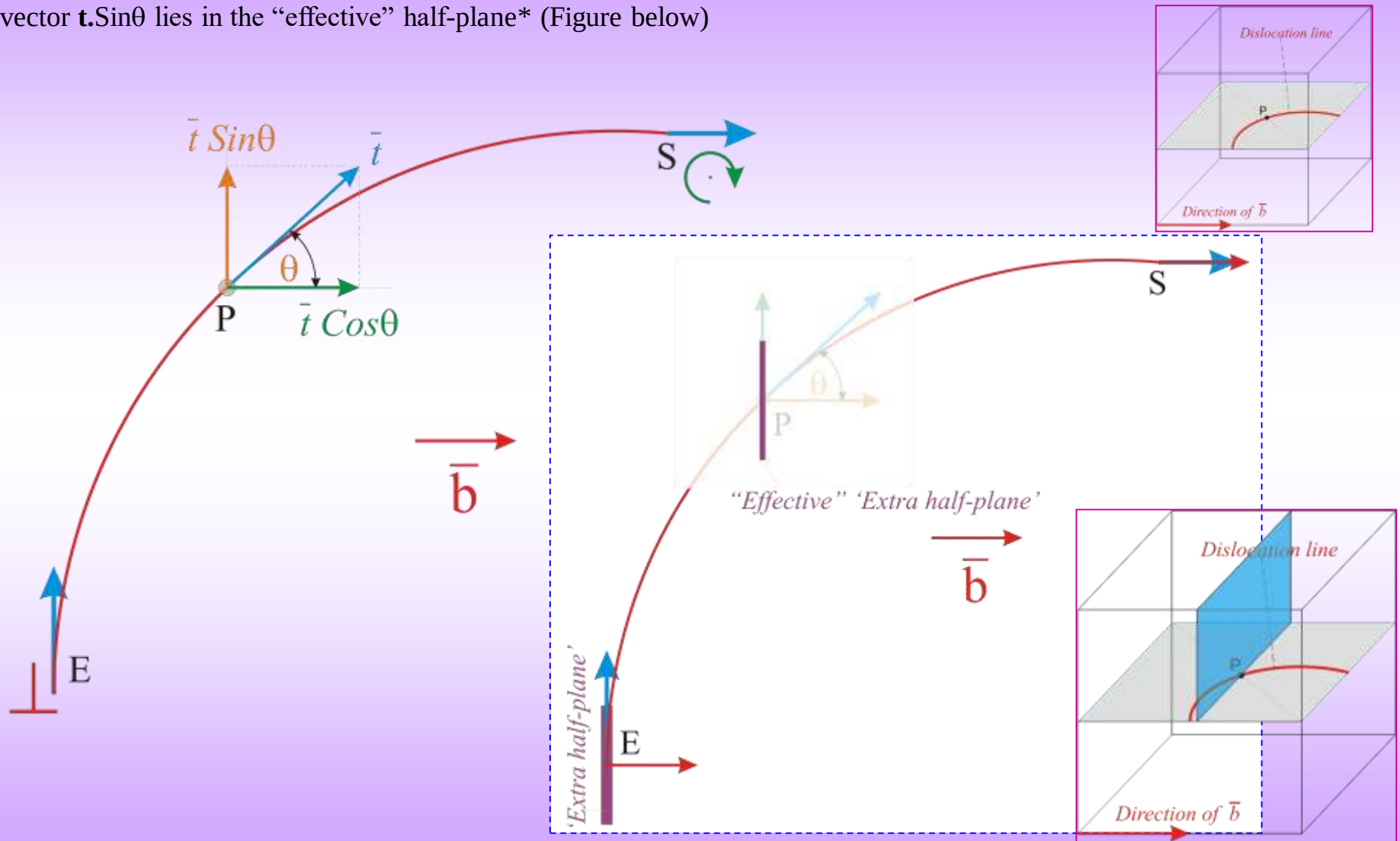
'perpendicular to $\mathbf{t}$ ' $\rightarrow$ edge component



Edge and Screw components: different way to visualize the orientation of the effective half-plane

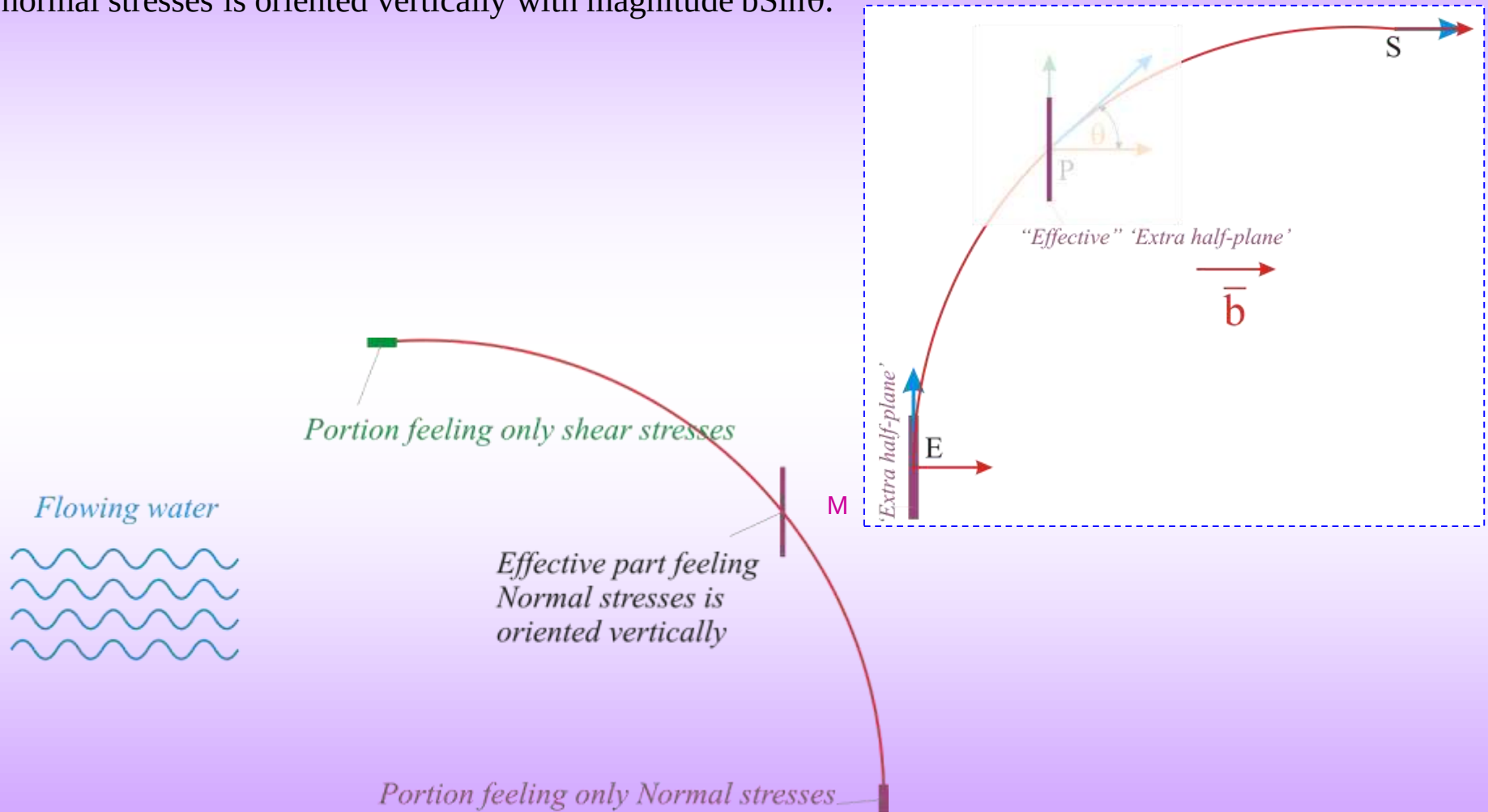
Instead of resolving the $\mathbf{b}$ vector if the $\mathbf{t}$ vector is resolved to find the edge and screw components

For an edge dislocation the extra half-plane contains the $\mathbf{t}$ vector $\rightarrow$ by resolving the $\mathbf{t}$ vector the edge component of the $\mathbf{t}$ vector $t \sin \theta$ lies in the “effective” half-plane* (Figure below)



**Note: For a mixed dislocation there is no distinct 'half-plane'*

Assume water is flowing from left to right onto a rigid curved wall (in red colour below). The green portion of the wall 'feels' only shear stresses, while the maroon portion feels only normal stresses (of magnitude b). A point M (in the curved segment) feels both normal and shear stresses. The effective part which feels normal stresses is oriented vertically with magnitude $b\sin\theta$.



Motion of Dislocations

- ❑ Two kinds of motion of a dislocation are possible: **Glide** and **Climb**.
- ❑ First we consider glide motion. Glide involves motion of dislocations *on* the slip plane.
- ❑ Dislocations *may* move under an externally applied force (resulting in stress inside the material- often casually referred to ‘applied stress’).
- ❑ At the local level shear stresses on the slip plane can only drive dislocations.
- ❑ The minimum stress required to move a dislocation is called the **Peierls-Nabarro** (PN) stress or the **Peierls stress** or the **Lattice Friction** stress (*i.e the externally ‘applied stress’ may even be purely tensile but on the slip plane shear stresses must act in order to move the dislocation*).
- ❑ Dislocations may also move under the influence of other internal stress fields (*e.g. those from other dislocations, precipitates, those generated by phase transformations etc.*).
- ❑ Dislocations are attracted to free-surfaces (and interfaces with softer materials) and may move because of this attraction → this force is called the **Image Force**.
- ❑ *In any case the Peierls stress must be exceeded for the dislocation to move.*
- ❑ The value of the Peierls stress is different for the edge and the screw dislocations.
- ❑ The first step of plastic deformation can be considered as the step created when the dislocation **moves and leaves the crystal**.
→ *“One small step for the dislocation, but a giant leap for plasticity”.*
- ❑ When the dislocation leaves the crystal a step of height ‘b’ is created → with it all the stress and energy stored in the crystal due to the dislocation is relieved.

- ❑ We saw that shear stresses are required for the motion of dislocations.
- ❑ In general stresses (tensile/compressive/shear) can arise from three sources as in the figure below. Further, these stresses may give rise to shear stresses at the level of the slip plane.
- ❑ External loading is the easy part to understand (like pulling the specimen). To understand constraints let us consider the example of a specimen between rigid walls, which is heated. This specimen will try to expand and the constraint of the walls will lead to stresses in the body.
- ❑ Internal stress fields can arise from ‘*microstructural features/effects*’ like coherent precipitates, dislocations, phase transformations, etc. These stresses can lead to shear stresses on the slip plane.
- ❑ The interaction of internal stress fields (e.g. that associated with a dislocation) with interfaces (across which region of higher or lower modulus exists), can lead to the modification of the inherent stress field associated with the ‘defect*’ → which can result in an ‘*effective stress*’ on the defect itself. In the case of a dislocation, a ‘free-surface’ will attract the dislocation and an interface with a harder material will repel it. [If the force of attraction exceeds the Peierls force**, then the dislocation will spontaneously move and can leave the crystal.

Origin of shear stresses

1) From external loading and constraints

2) From internal stress fields

3) From ‘*interaction*’ interfaces of higher or lower compliance

Motion of Edge dislocation

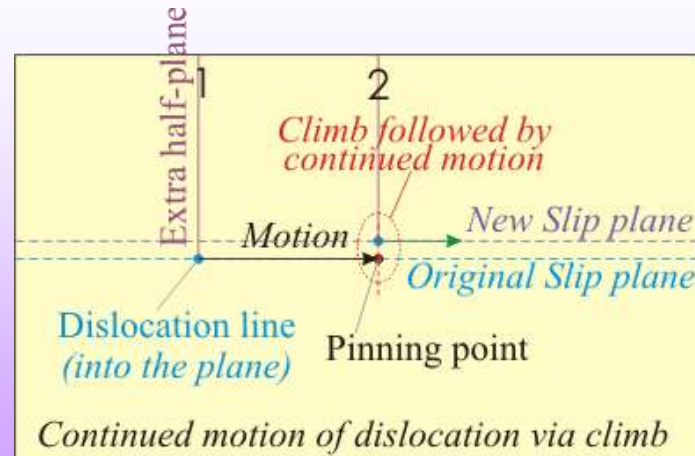
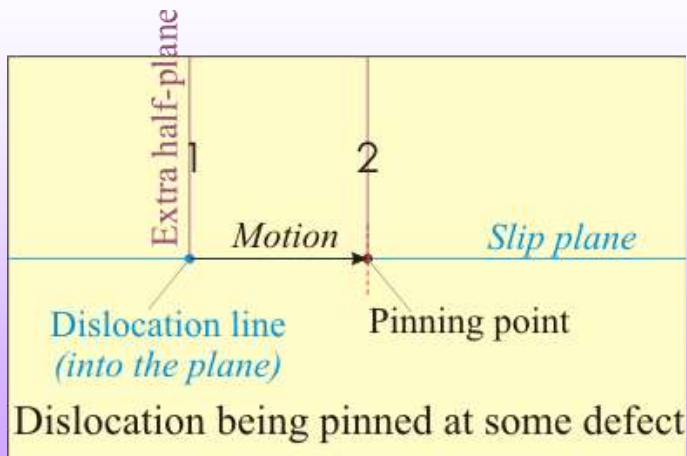
Conservative
(Glide)

Motion of dislocations on the slip plane

Non-conservative
(Climb*)

Motion of dislocation $\perp$ to the slip plane

- ❑ For edge dislocation: as $\mathbf{b} \perp \mathbf{t} \rightarrow$ they define a plane $\rightarrow$ the *slip plane*.
- ❑ Climb involves addition or subtraction of a row of atoms below the half plane
 - ▶ +ve climb = climb up $\rightarrow$ removal of a row of atoms
 - ▶ -ve climb = climb down $\rightarrow$ addition of a row of atoms.
- ❑ Importance of climb: climb plays an important role in many ways.
 - If an edge dislocation is 'stuck' at some obstacle on a *particular slip plane*, then it can continue to glide if climb can climb 'above' that slip plane. This way climb plays an important role in facilitating continued slip.
 - Vacancy concentration in the crystal can decrease due to +ve climb.

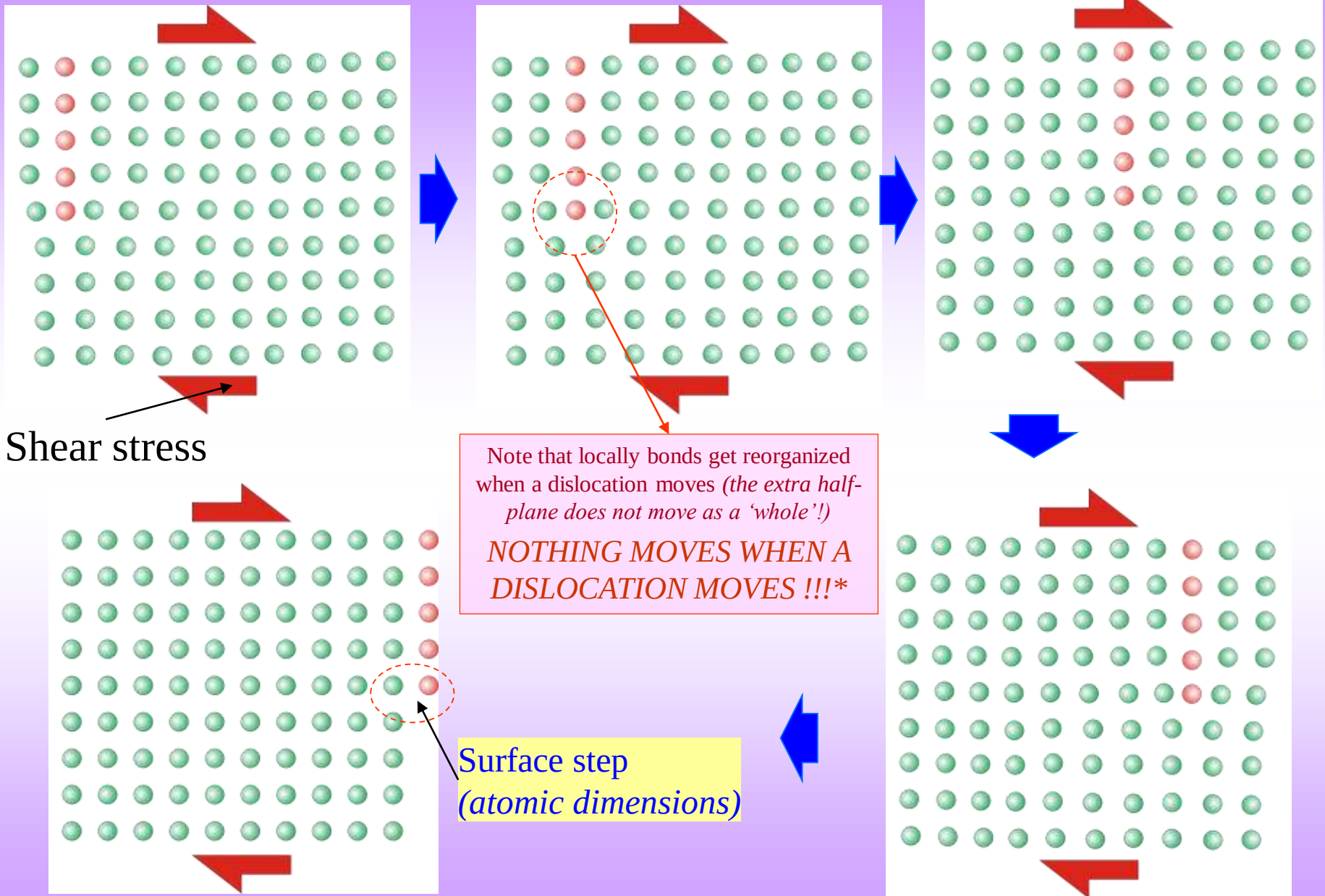


* There is even an interesting phenomenon called conservative climb!!

(Also, non-conservative does not mean that 'conservation of mass' is not valid. Just that, mass at the defect is not conserved.)

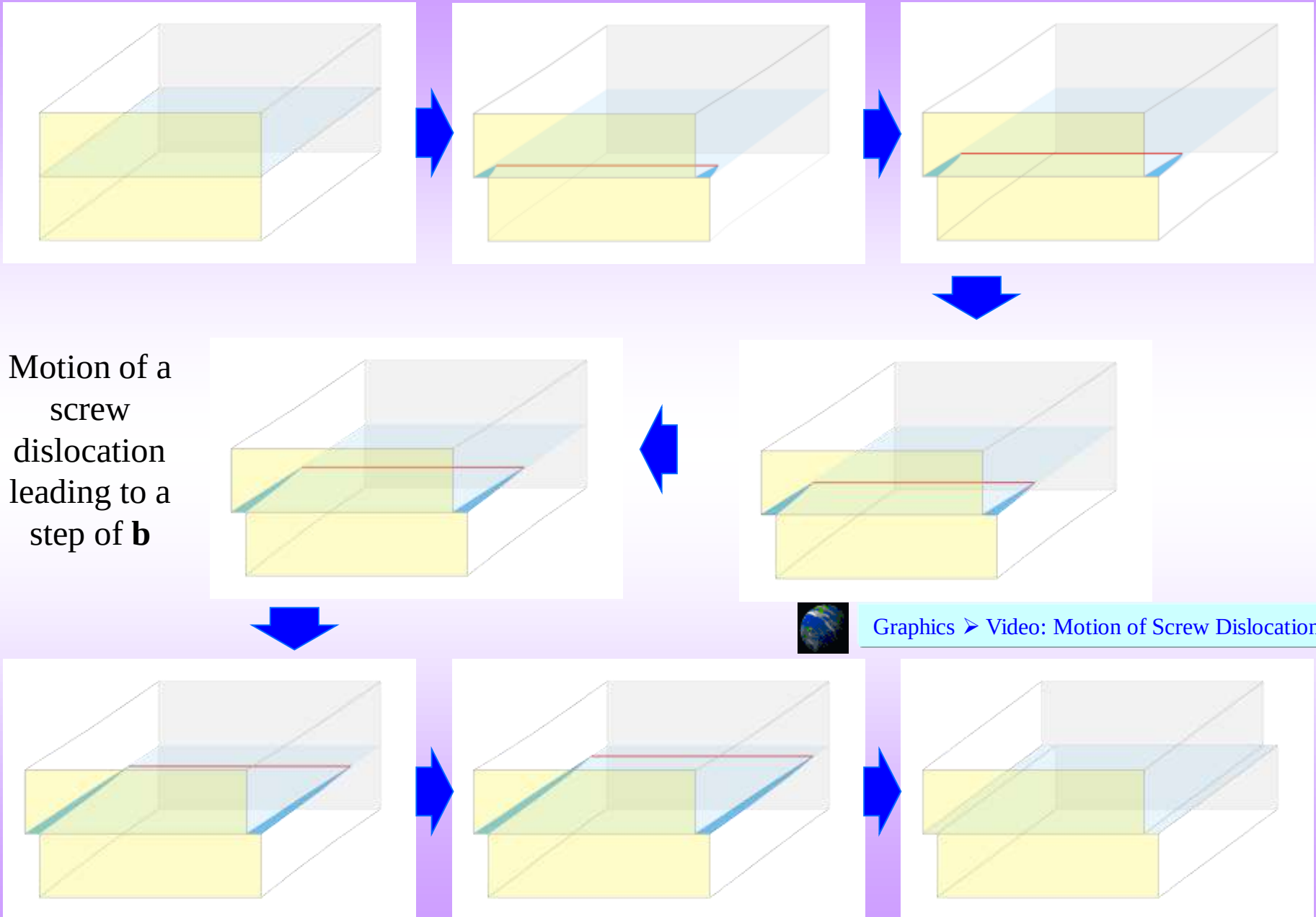
Edge Dislocation Glide

Motion of an edge dislocation leading to the formation of a step (of 'b')

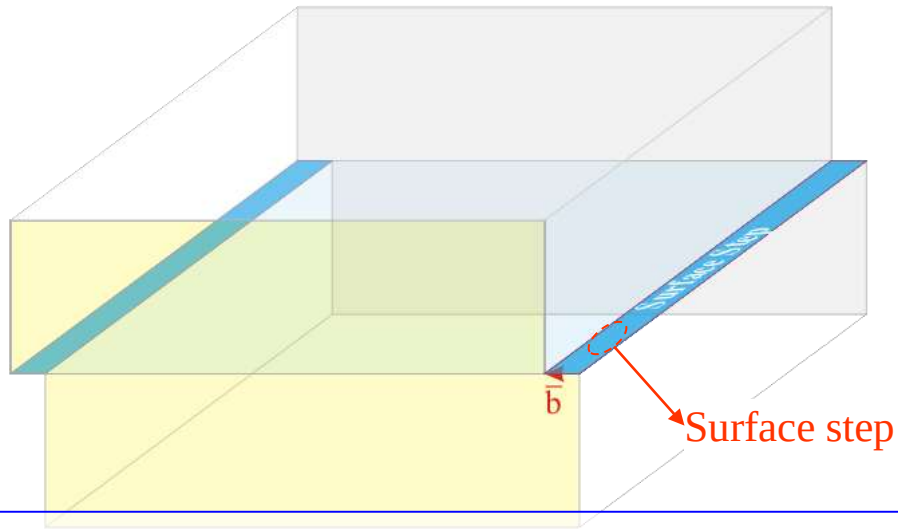


* Like in the propagation (motion) of a transverse wave, “nothing” (no matter) moves in the direction of the wave.

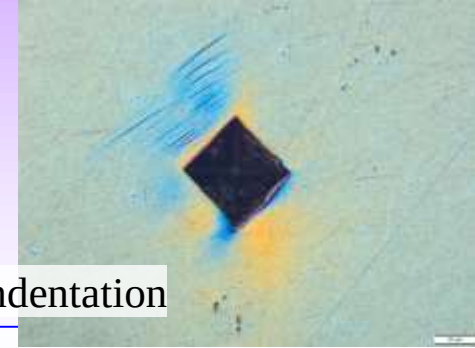
Screw Dislocation Glide



Note: Schematic diagrams



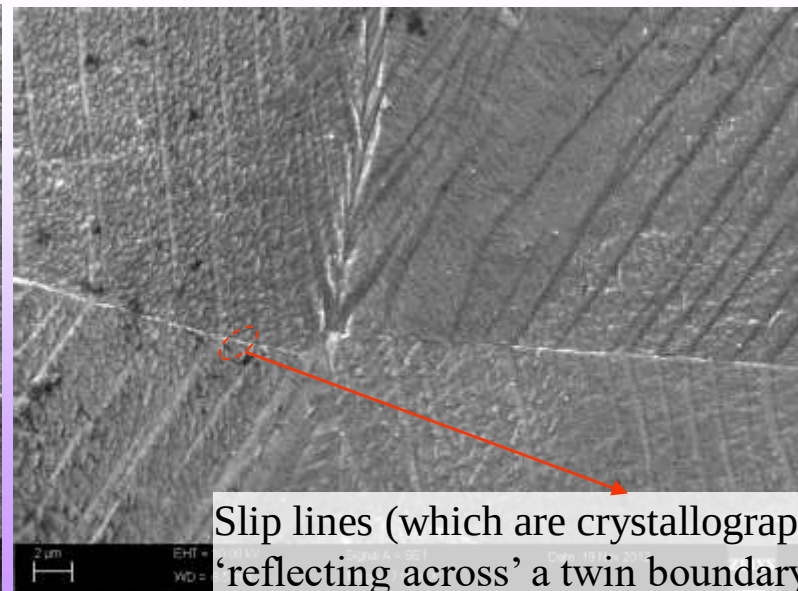
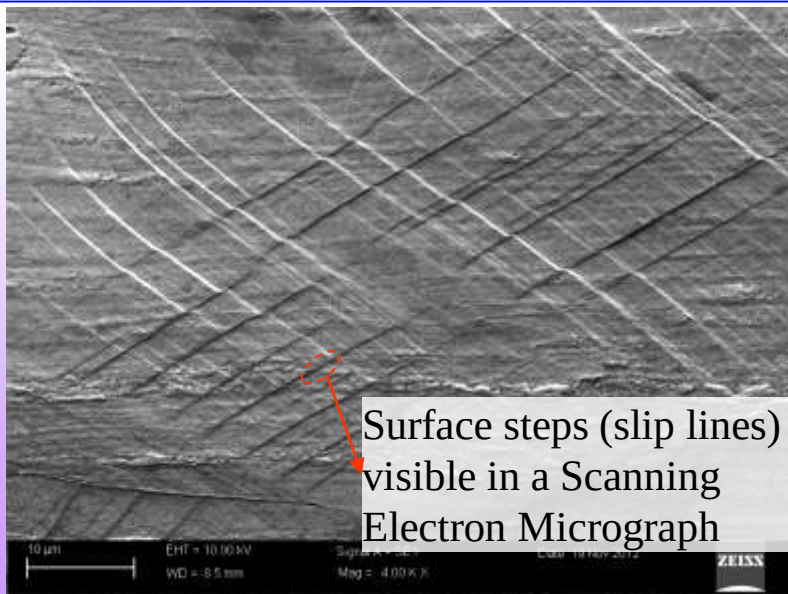
- When the dislocation leaves the crystal, the stress field associated with it is relieved.
- However, it costs some energy to create the extra surface corresponding to the step.



Slip lines around an indentation

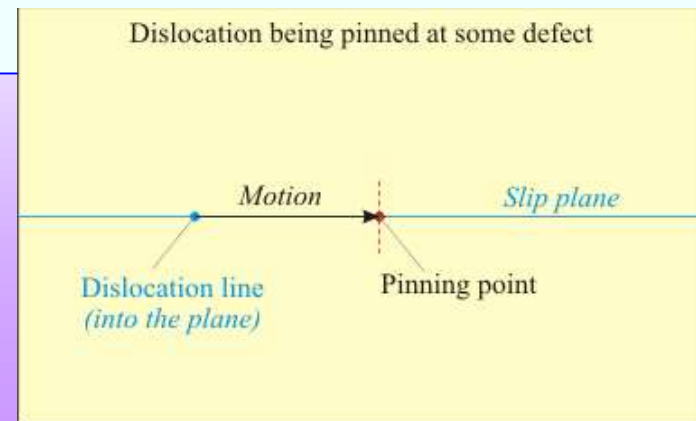
■ Are these steps visible?

These steps being of atomic dimensions are not visible in optical microscopes. However, if many dislocations operate on the same slip plane then a step of nb ($n \sim 100\text{s}-1000\text{s}$) is created which can even be seen in an optical microscope (*called the slip lines*).



Dislocations leaving the slip plane

- ❑ As it was observed the ‘first step’ of plastic deformation is the motion of a dislocation leaving the crystal (or to some other interface bounding the crystal) → leading to the formation of a step.
- ❑ For continued plastic deformation it is necessary that dislocations continue to move and leave the crystal. Hence, any impediments to the motion of a dislocation will lead to ‘hardening’ of the crystal and would ‘stall’ plastic deformation (*the pinning of a dislocation*).
- ❑ Once a dislocation has been pinned it can either ‘break down the barrier’ or ‘bypass’ the barrier.
- ❑ Bypassing the barrier can take place by mechanisms like:
 - Climb ➤ Cross Slip ➤ Frank-Read mechanism
- ❑ In climb and cross slip the dislocation leaves/changes its ‘current’ slip plane and moves to another slip plane thus avoiding the barrier
- ❑ However, these processes (climb and cross slip) can occur independent of the pinning of the dislocation!



Dislocation leaving/changing the slip plane

Edge dislocation

Climb

Non-conservative:
involves mass transport*

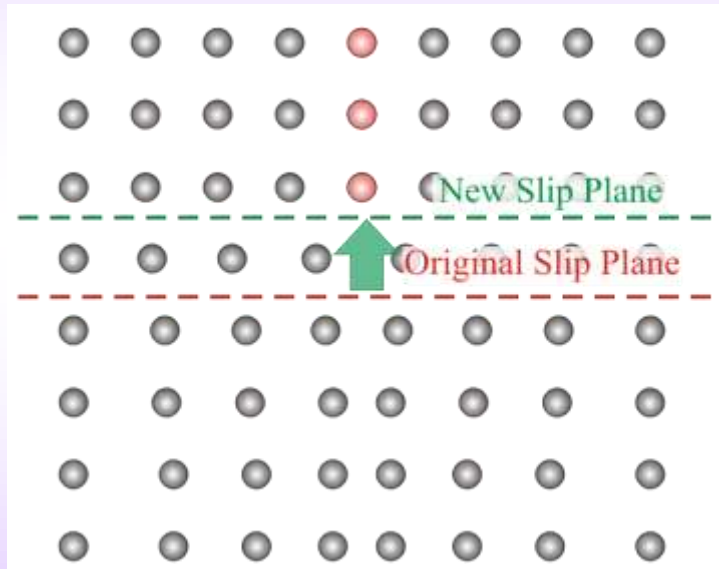
Screw dislocation

Cross Slip

Conservative

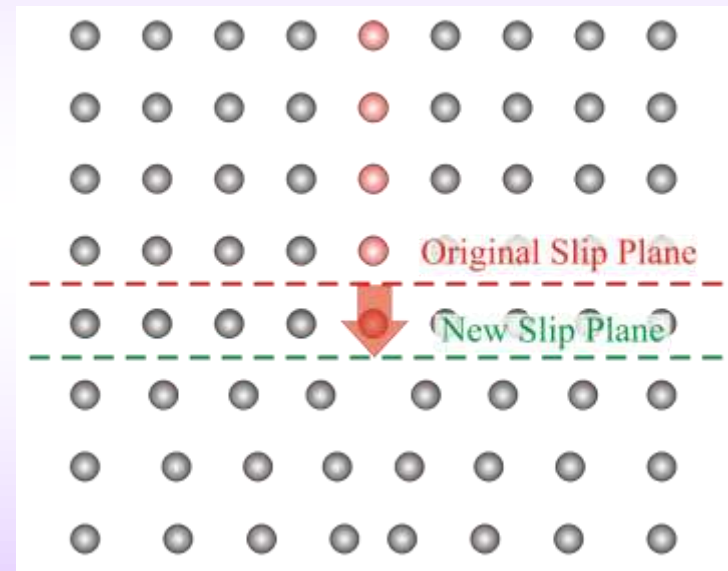
In climb an edge dislocation moves to an adjacent parallel plane, but in cross slip a screw dislocation moves to a plane inclined to the original plane.

Climb of an Edge Dislocation



Positive climb

Removal of a row of atoms



Negative climb

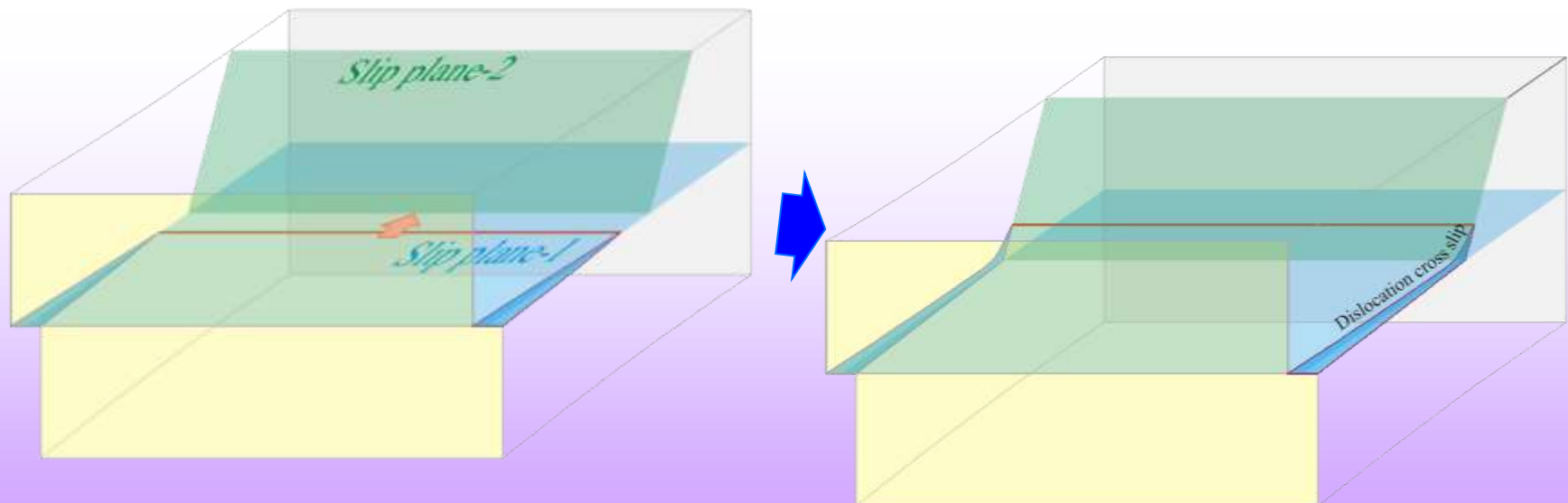
Addition of a row of atoms

Removal of a row of atoms leads to a decrease in vacancy concentration in the crystal and negative climb leads to an increase in vacancy concentration in the crystal.

**Conservative climb is also possible!! → by motion of prismatic edge loop on the slip plane*

Screw dislocation: Cross Slip

- ❑ Let the dislocation be moving on SP1 (as the resolved shear stress is maximum on Slip Plane-1 (SP1)). *Note: SP1 and SP2 are equivalent and are crystallographically determined slip planes.*
- ❑ The figures below show the cross slip of a screw dislocation line from SP1 to Slip plane-2 (SP2). This may occur if the dislocation is 'pinned' in slip plane-1.
- ❑ For such a process to occur the Resolved Shear Stress on SP2 should be at least greater than the Peierls stress
(often stresses higher than the Peierls stress has to be overcome due to the presence of other stress fields).
- ❑ It is to be noted that SP1 & SP2 are (usually) crystallographically equivalent, i.e. if SP1 is $(111)_{\text{CCP Crystal}}$ then SP2 can be $(\bar{1}\bar{1}1)_{\text{CCP Crystal}}$.



The dislocation is shown cross-slipping from the blue plane to the green plane

➤ As we have seen *slip* (a technical term) is one of the many mechanisms by which plastic deformation can occur.

The first step of plastic deformation by slip (at the fundamental level) is the motion of a dislocation **leaving the crystal**.

- ☐ By externally applied force (or some other means!) stress has to be ‘generated’ within the crystal.
- ☐ The slip plane should feel **shear stresses**.
- ☐ The shear stress should exceed the ‘Critical Resolved Shear Stress (CRSS)’ or equivalently its atomic level counterpart, the Peierls stress.
- ☐ The dislocation should leave the crystal creating a surface step of height ‘ b ’.

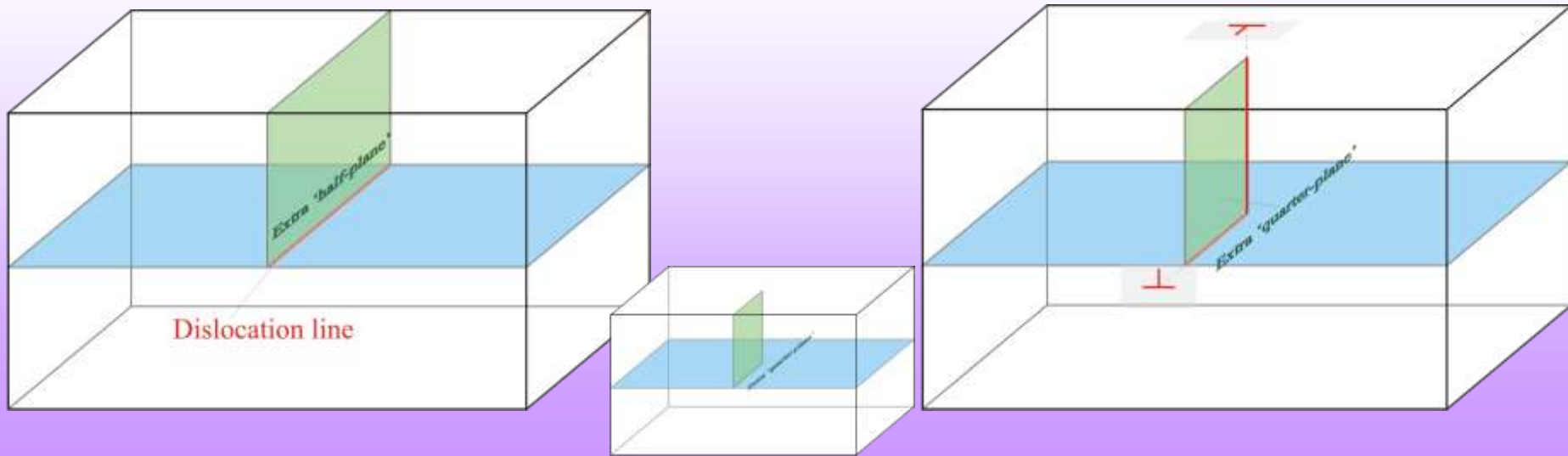
The process ahead of this which leads to an arbitrary shape change is rather complicated and we will deal with a part (a very little part) of it later.

Where can a dislocation line end?

- ❑ Dislocation line cannot end inside the crystal (*abruptly*)
- ❑ The dislocation line:
 - Ends on a free surface of the crystal
 - Ends on an internal surface or interface
 - Closes on itself to form a loop
 - Ends in a *node*
- ❑ A *node* is the intersection point of more than two dislocations
- ❑ The vectoral sum of the Burgers vectors of dislocations meeting at a node = 0

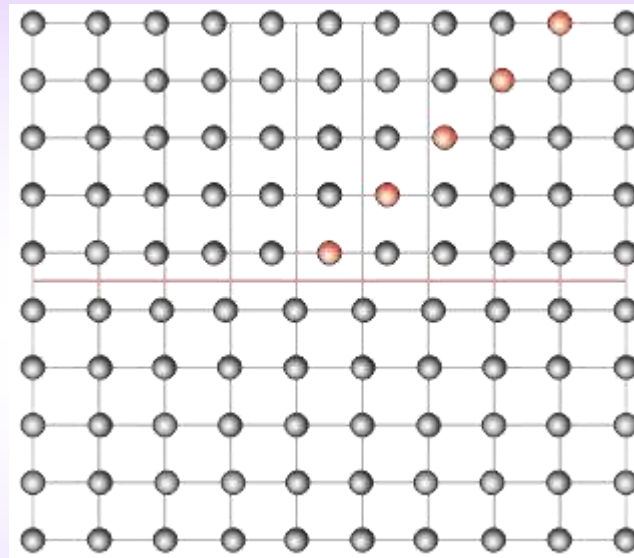
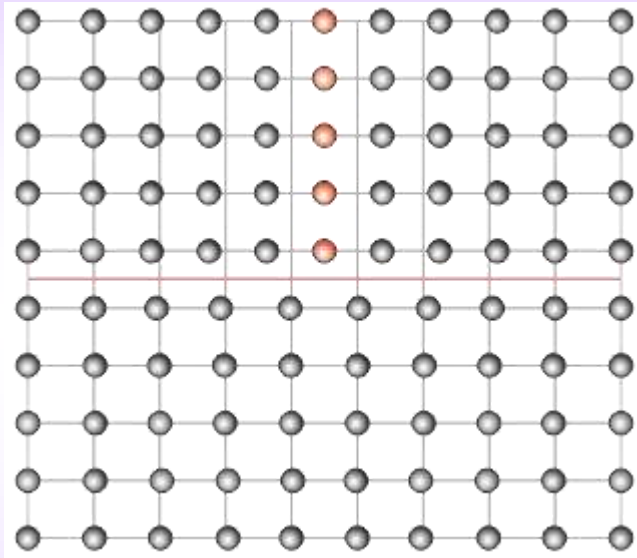
Funda Check  What about the introduction of a quarter plane of atoms- doesn't the dislocation line end inside the crystal?

- As seen in the figure below there are two sections to the dislocation line ending on free surface of the crystal and hence not inside the crystal.



Where is the extra half-plane in an edge dislocation?

- Multiple planes can be visualized as the extra half plane (two examples in the figures below). Based on the imaging conditions in a high-resolution lattice fringe imaging system, any of these may be highlighted.



Positive and Negative dislocations

- ❑ As we have seen when there are two or more EDGE dislocations in a slip plane one of them is assigned a +ve sign and the other one a -ve sign (*done arbitrarily*)
- ❑ In the case of screw dislocations the Right Handed Screw (RHS) Dislocation is *Structurally Distinct* from the **Left Handed Screw** (LHS) Dislocations
 - In the case of RHS dislocation as a clockwise circuit (Burgers) is drawn then a helical path leads into the plane of the

Energy of dislocations

- ❑ The presence of a dislocation distorts the bonds and costs energy to the crystal. Hence, dislocations have distortion energy associated with them
- ❑ The energy is expressed as Energy per unit length of dislocation line
→ Units: [J/m]
- ❑ Edge → Compressive and tensile stress fields
Screw → Shear stress fields
- ❑ The energy of a dislocation can approximately be calculated from linear elastic theory. The distortions are very large near the dislocation line and the linear elastic description fails in this region → called the **Core** of the dislocation (*estimates of this region range from b to $5b$ depending on the crystal in question*). The structure and energy of the core has to be computed through other methods and the energy of the core is about **1/10** the total energy of the dislocation.
- ❑ The formula given below gives reasonable approximation of the dislocation energy.

Energy of dislocation

E

Elastic

Non-elastic (Core)

~E/10

The core energy is about 1/10th the total energy of the dislocation

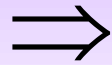
Elastic Energy of a dislocation / unit length →

$$E_d \sim \frac{1}{2} G b^2$$

$G \rightarrow (\mu)$ shear modulus
 $b \rightarrow |\mathbf{b}|$

- ❑ As it costs energy to put a dislocation in a crystal:
 - Dislocations tend to have as small a **b** as possible
 - There is a **line tension** associated with the dislocation line
 - Dislocations may dissociate into **Partial Dislocations** to reduce their energy

$$E_d \sim \frac{1}{2} Gb^2$$



Dislocations will have as small a **b** as possible

Dislocations
(in terms of lattice translation)

Full

b → Full lattice translation

Partial

b → Fraction of lattice translation

Another formula for the energy (Edge dislocation)

$$E_d^{edge} \sim \frac{Gb^2}{4\pi(1-\nu)} \left[2 + \ln\left(\frac{\gamma_0}{b}\right) \right]$$

γ_0 - size of the control volume $\sim 70b$

Core contribution

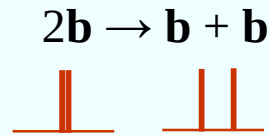
$$E_d^{screw} \sim \frac{Gb^2}{4\pi} \left[2 + \ln\left(\frac{\gamma_0}{b}\right) \right]$$

$$E_d^{mixed} \sim \frac{Gb^2}{4\pi} \left[\frac{\sin^2 \theta}{(1-\nu)} + \cos^2 \theta \right] \ln\left(\frac{\gamma_0}{b}\right)$$

Dissociation of dislocations

- ❑ As the dislocation line energy varies as b^2 (per unit length of the dislocation line), dislocations may dissociate to reduce their energy.
- ❑ The dissociation of a dislocation (or for that matter the (re)combination can be written as a reaction. E.g. in the example considered below, $2b \rightarrow (b+b)$.

Consider the reaction*:



Change in energy:

Initial energy before splitting into partials: $G(2b)^2/2 = 2Gb^2$

Energy after splitting into partials: $2[G(b)^2/2] = G(b)^2$

Reduction in energy = $2Gb^2 - Gb^2 = Gb^2$.

$\Rightarrow$ The reaction would be favorable.

* Note that this example is considered for illustration purpose only (here a full dislocation is not splitting into partials).

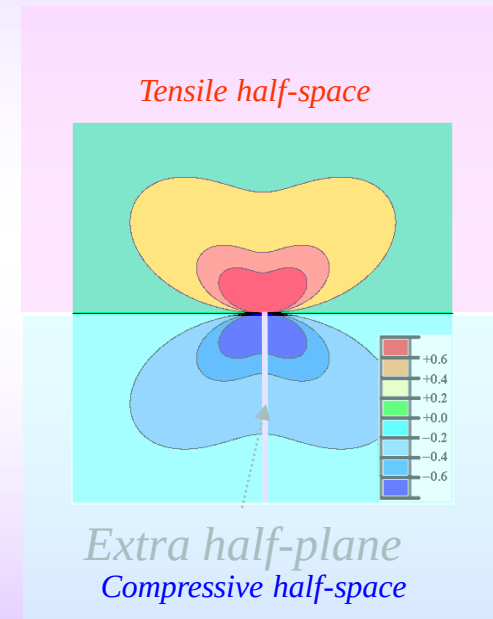
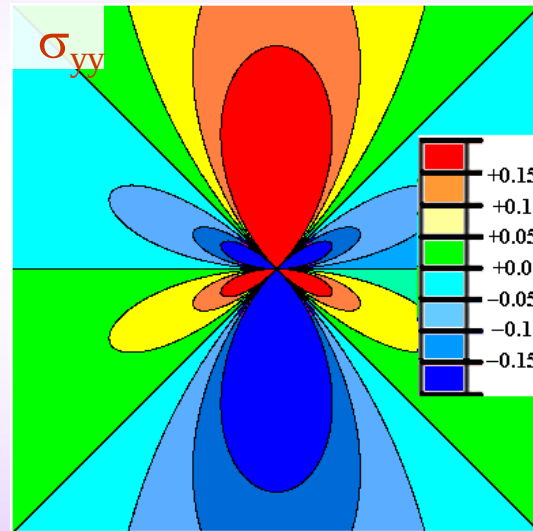
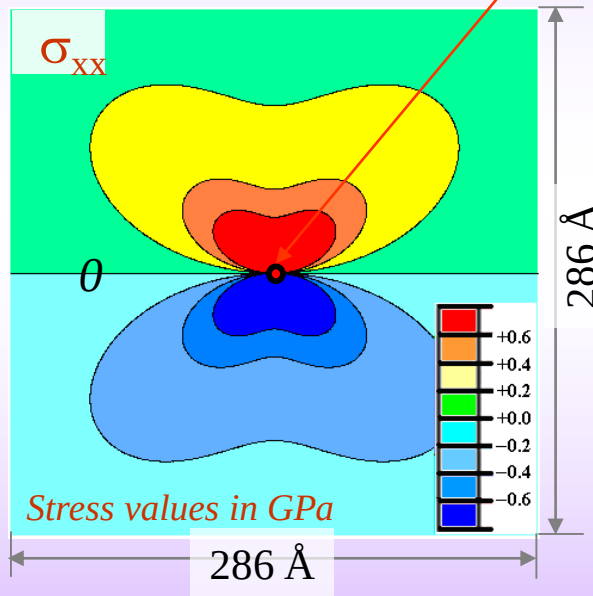
- ❑ An **edge dislocation** in an infinite body has compressive stress field (–ve sign) in the *region of the ‘extra’ half-plane* and tensile stress fields (+ve sign) in the *region of the ‘missing’ half-plane*. The slip plane divides these two regions and corresponds to zero stresses.
- ❑ These stress fields will be altered in a finite body.
- ❑ Asymmetric position of the dislocation in the crystal will also alter the stress field described by the standard equations (*as listed below*).
- ❑ The core region is ignored in these equations (*which hence have a singularity at $x = 0, y = 0$*) (Core being the region where the linear theory of elasticity fails).
- ❑ Obviously a real material cannot bear such ‘singular’ stresses.
- ❑ The interaction of the stress fields of the dislocations with: (i) those originating from externally applied forces and (ii) other internal stress fields → determines the motion of dislocation → leading to many aspects of **mechanical behaviour of materials**.

$$\left. \begin{aligned} \sigma_{xx} &= \frac{Gb}{2\pi(1-\nu)} \frac{y(3x^2 + y^2)}{(x^2 + y^2)^2} \\ \sigma_{yy} &= \frac{-Gb}{2\pi(1-\nu)} \frac{y(x^2 - y^2)}{(x^2 + y^2)^2} \\ \sigma_{xy} = \tau_{xy} &= \frac{-Gb}{2\pi(1-\nu)} \frac{x(x^2 - y^2)}{(x^2 + y^2)^2} \end{aligned} \right\} \text{stress fields}$$

The material is considered **isotropic** (two elastic constants only – E & ν or G & ν)
 → in reality crystals are **anisotropic** w.r.t to the elastic properties.

- ❑ Note that the region near the dislocation has stresses of the order of GPa.
- ❑ These stresses are the self stresses of the dislocation and a straight dislocation line cannot move under the action of self stresses alone (*in an infinite body*).
- ❑ *A dislocation can interact with other defects (which are associated with their own stress fields) in the material, via these 'long range' stress fields.*

Position of the Dislocation line → into the plane



$$\sigma_{xx} = \frac{Gb}{2\pi(1-\nu)} \frac{y(3x^2 + y^2)}{(x^2 + y^2)^2}$$

$$\sigma_{yy} = \frac{-Gb}{2\pi(1-\nu)} \frac{y(x^2 - y^2)}{(x^2 + y^2)^2}$$

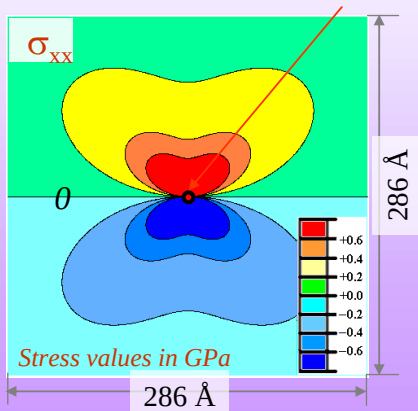
Assumptions:

Iso-stress contours, infinite body, straight edge dislocation along 'z', plane strain condition, core of dislocation ignored (i.e. the equations should not be used close to the dislocation line– in the range of 'b' to '5b')

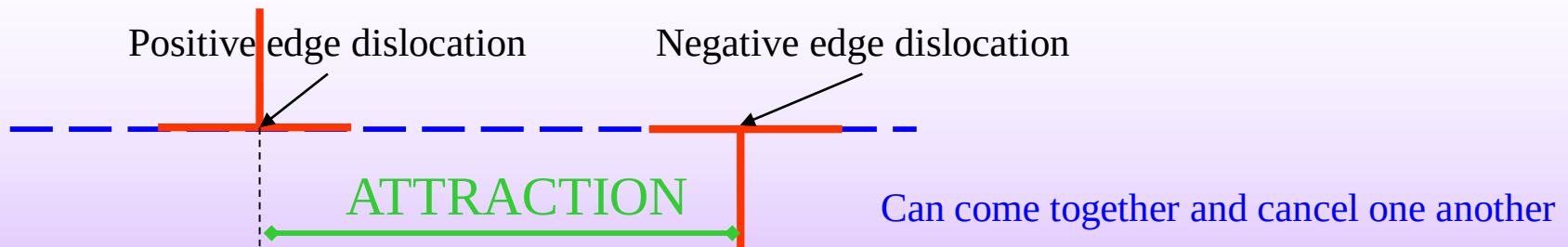
➤ Material properties used in the plots are in the last slide

Note that the stresses near the dislocation line reaches values in GPa. Yield stress is usually in 100s of MPa. How does a material not yield under the effect of such high stresses?

- ❑ The predominant mechanism of plastic deformation is slip, which involves motion of a large number of dislocations (ultimately leaving interfaces).
- ❑ The stresses described here are due to the ‘dislocation itself’ (the causative agent for plastic deformation by slip) and they are the *elastic stress fields* associated with the dislocation.



- ❑ Here we only consider elastic interactions between edge dislocations on the same slip plane.
- ❑ This can lead to Attractive and Repulsive interactions.
- ❑ To understand these interaction we need to consider Positive and Negative edge dislocations. If a single edge dislocation is present in a material it can be called either positive or negative. If two (or more) dislocations are present on the same slip plane, with the extra half-plane on two different sides of the slip plane, then one of them is positive and the other negative.
- ❑ The picture (region of attraction and repulsion) gets a little ‘detailed’ if the two dislocations are arbitrarily oriented. [See [Stress Fields of Dislocations](#)]



- ❑ Slip system (family) $\rightarrow \langle 110 \rangle, \{111\}$. E.g. of a specific slip system: $[-110], (111)$.
 - The correct combination of the plane and direction has to be chosen for a particular dislocation under consideration. E.g., $[1-10]$ direction along with the (111) plane (as this direction lies on the given plane).
- ❑ Perfect dislocations can split into partials (*Shockley partials considered first*) to reduce their energy. As we shall see, this can be best understood with edge dislocation, where two atomic planes form the perfect (full) dislocation and when these two atomic planes separate we form partials (each partial has one atomic plane).
- ❑ The dissociation into partials leaves a **Stacking Fault*** between the two partials on the slip plane. Hence, in materials with **low stacking fault energy***, the perfect dislocation will tend to split into Shockley partials.
- ❑ The two partials repel each other and want to be as far as possible $\rightarrow$ but this leads to a larger faulted area (leading to an increase in energy) $\rightarrow$ depending on the stacking fault energy there will be an equilibrium separation between the partials.
- ❑ The Shockley partial has Burgers vector of the type: $(1/6)[211]$ type.
This is an important vector in the CCP crystals, as vectors of this family connect B site to C site and vice-versa.
- ❑ For a pure edge dislocation in a CCP crystal the 'extra half-plane' consists of two atomic planes. The partial dislocations consist of one 'extra' atomic plane each (but the Burgers vector of the partial is not perpendicular to the dislocation line- as in the case of the perfect edge dislocation).

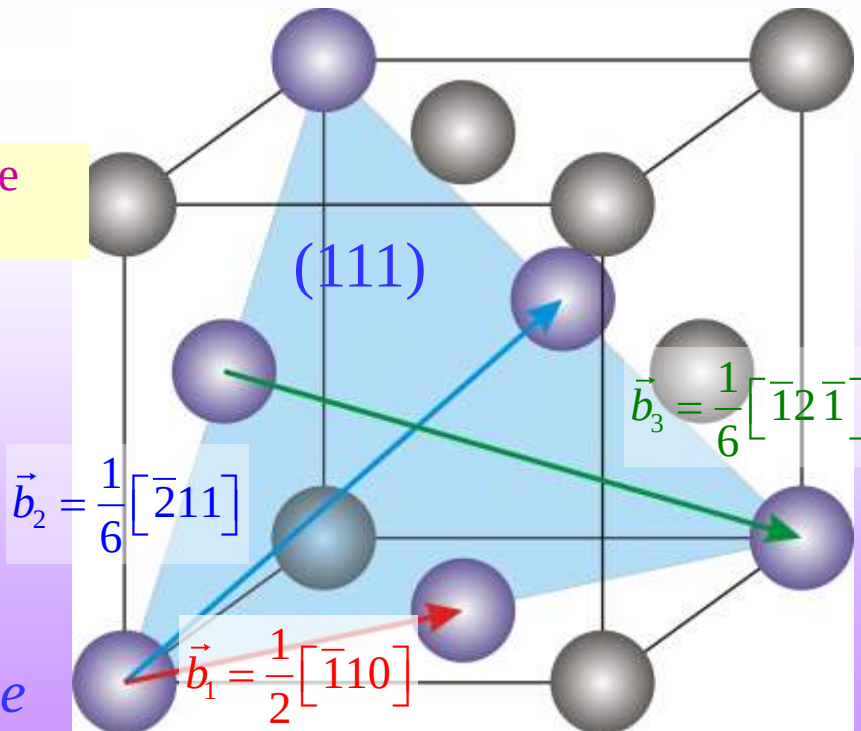
* Will be considered in the topic on 2D defects

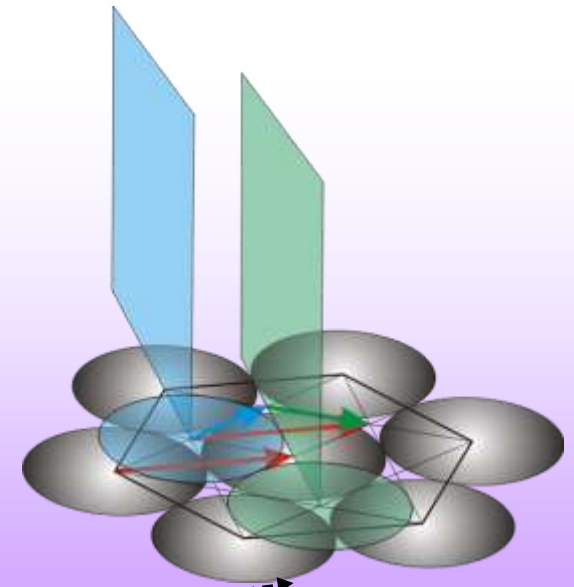
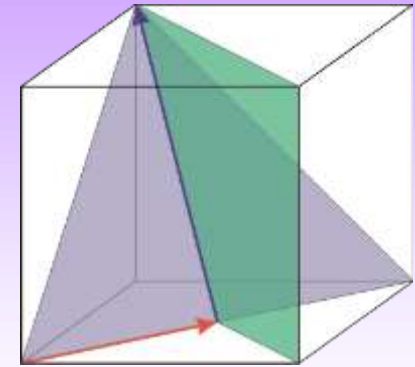
$$(\vec{b}_1)_s = \left(\frac{1}{2} [\bar{1}10] \right)_{(111)}$$

- ❑ Let us consider one member of the ‘family’ of perfect (full) dislocations:
- ❑ This geometry is explained in this and the next page.
- ❑ For the perfect edge dislocation, the extra half-plane comprises of two atomic planes.
- ❑ These two atomic planes can ‘separate out’ to give rise to Shockley partials (considered later).

Perfect Edge
Dislocation

(111)
Slip plane





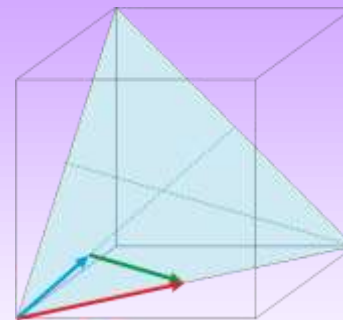
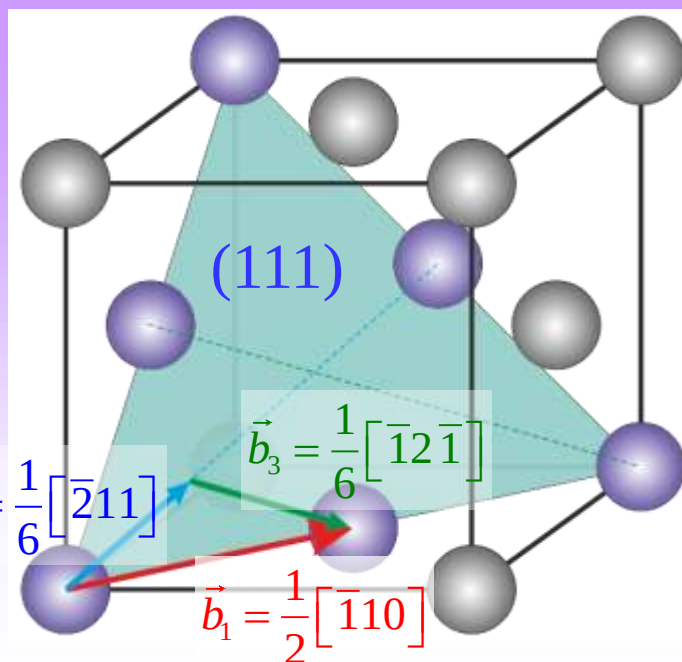
One crystal plane is 'lattice plane' decorated with two atomic planes.

CCP: Shockley Partial Dislocations

- ❑ We had noted that the extra ‘half-plane’ consists of two atomic planes.
- ❑ These planes can separate out to give rise to Shockley partial dislocations. This happens so as to reduce the strain energy associated with the dislocation.
- ❑ The Burgers vector of the partial dislocations are not full lattice translation vectors. In the case of Shockley partial dislocation, the length is $1/3^{\text{rd}}$ of the $\langle 112 \rangle$ type vector.
- ❑ These partials lie on the $\{111\}$ slip plane and are glissile (i.e. can move).
- ❑ The region between the two partial dislocations is a stacking fault in the CCP crystal (i.e. the stacking fault is bounded by the two partial dislocations). This implies that in the formation of the partial dislocations, not only do we have to consider the strain energy associated with the dislocation, but also the stacking fault energy (SFE) of the crystal.
- ❑ The separation between the partials, which is the width of the stacking fault ‘ribbon’, is determined by the SFE of the crystal. In crystals with low SFE like Cu, the partials are widely separated and in crystals with high SFE (like Al) the separation between the partials is small (and in fact can be considered a perfect dislocation for ‘practical purposes’).

CCP

Shockley Partial



Some of the atoms are omitted for clarity, Full vectors $[\bar{2}11]$ (blue vector) & $[\bar{1}2\bar{1}]$ (green vector) shown in the left figure

$$\left(\frac{1}{2}[\bar{1}10]\right)_{(111)} \rightarrow \left(\frac{1}{6}[\bar{1}2\bar{1}]\right)_{(111)} + \left(\frac{1}{6}[\bar{2}11]\right)_{(111)}$$

Shockley Partial

Reduction in the strain energy of the dislocation on splitting into Shockley partials

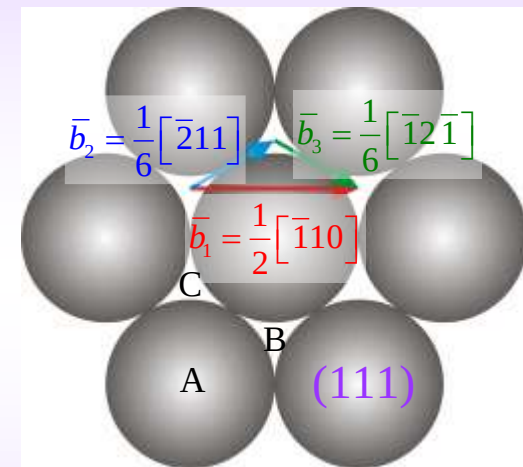
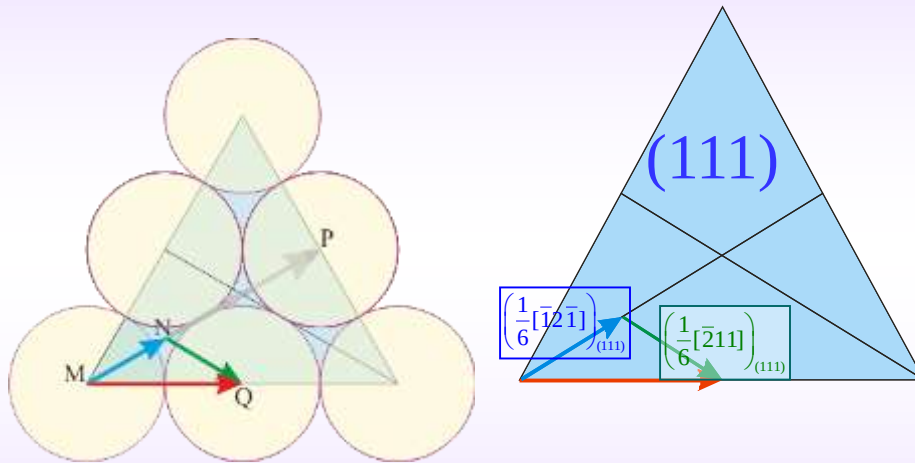
$$|\vec{b}_2| = \left(\frac{\sqrt{1^2 + 2^2 + 1^2}}{6}\right)^2 = \left(\frac{\sqrt{6}}{6}\right)^2 = \frac{1}{6}$$

$$(b_2^2 + b_3^2) = 1/6 + 1/6 = 1/3$$

$$b_1^2 > (b_2^2 + b_3^2)$$

$$\frac{1}{2} > \frac{1}{3}$$

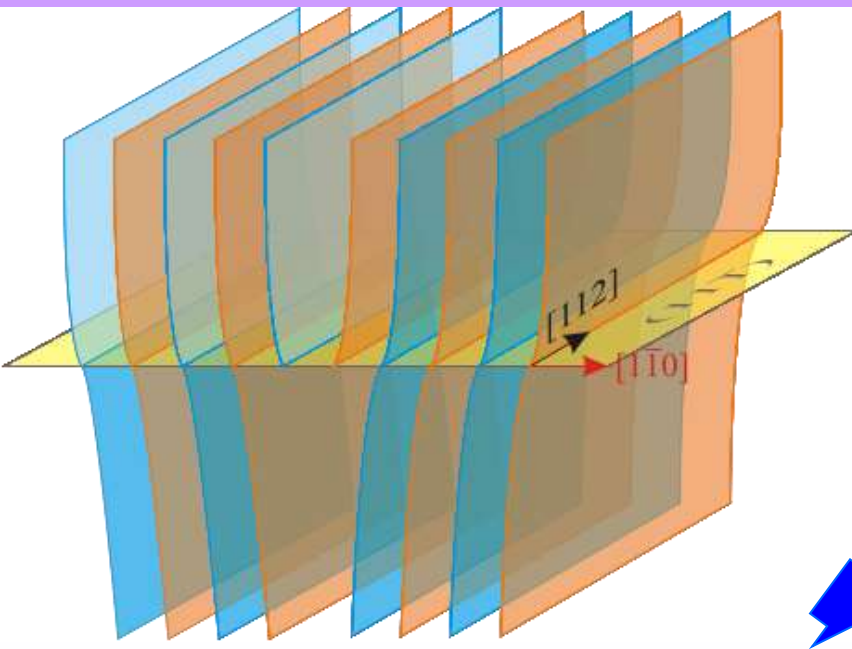
- ❑ The vectors associated with the partials (blue and green vectors), connect the 'B' position to the 'C' position (MN in figure below) & 'C' position to the 'B' position (NQ) in the {111} close packed plane (i.e. this is their significance).
- ❑ The dislocations with these Burgers vectors are partials because, the lattice translation vector along this direction is $\frac{1}{2}\langle 112 \rangle$ and the vectors MN & NQ are $\frac{1}{3}$ of the lattice translation vectors (i.e. $\frac{1}{3} \cdot \frac{1}{2}\langle 112 \rangle = \frac{1}{3}(\text{MP})$).



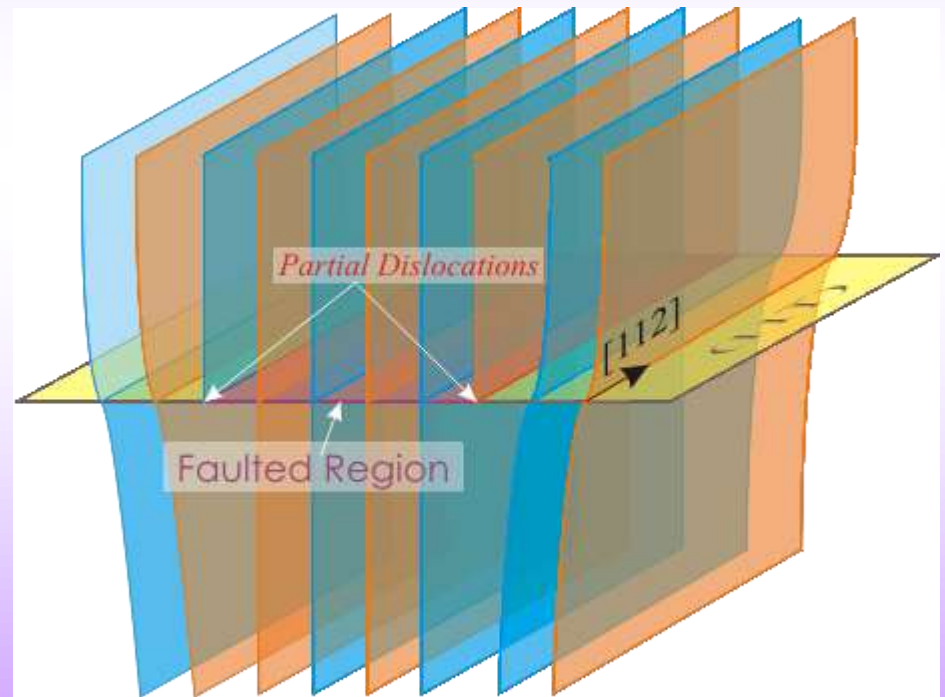
Energy of the dislocation is proportional to b^2 .

As the energy of the system is reduced on dissociation into partials the perfect dislocation will split into two partials.

Shockley Partials



Perfect edge dislocation ('full' Burgers vector) with two atomic 'extra-half' planes



Partial dislocations: each with one atomic 'extra-half' plane

Here one dislocation splits into two partials

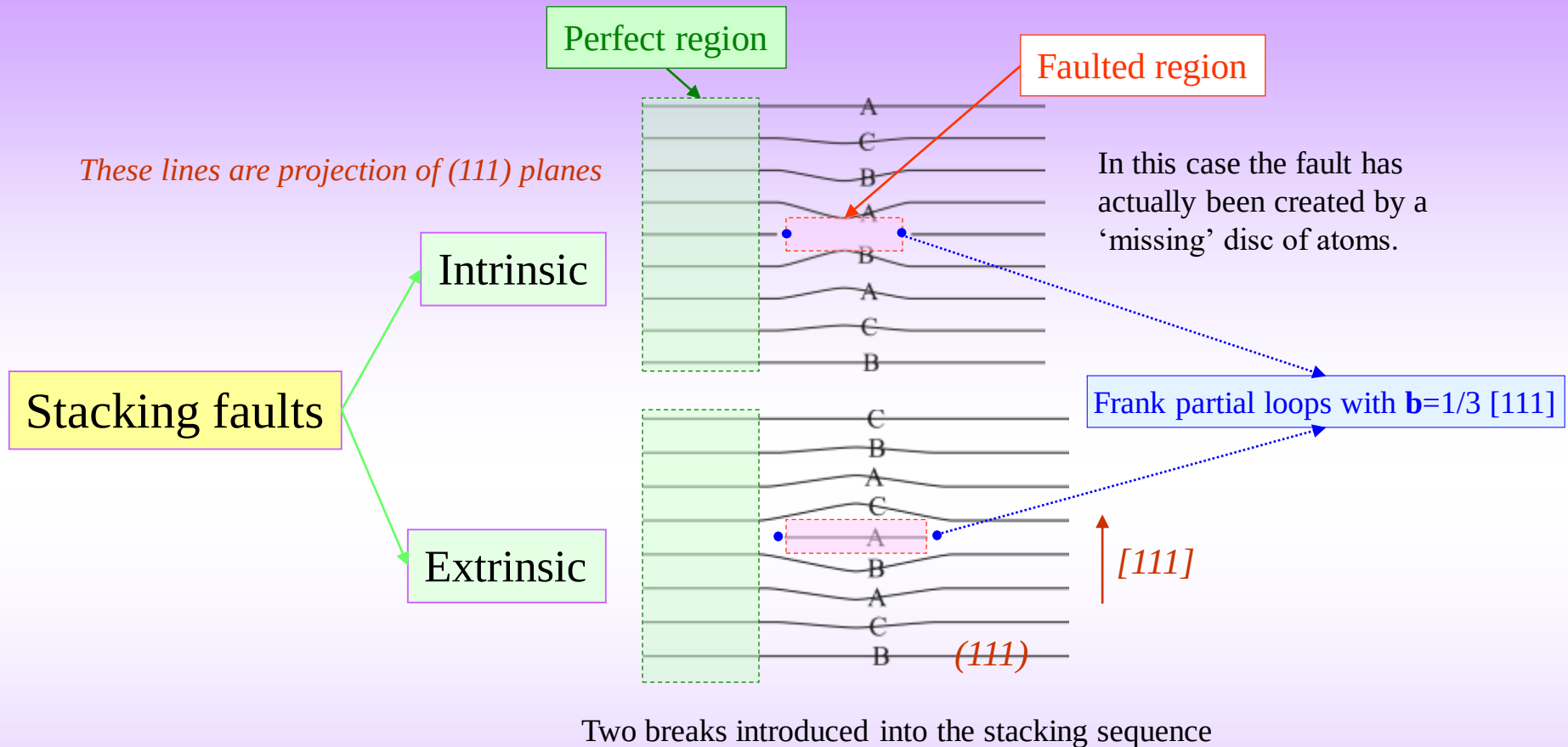
[Click here to see how two dislocations give rise to one dislocation](#)

Dislocation loops and Frank Partial dislocations

- ❑ Formed by insertion or removal of a disc of atoms from the (111) plane.
- ❑ The (111) crystal plane consists of three atomic planes and the lattice translation vector along [111] has a magnitude of $\sqrt{3}$ (the distance between the atomic planes along [111] is $\sqrt{3}/3$). The packing along this direction is: ABCABCABC...
- ❑ ➤ Removal of a disc → Intrinsic fault
➤ Insertion of a disc → Extrinsic fault
- ❑ $\mathbf{b}_{\text{Frank Partial}} = [111]/3$ (this is not a full lattice translation vector). The full lattice translation vector = [111].
- ❑ As $\mathbf{b}$ is not on a slip plane (a member of the {111} family) the dislocation cannot move conservatively (i.e. without mass transport) → is a **Sessile Dislocation** (as opposed to a **Glissile dislocation** (which can move, e.g. the Shockley partials)).
- ❑ Excess vacancies (quenched-in or formed by irradiation) can form an intrinsic fault (these may have hexagonal shape in some cases).
- ❑ This shows that a dislocation loop can have a completely edge character; but never a completely screw character.

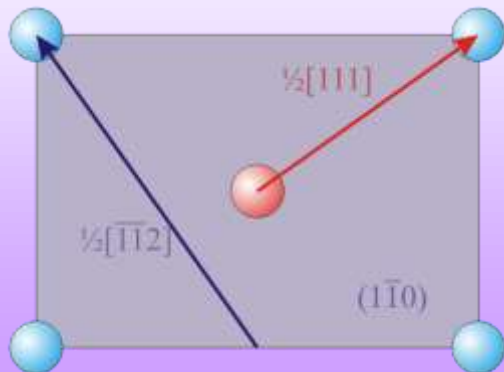
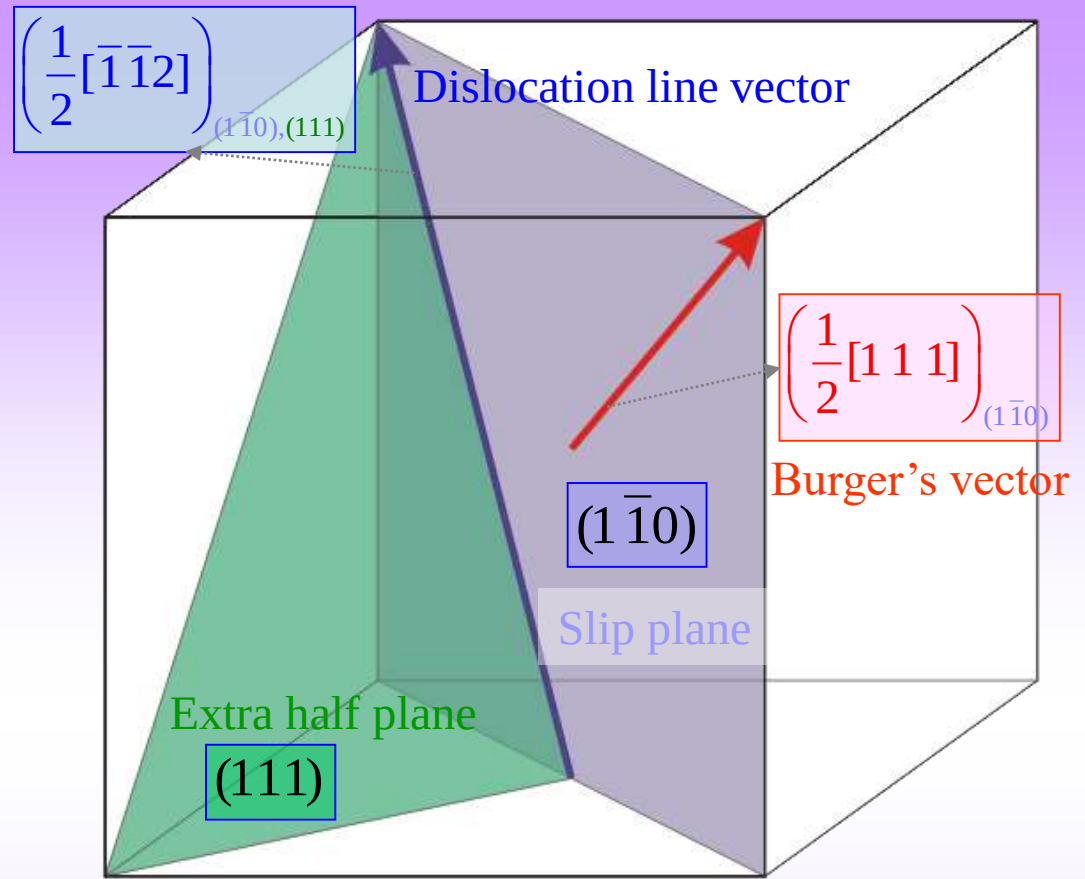
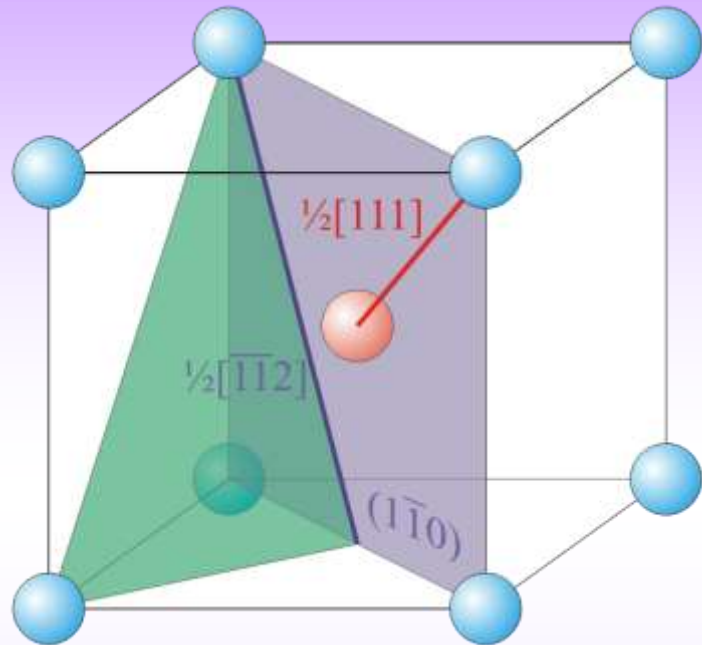
Frank partial dislocation loop bounding a stacking fault in CCP crystal

This shows that pure edge dislocation loop can exist (but a pure screw loop cannot exist)



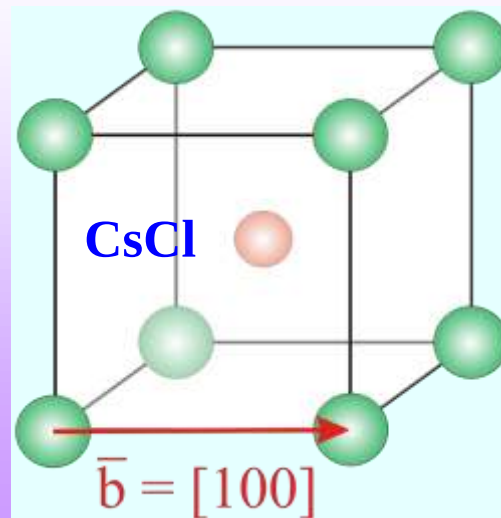
BCC

Pure edge dislocation



Dislocations in Ionic crystals

- ❑ In ionic crystals if there is an extra half-plane of atoms contained only atoms of one type then the charge neutrality condition would be violated → this is an unstable condition.
- ❑ This implies that Burgers vector has to be a full lattice translation vector:
 $\text{CsCl} \rightarrow \mathbf{b} = \langle \mathbf{100} \rangle$ *Cannot be $\frac{1}{2}\langle 111 \rangle$*
 $\text{NaCl} \rightarrow \mathbf{b} = \frac{1}{2} \langle \mathbf{110} \rangle$ *Cannot be $\frac{1}{2}\langle 100 \rangle$.*
- ❑ This makes Burgers vector large in ionic crystals:
 $\text{Cu} \rightarrow |\mathbf{b}| = 2.55 \text{ \AA}$
 $\text{NaCl} \rightarrow |\mathbf{b}| = 3.95 \text{ \AA}$.
- ❑ A large value for Burgers vector implies a higher Peierls stress. It so happens that the stress required to propagate cracks in such materials is lower (i.e. $\sigma_f < \sigma_y$) and hence ionic materials are very brittle (at low temperatures).



Formation of dislocations (in the bulk of the crystal)

- ❑ Due to accidents in crystal growth from the melt.*
- ❑ Mechanical deformation of the crystal.
- ❑ Nucleation of dislocation.
 - Homogenous nucleation of a dislocation required high stresses ($\sim G/10$).
 - Stress concentrators in the crystal can aid the process.
- ❑ Dislocation density increases due to plastic deformation mainly by multiplication of pre-existing dislocations.

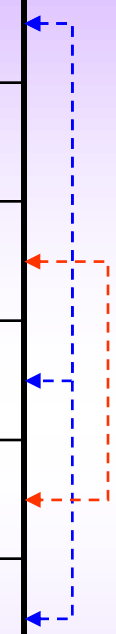
Typical values of Dislocation Density

- ❑ Dislocation density refers to the *length of dislocation lines in a volume of material* → hence the units are $[m/m^3]$
(it is better not to cancel the 'm' in the numerator and the denominator and write as $/m^2$ as the units m/m^3 is more physical!).
- ❑ Annealed crystal: dislocation density (ρ) $\sim 10^6 - 10^{10} \text{ m/m}^3$.
- ❑ Cold worked crystal: $\rho \sim 10^{12} - 10^{14} \text{ m/m}^3$.
- ❑ As the dislocation density increases the crystal becomes stronger (more about this later).

Note: in this context it is noteworthy that screw dislocations can actually play a role in crystal growth → Constructive role of dislocations.

Burgers vectors of dislocations in cubic crystals

Crystallography determines the Burgers vector
fundamental lattice translational vector lying on the slip plane

Monoatomic FCC	$\frac{1}{2}\langle 110 \rangle$	
Monoatomic BCC	$\frac{1}{2}\langle 111 \rangle$	
Monoatomic SC	$\langle 100 \rangle$	
NaCl type structure	$\frac{1}{2}\langle 110 \rangle$	
CsCl type structure	$\langle 100 \rangle$	
DC type structure	$\frac{1}{2}\langle 110 \rangle$	

A rule of thumb can be evolved as follows:

"Close packed volumes tend to remain close packed,
close packed areas tend to remain close packed &
close packed lines tend to remain close packed"

Close packed in this context implies 'better bonded'

Slip systems

- ❑ A combination of a **slip direction** (**b**) lying on a **slip plane** is called a slip system. This is described in terms of a family of directions and a family of planes.
- ❑ In close packed crystals it is a close packed direction lying on a close packed plane.
- ❑ In BCC crystals there are many planes with similar planar atomic density → there is no clear choice of slip plane. Hence, the slip lines are wavy.
- ❑ There might be more than one *active* slip system in some crystals (*e.g. BCC crystals below*).
→ *the active slip system gives rise to plastic deformation by slip.*
- ❑ Even if there is only one slip system is active at low temperature, more slip systems may become active at high temperatures → polycrystalline materials which are brittle at room temperature may become ductile at high temperatures.

Crystal	Slip plane(s)	Slip direction
FCC	{111}	<110>
HCP*	(0001), (0002)	<11 $\bar{2}$ 0>
BCC <i>Not close packed</i>	{110}, {112}, {123}	<111>

Anisotropic slip behaviour

Planar density:

$$\{110\} \rightarrow \frac{\sqrt{2}}{a^2} = \frac{1.414}{a^2}$$

$$\{112\} \rightarrow \frac{4}{a^2 \sqrt{6}} = \frac{1.633}{a^2}$$

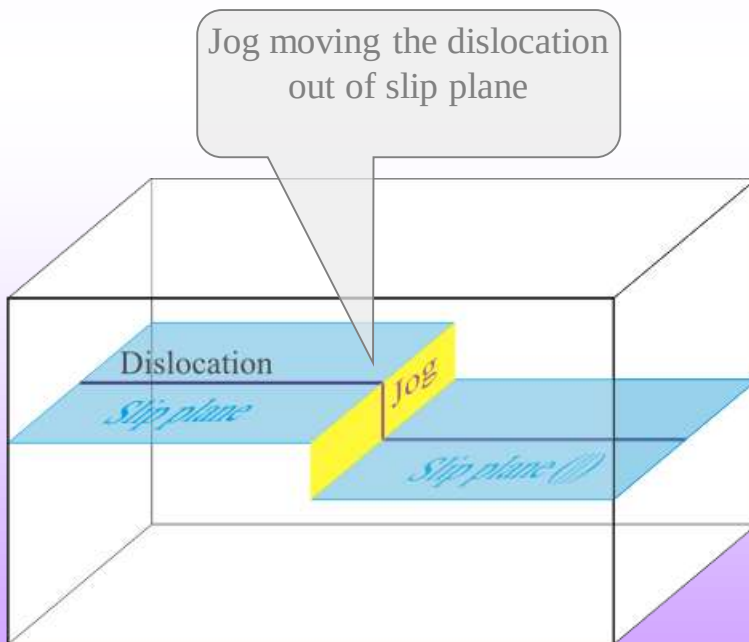
No clear choice of slip plane
⇒ Wavy slip lines

* In HCP metals, the c/a ratio will determine the slip plane. If c/a ratio is $> (c/a)_{ideal}$ then basal slip will be preferred.

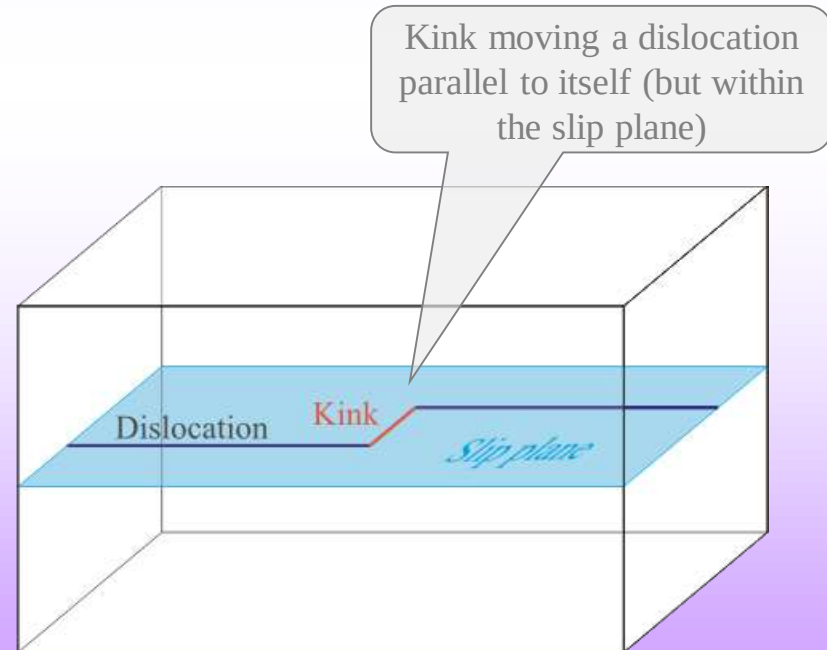
Jogs and Kinks

Defect in a defect!

- ❑ A straight dislocation line can have a break in it of two types:
 - A jog moves it out of the current slip plane (→ *to a parallel one*)
 - A kink leaves the dislocation on the slip plane.
- ❑ The **Jog** and the **Kink** can be considered as a defect in a dislocation line (*which itself is a defect* → *hence these are defects in a defect*).
- ❑ Jogs and Kinks can be produced by intersection of straight dislocations.



Jogs



Kinks

- ❑ Jogs and Kinks in a screw dislocation will have edge character.
- ❑ Jog in a Edge dislocation has Edge character and Kink in a edge dislocation has screw character.

Jogs and Kinks: Character Table

	Edge dislocation	Screw Dislocation
Jog	Edge character	Edge character
Kink	Screw Character	Edge character

Jogs

- ❑ The presence of a jog in a dislocation line increases the energy of the crystal.
- ❑ The energy of a jog per unit length is less than that for the dislocation (as this lies in the distorted region near the core of the dislocation).
- ❑ This energy is about 0.5-1.0 eV ($\sim 10^{-19}$ J) for metals.

$$E_{Jog} = \alpha G b_1^2 b_2$$

- $b_1 \rightarrow$ Burgers vector of the dislocation
- $b_2 \rightarrow$ Length of the jog
- $\alpha \rightarrow$ Constant with value $\in (0.5-1.0)$

Dislocation-Dislocation Interactions

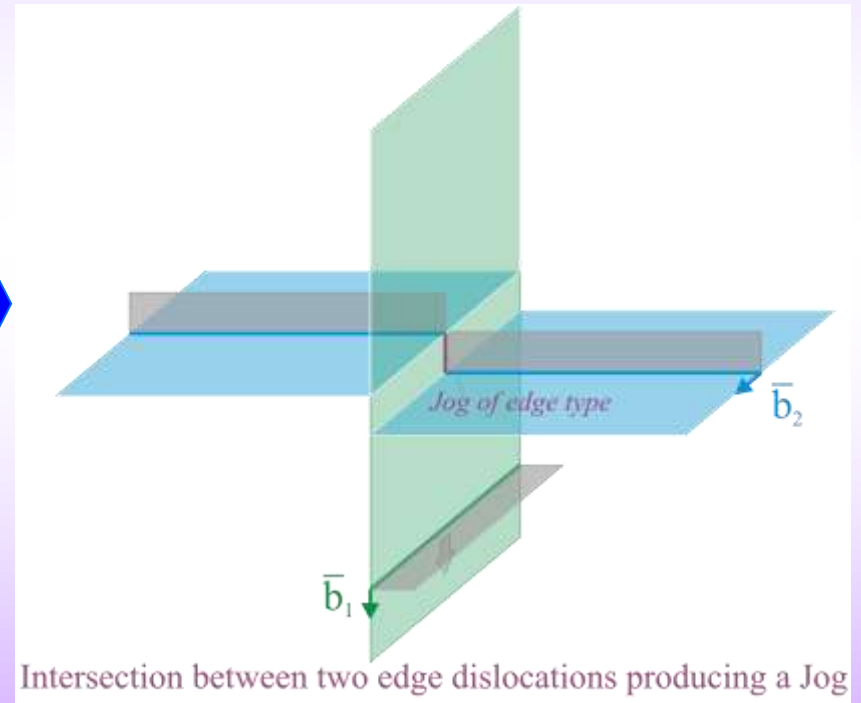
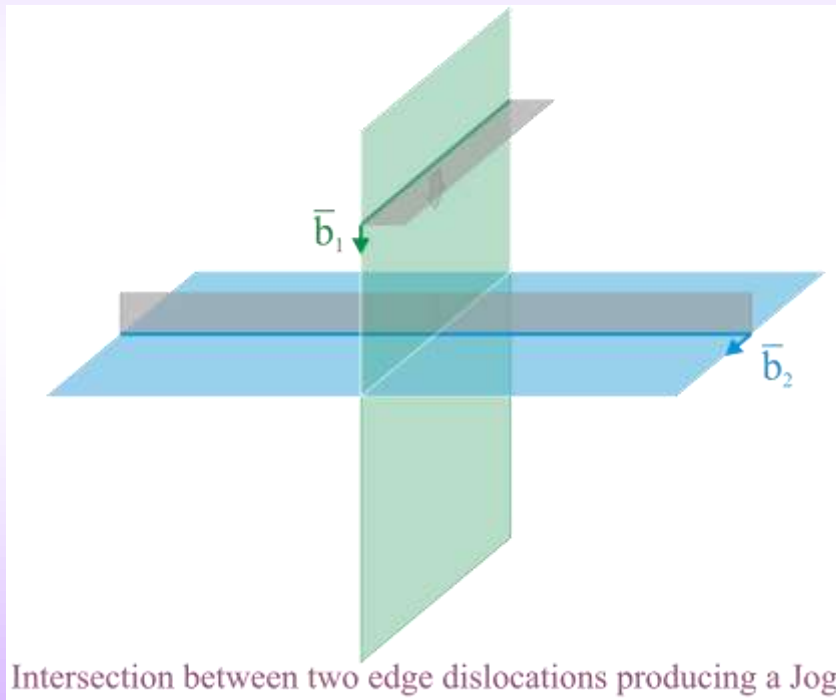
- ❑ Two straight dislocation can intersect to leave Jogs and Kinks in the dislocation line.
- ❑ These extra segments in a dislocation line cost energy and hence require work done by the external force $\Rightarrow$ lead to hardening of the material.
(Additional stress as compared to the stress required to glide the dislocation line is required to form the Jog/Kink).
- ❑ Four types of interactions are considered in the next few pages.
- ❑ One additional type of dislocation reaction important in CCP crystals is the formation of the Lomer-Cottrell lock by the intersection of two dislocations.

[Click here to know more about Lomer-Cottrell locks](#)

1 Edge-Edge Intersection

Perpendicular Burgers vector

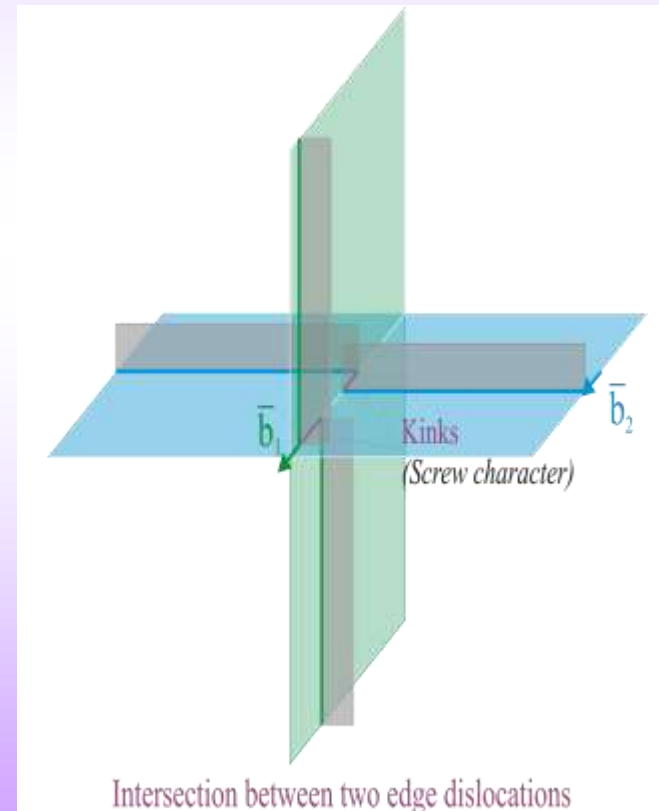
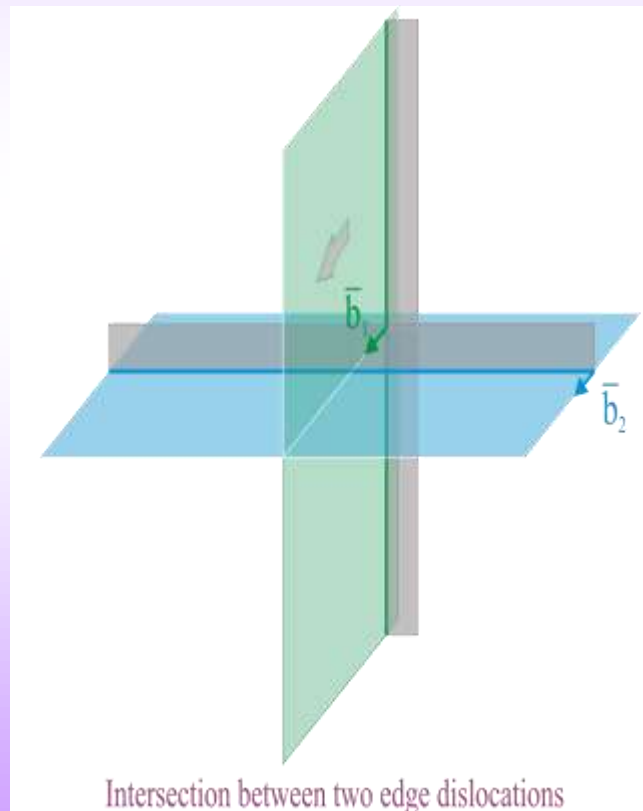
- ❑ The jog has edge character and can glide (with Burgers vector = $\mathbf{b}_2$).
- ❑ The length of the jog = $\mathbf{b}_1$.
- ❑ Edge Dislocation-1 (Burgers vector $\mathbf{b}_1$) → Unaffected as $\mathbf{b}_2$ is $\parallel \mathbf{t}_1$ (line vector) .
- ❑ Edge Dislocation-2 (Burgers vector $\mathbf{b}_2$) → Jog (Edge character) → Length $|\mathbf{b}_1|$.



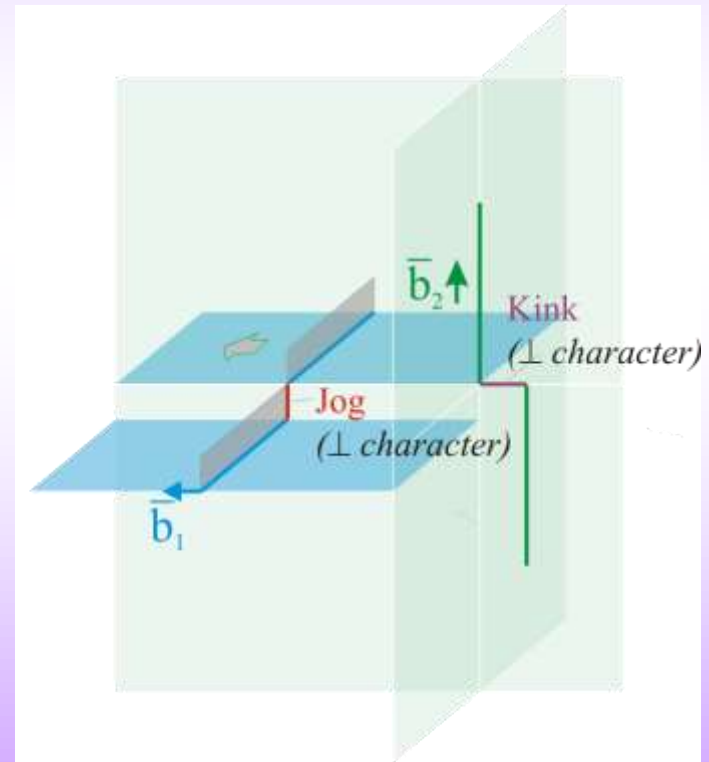
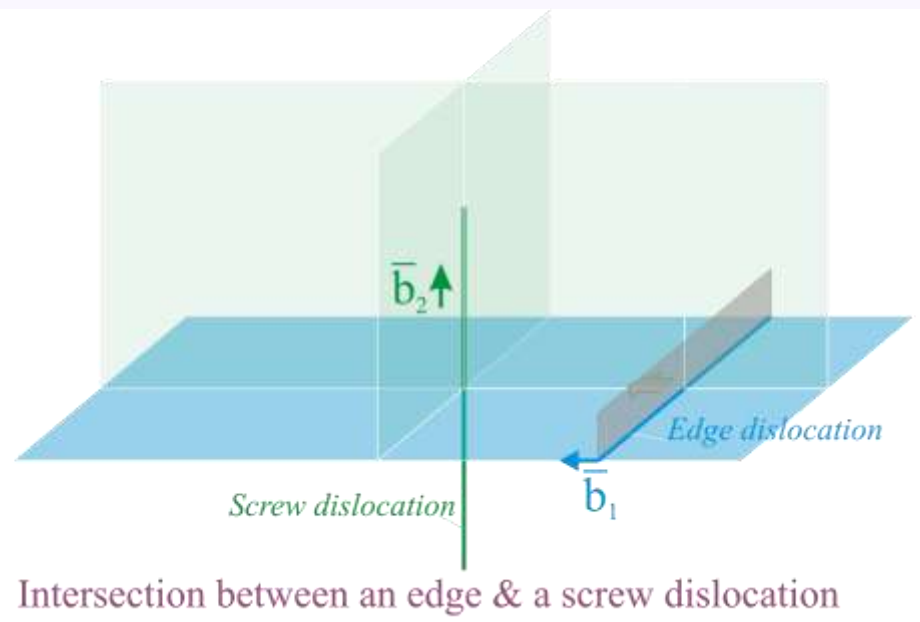
2 Edge-Edge Intersection

Parallel Burgers vector

- ❑ Both dislocations are kinked.
- ❑ Edge Dislocation-1 (*Burgers vector $\mathbf{b}_1$*) $\rightarrow$ Kink (Screw character) $\rightarrow$ Length $|\mathbf{b}_2|$.
- ❑ Edge Dislocation-2 (*Burgers vector $\mathbf{b}_2$*) $\rightarrow$ Kink (Screw character) $\rightarrow$ Length $|\mathbf{b}_1|$.
- ❑ The kinks can glide.



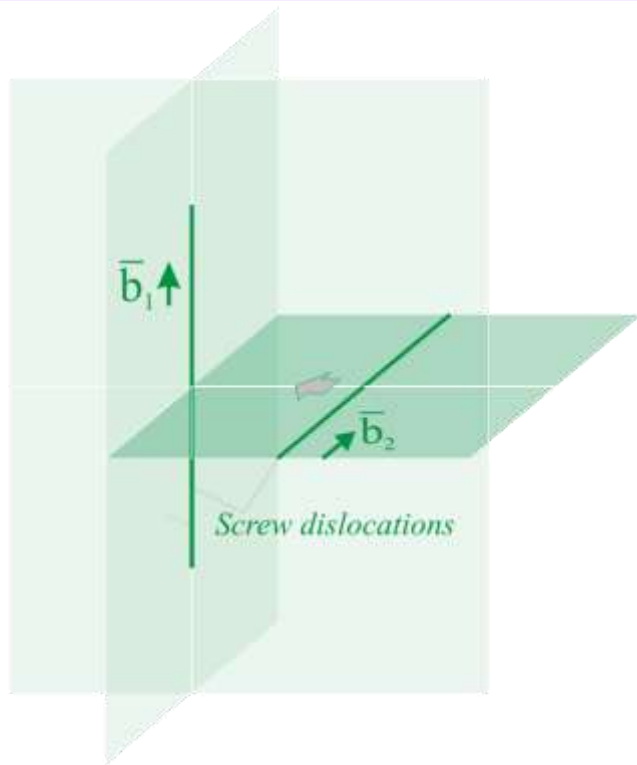
- ❑ Edge Dislocation (*Burgers vector* $\mathbf{b}_1$) $\rightarrow$ Jog (Edge Character) $\rightarrow$ Length $|\mathbf{b}_2|$.
- ❑ Screw Dislocation (*Burgers vector* $\mathbf{b}_2$) $\rightarrow$ Kink (Edge Character) $\rightarrow$ Length $|\mathbf{b}_1|$.



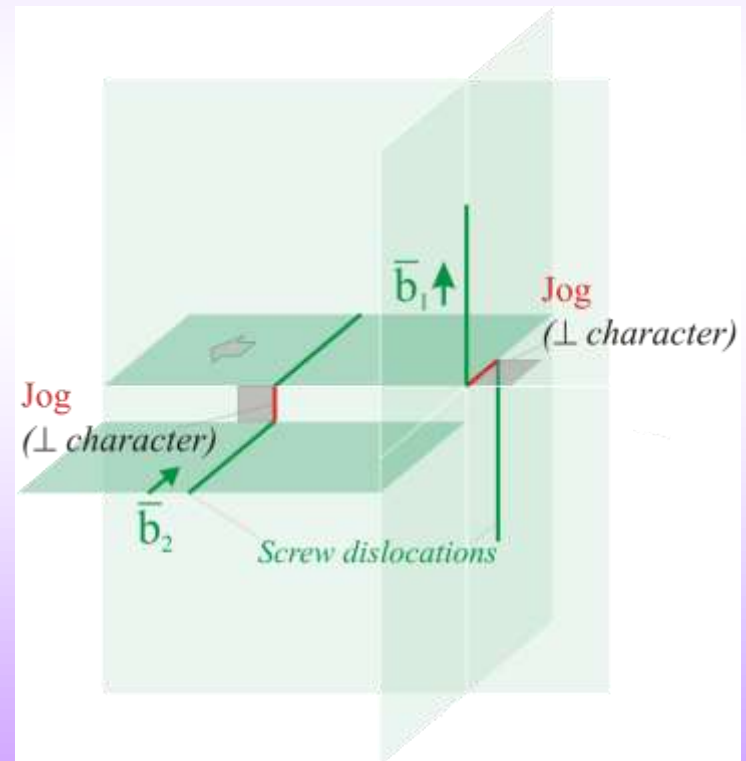
4 Screw -Screw Intersection

Perpendicular Burgers vector

- ❑ Important from plastic deformation point of view.
- ❑ Screw Dislocation (*Burgers vector* $\mathbf{b}_1$) $\rightarrow$ Jog (Edge Character) $\rightarrow$ Length $\mathbf{b}_2$.
- ❑ Screw Dislocation (*Burgers vector* $\mathbf{b}_2$) $\rightarrow$ Jog (Edge Character) $\rightarrow$ Length $\mathbf{b}_1$.
- ❑ Both the jogs are non conservative
(i.e. cannot move with the dislocations by glide).



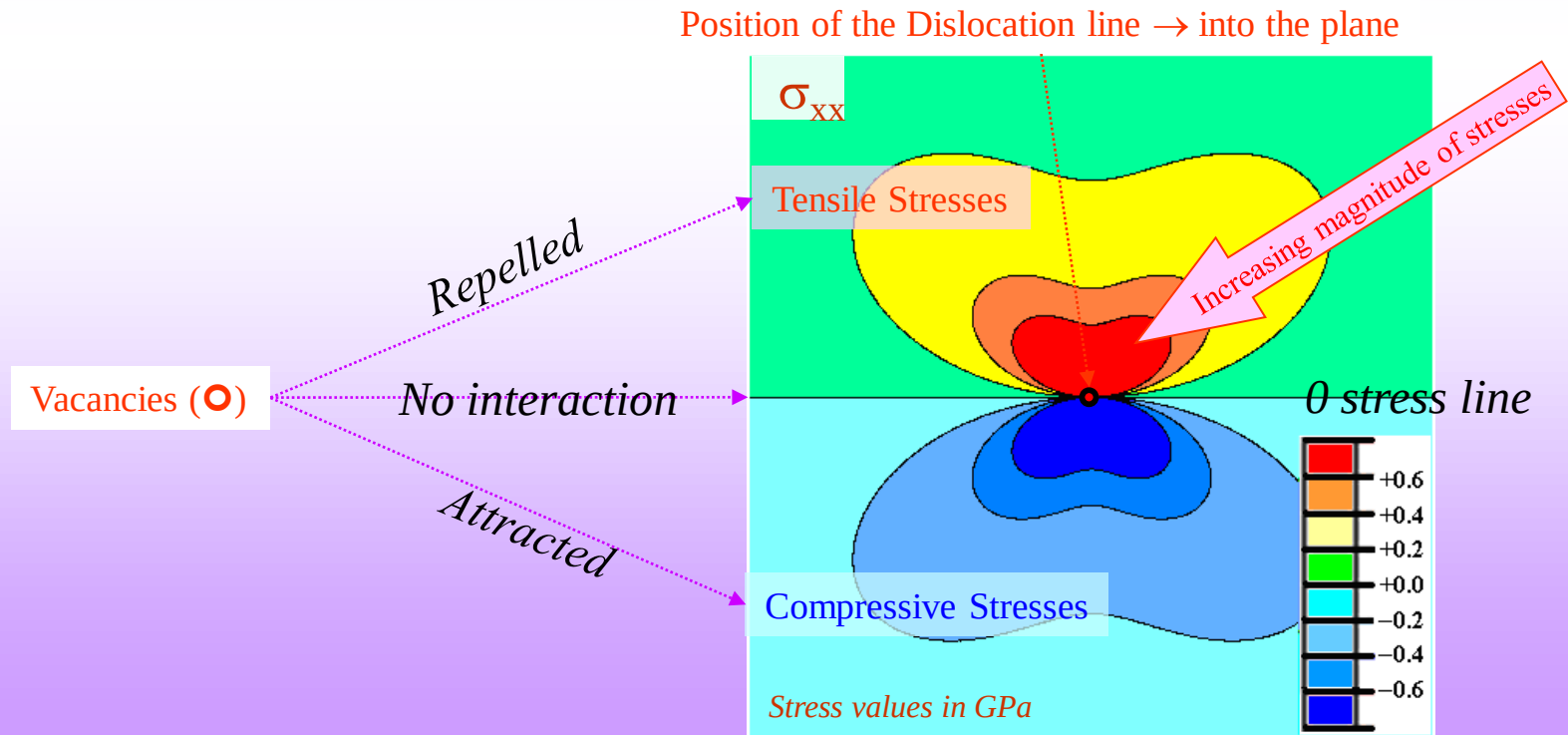
Intersection between two screw dislocations



Dislocation-Point defect Interactions

- ❑ The stress field of a dislocation can interact with the stress field of point defects.
- ❑ Defects associated with tensile stress fields are attracted towards the compressive region of the stress field of an edge dislocation (*and vice versa*).
- ❑ Solute atoms can segregate in the core region of the edge dislocation (*formation of the Cottrell atmosphere*) → higher stress is now required to move the dislocation (*the system is in a low energy state after the segregation and higher stress is required to 'pull' the dislocation out of the energy well*).
- ❑ Higher free-volume at the core of the edge dislocation aids this segregation process.
- ❑ Defects associated with shear stress fields (having a non-spherical distortion field; e.g. interstitial carbon atoms in BCC Fe) can interact with the stress field of a screw dislocation.

- ❑ Vacancies are attracted to the compressive regions of an edge dislocation and are repelled from tensile regions. This is due to stress gradients (i.e. increasing stress as we move closer to the dislocation line).
- ❑ The behaviour of substitutional atoms smaller than the parent atoms is similar to that of the vacancies.
- ❑ Larger substitutional atoms are attracted to the tensile region of the edge dislocation and are repelled from the compressive regions.
- ❑ Interstitial atoms (associated with compressive stress fields) are attracted towards the tensile region of the edge dislocation and are repelled from the compressive region of the stress field.

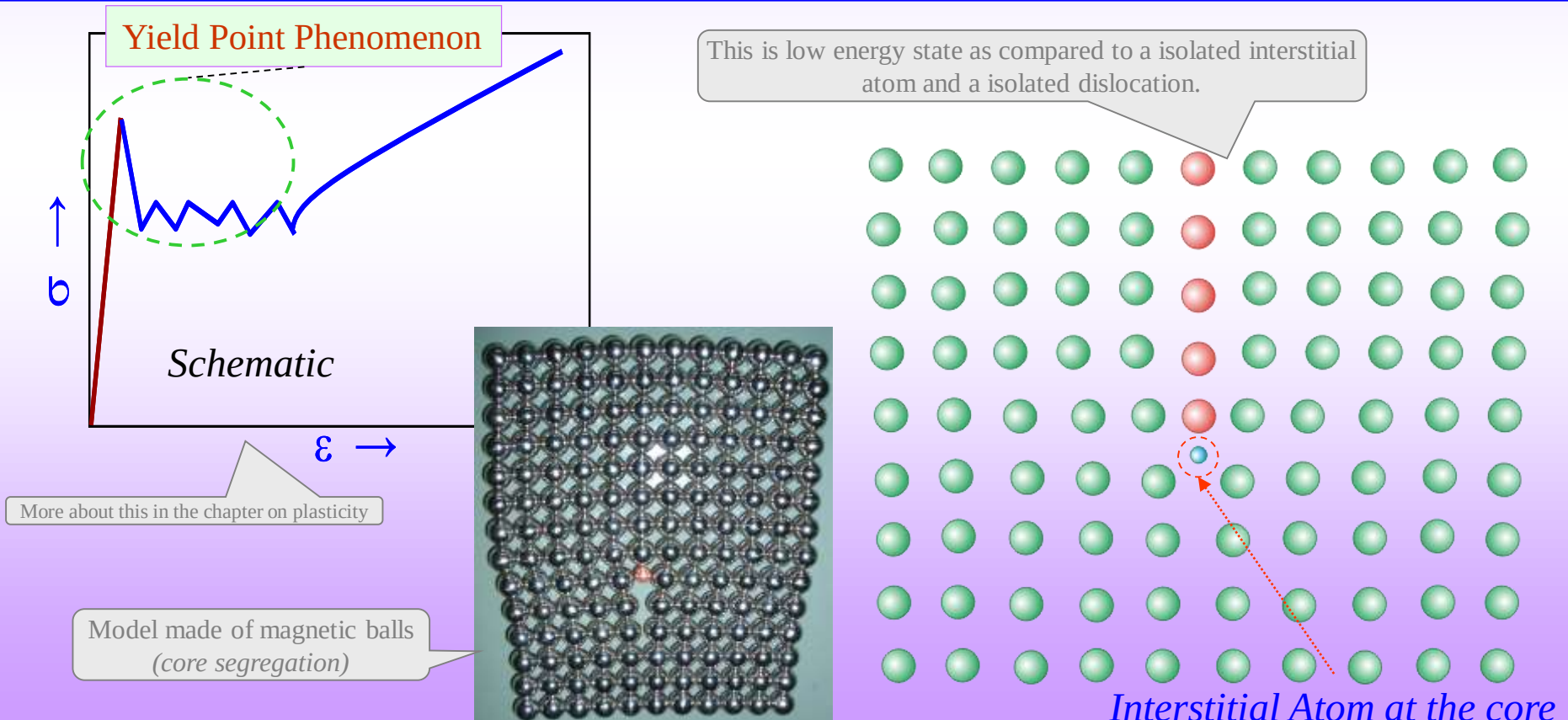


Summary of edge dislocation - point defect interactions

Point Defect	Tensile Region	Compressive Region
Vacancy	Repelled	Attracted
Interstitial <i>(of size larger than the void)</i>	Attracted	Repelled
Smaller substitutional atom	Repelled	Attracted
Larger Substitutional atoms	Attracted	Repelled

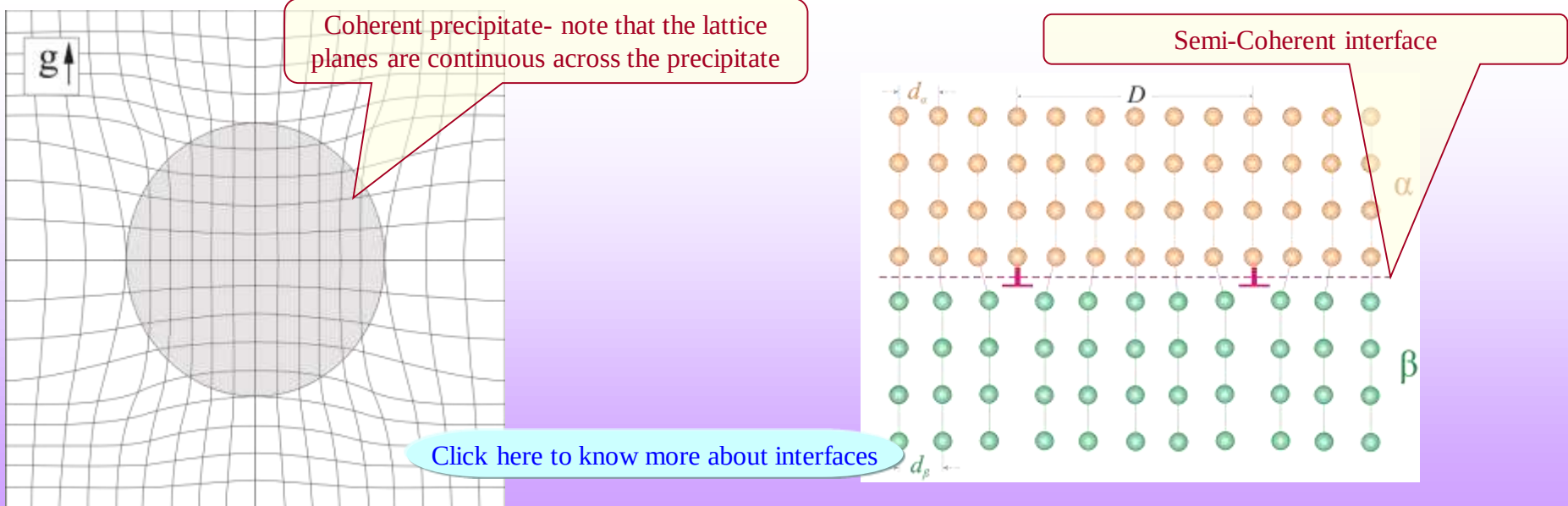
Yield Point Phenomenon

- ❑ The interaction of interstitial carbon atoms with edge dislocations (*interaction of stress fields of dislocations with solute atoms*) → leading to their segregation to the core of the edge dislocations (forming a Cottrell atmosphere) is responsible for the Yield Point Phenomenon seen in the tensile test of mild steel specimens.
- ❑ In Yield point phenomenon, there is a yield drop (when the dislocation breaks free of Cottrell atmosphere) followed by serrated yielding (due to repeated ‘pinning’ of edge dislocations by carbon atoms).



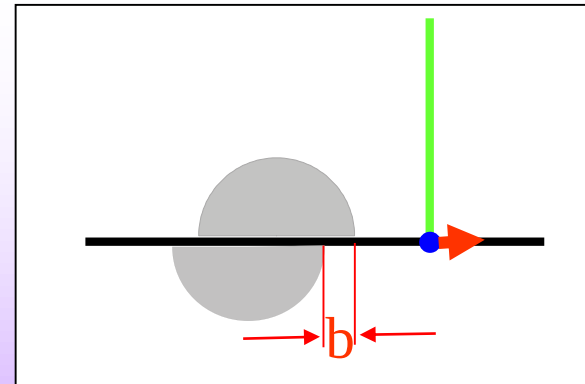
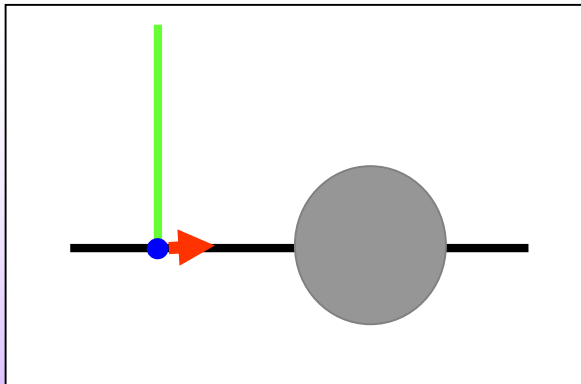
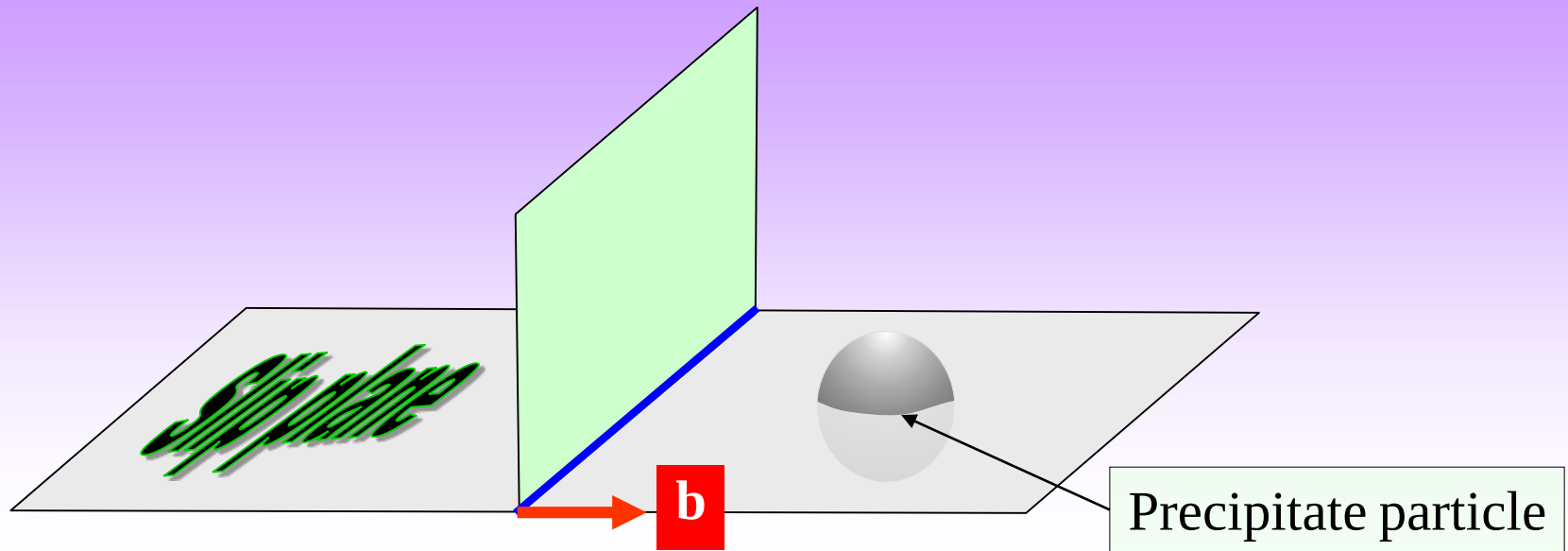
Dislocation-Precipitate Interactions

- ❑ Dislocations can interact with the stress fields of precipitates.
- ❑ Moving dislocations can:
 - A ■ glide through **coherent precipitates*** → shearing the precipitate.
 - B ■ bow around **incoherent precipitates**, leaving loops as they bypass two precipitates which act like pinning centres for the dislocations (→ *leading to an increase in the dislocation density*).
- ❑ Both these processes need an application of higher stresses (assuming an harder precipitate) → lead to the strengthening of the material.
- ❑ Semi-coherent precipitates have interfacial misfit dislocations which partially relieve the coherency strains. (*These dislocations are structural dislocations*).



* Though the word coherent is used as an adjective for the precipitate- actually what is meant is that the interface is coherent (or semi-coherent if we talk about a semi-coherent precipitate)

A ■ Glide through coherent precipitates

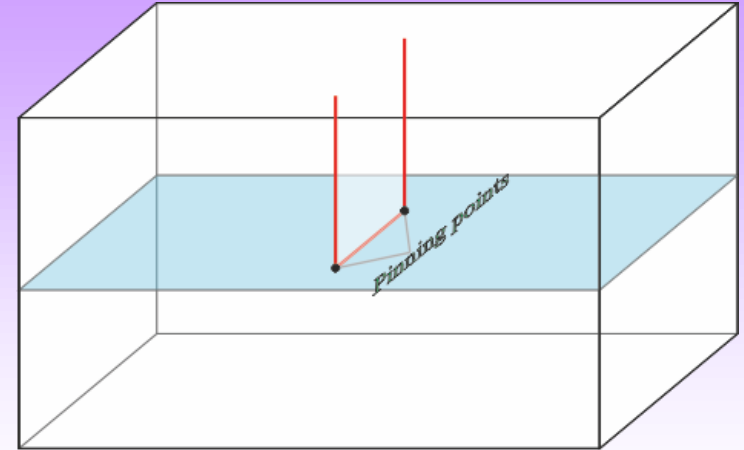


- ❑ A dislocation can be pinned between two points*, thus hindering the motion of the dislocation.
- ❑ For motion of the dislocation, leading to plasticity the dislocation has to bypass the pinned segment, under the action of the applied stress (shear stress on the slip plane drives the motion).
- ❑ The dislocation takes a series of configurations (as shown the in the figures) under the action of the applied stress → leading to the formation of a dislocation loop (*leaving the original pinned segment*).
- ❑ This leads to an **increase in dislocation density** (*one of the mechanisms by which dislocation density increases with plastic deformation*).
- ❑ As the original segment is retained the ‘source’ (Frank-Read source) can operate repeatedly forming a loop each time.
- ❑ As the preexisting loops would oppose the formation of the next loop (repulsive stresses-dislocations of the same sign), **higher stresses** are required to operate the source each time.
- ❑ Till the formation of the half-loop (semi-circle), increasingly higher stresses are required. After this the process occurs downhill in stresses.
- ❑ The maximum stress ($\tau_{\max}$) required to operate the source thus corresponds to the formation of the half-loop (with radius $r_{\min}$).

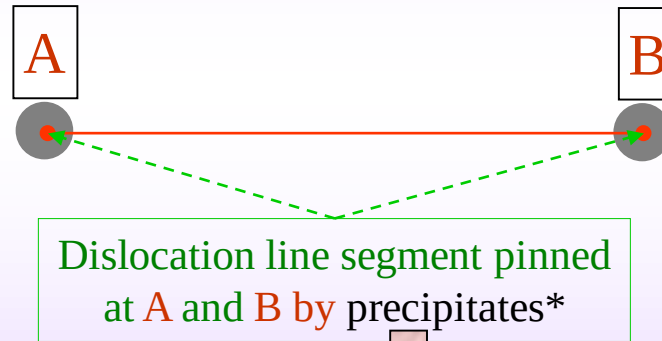
$$\tau_{\max} \sim \frac{Gb}{2r_{\min}}$$

* The pinning points could be the points at which the dislocation leaves the slip plane (or could be due to incoherent precipitates etc.)

One example of 'pinning points' where the dislocation leaves the slip plane



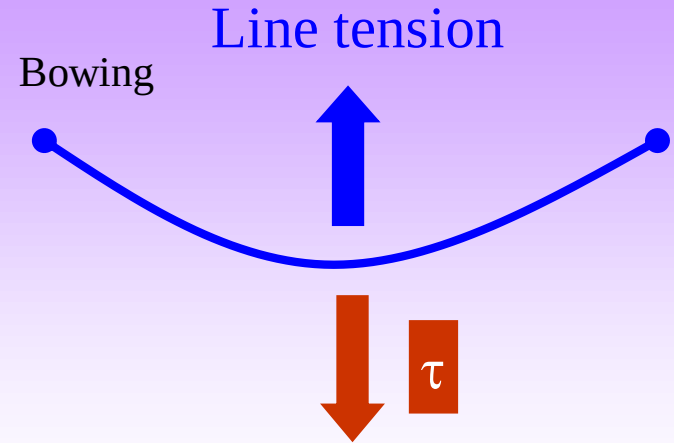
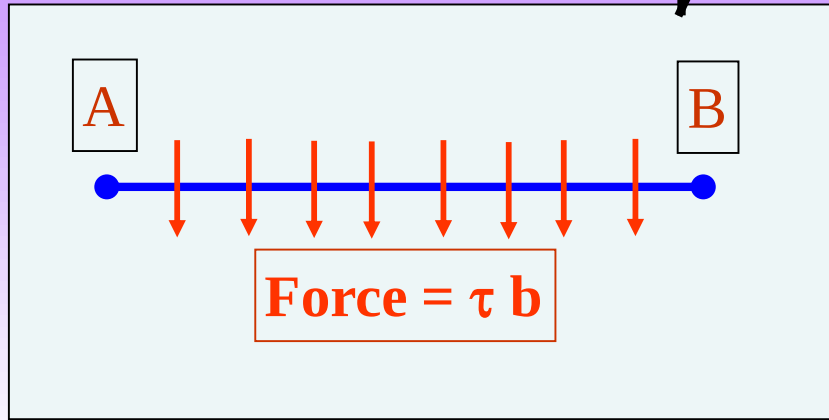
Initial configuration



*** Pinning could (also) be caused by:**

- Dislocation in the plane of the paper intersects dislocations in other planes
- Anchored by impurity atoms or precipitate particles
- Dislocation leaves the slip plane at A and B

Application of stress on dislocation segment



- As the dislocation line gets curved the energy of the system increases $\Rightarrow$ work has to be done by external stresses to cause this extension.
- Line tension (opposes the shear stress on the slip plane (τ). At a given stress there might be a balance of forces leading to a curved geometry of the dislocation line.
- Further extension of the dislocation line occurs by increasing the stress.

Force on dislocation line $= \tau b ds$

Line tension force $= 2 \gamma \sin\left(\frac{d\theta}{2}\right) \sim \gamma d\theta$

Dislocation energy / length $= \gamma \sim \frac{1}{2} Gb^2$

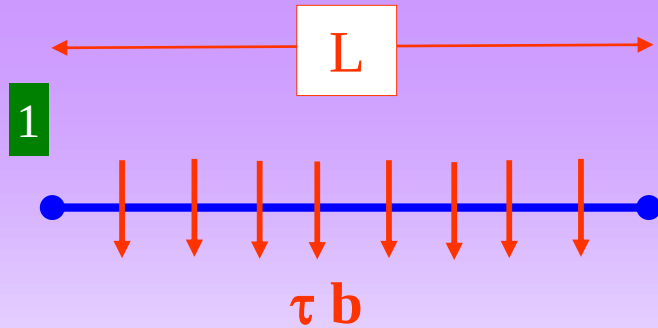
$$\tau \sim \frac{Gb}{2r}$$

For equilibrium in curved configuration

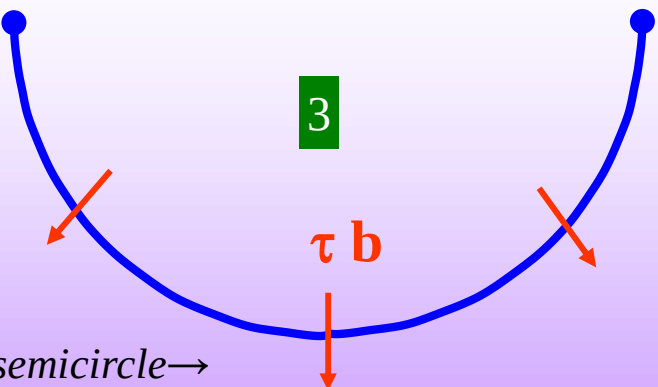
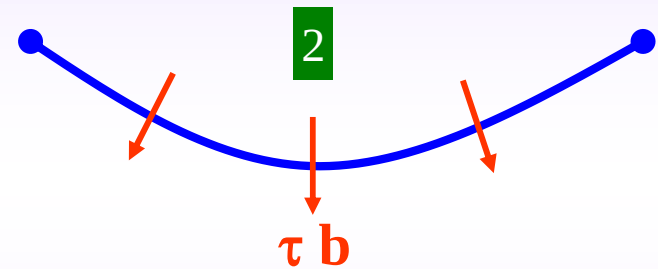
$$\gamma d\theta = \tau b ds$$

$$\tau_{\max} \Rightarrow r_{\min}$$

Increasing stress



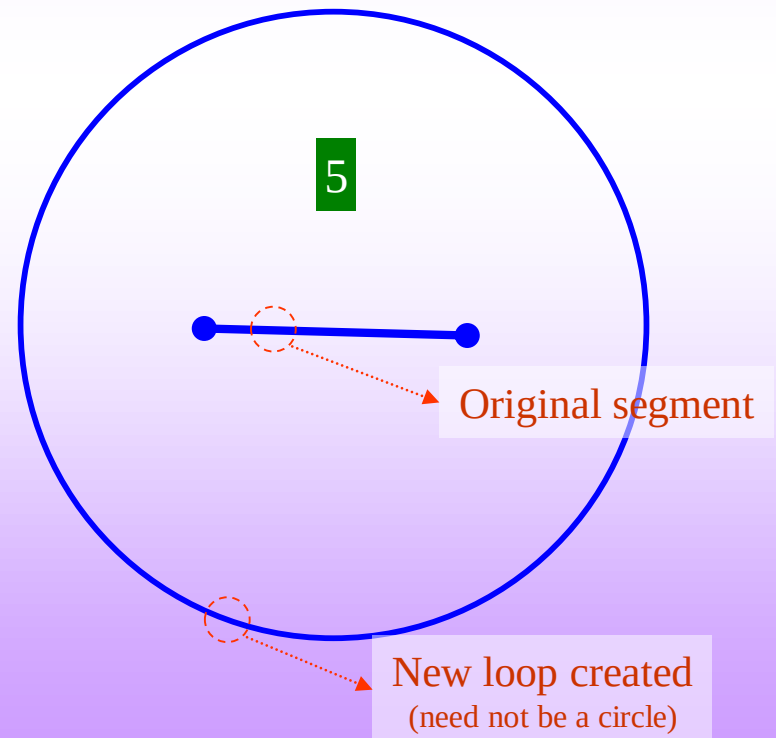
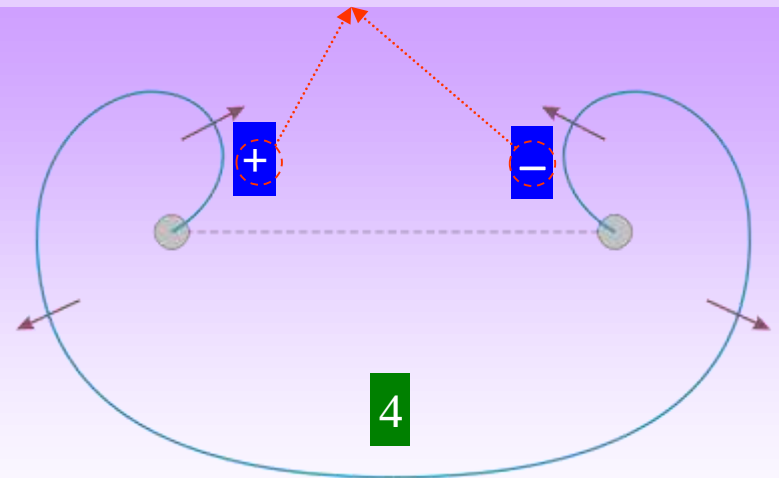
Direction of dislocation motion is $\perp$ to the dislocation line (except at A and B)



semicircle $\rightarrow$ corresponds to **maximum stress** required to expand the loop

After this decreasing stress is required to expand the loop

+ and - segments come together and annihilate each other



Frank-Read dislocation source →

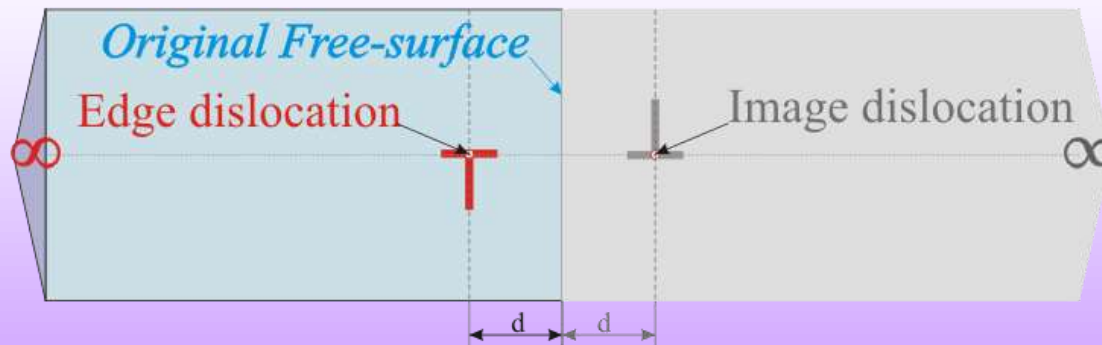
- Can operate from a single source producing a loop each time
- This loop produces a slip of $1b$ each time on the slip plane
- The maximum value of shear stress required is when the bulge becomes a semi-circle ($r_{\min} = L/2$) → $\tau_{\max} \sim Gb/L$
 - ⇒ $\tau \downarrow$ as $L \uparrow$ i.e. The longest segments operate first
 - ▶ When the long segments get immobilized shorter segments operate with increasing stress
⇒ *work hardening*
- If the dislocation loops keep piling up on the slip plane the back stress will oppose the applied stress
- When the *back-stress* $> \tau_{\max}$ the source will cease to operate
- Double ended F-R sources have been observed experimentally they are not frequent ⇒ other mechanisms must exist

$$\tau_{\max} = \alpha Gb / L$$

$\alpha = 0.5$ for edge dislocations and $\alpha = 1.5$ for screw dislocations.

Dislocation- Free surface Interaction → Concept of Image Forces

- ❑ A dislocation near a free surface (in a semi-infinite body) experiences a force towards the free surface, which is called the image force. This is a type of **Configurational Force** (*i.e. force experienced when the energy is lowered by a change in configuration of a system*)
- ❑ The force is called an ‘image force’ as the force can be calculated assuming an negative hypothetical dislocation on the other side of the surface (*figure below*). The force of attraction between the dislocations (+ & –) is gives the image force. The material properties are identical throughout.
- ❑ If the image force exceeds the Peierls stress then the dislocation can spontaneously leave the crystal, without application of external stresses!
- ❑ Hence, regions near a free surface and nano-crystals can become **spontaneously dislocation free**. In nanocrystals due to the proximity of more than one surface, many images have to be constructed and the net force is the superposition of these image forces.

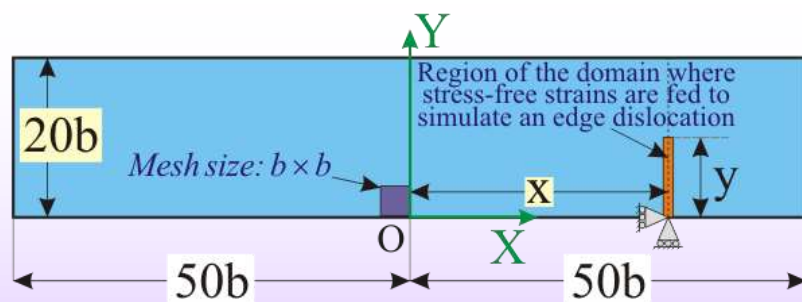


$$F_{image} = \frac{-Gb^2}{4\pi(1-\nu)d}$$

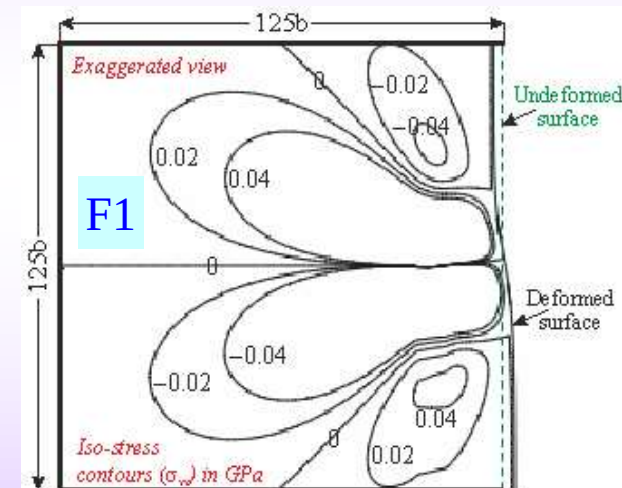
A hypothetical negative dislocation is assumed to exist across the free-surface for the calculation of the force (attractive) experienced by the dislocation in the proximal presence of a free-surface

Domain deformations in Nanocrystals in the presence of dislocations

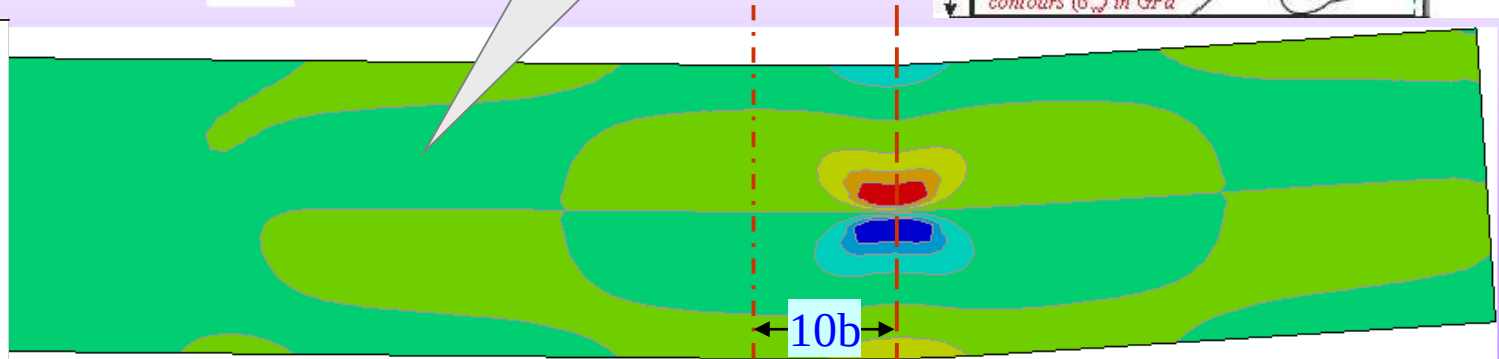
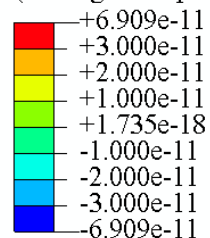
- ❑ Dislocation near a free surface in a semi-infinite body can deform the surface. This is a small deformation as shown in the figure below (F1) for the case of an edge dislocation.
- ❑ In nanocrystals (*e.g. the kind shown in the figure below*) the domain can bend/buckle in the presence of dislocations (*the figure shows the effect of an edge dislocation in a plate → a screw dislocation leads to the twisting of - for example - a cylindrical domain*)
- ❑ This is elastic deformation in the presence of dislocations in a nanomaterial!
 - Hence, we can have *reversible plastic deformation due to elasticity!!!*



Dislocation in a thin 'plate' of Al leading to its bending



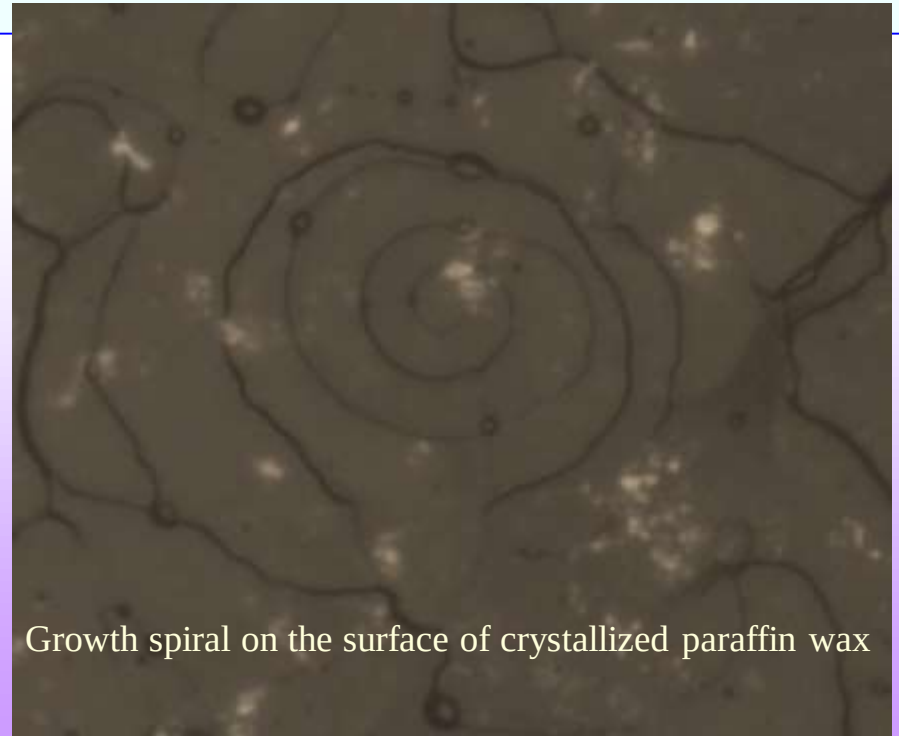
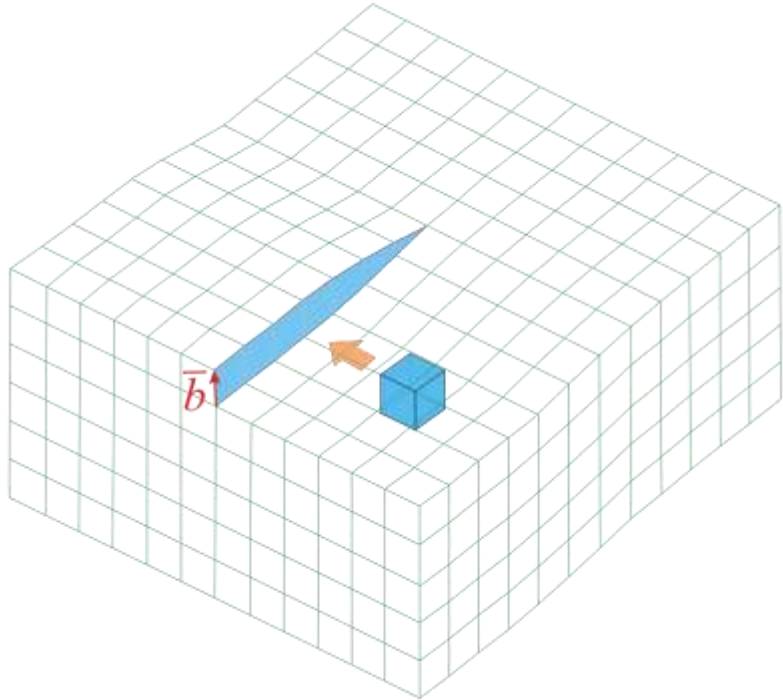
S, S11
(Average-compute)



Role of dislocations in crystal growth

Constructive role of dislocations!

- ❑ Crystals grown under low supersaturation ($\sim 1\%$) the growth rate is considerably faster than that calculated for an ideal crystal
- ❑ In an ideal crystal surface the difficulty in growth arises due to difficulty in the nucleation of a new monolayer
- ❑ If a screw dislocation terminates on the surface of a crystal then addition of atoms can take place around the point where the screw dislocation intersects the surface (*the step*) $\rightarrow$ leading to a spiral (actually spiral helix) growth staircase pattern



Growth spiral on the surface of crystallized paraffin wax

Role of dislocations in phase transformation

- ❑ Dislocations can act as heterogeneous nucleation sites during phase transformation.
 - Dislocations may dictate the orientation and morphology of the second phase.
- ❑ The strain associated with the dislocation may be partly relieved by the formation of a second phase.
- ❑ The strain associated with the transformation (the Eshelby strain) may be accommodated by plastic flow (mediated by dislocations).

Mental Picture of a dislocation

Let us consider the various ways of understanding the dislocation

→ The different perspectives

- ☐ Association with translational symmetry
- ☐ As a line defect
- ☐ Distortion of bonds → region of high energy
- ☐ Increase in entropy of the system
- ☐ Free volume at the core → pipe diffusion
- ☐ Core of dislocation & its geometry → Peierls Stress
- ☐ Stress & Strain fields
- ☐ Interaction with other defects
- ☐ Role in slip
- ☐ Role as a structural defect



Why are dislocations non-equilibrium defects?

$$\Delta G = \Delta H - T \Delta S \leftarrow \text{+ve for dislocations}$$

- It is clear from the above equation that if a configuration* gives an **entropy benefit** (i.e. ΔS is positive); then that state will be stabilized at some temperature (even if the **enthalpy cost** is very high for that configuration)
- In the present case: it **costs an energy** of $\sim Gb^2/2$ per unit length of dislocation line introduced into the crystal; but, this gives us a **configurational entropy benefit** (as this dislocation can exist in many *equivalent* positions in the crystal)
- This implies that there must be temperature where dislocations can become stable in the crystal (ignoring the change in the energy cost with temperature for now)
- *Unfortunately this temperature is above the melting point of all known materials*
- Hence, dislocations are not stable thermodynamic defects in materials

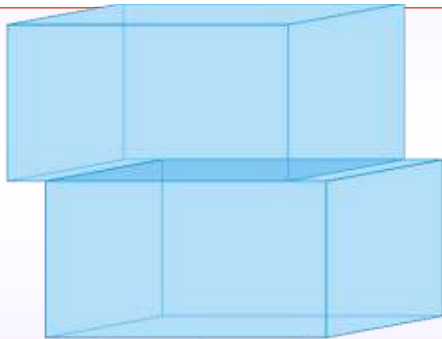
► The energy required to create **Kinks** and **Jogs** of length 'b' is $\sim Gb^3/10$
→ these can be created by thermal fluctuations

* Including positional, electronic, rotational & vibrational multiplicity of states

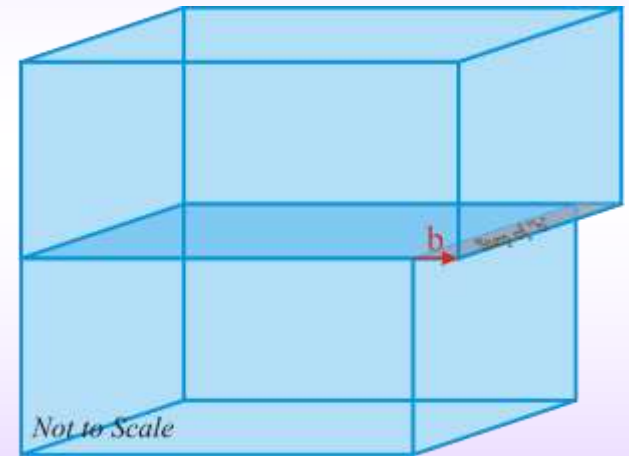
How are crystals weakened?

The two step process

- As we have seen the process of sliding (by shear) of an 'entire plane' of atoms can be reduced to a 'line-wise' process by **dislocations**
 - ▶ this leads to a shear stress reduction of a few orders of magnitude
- This problem can be further broken down (in dimension and energy) to the formation and migration of **kink pairs** along the dislocation line (*usually occurs to screw components of dislocations in BCC metals*)
 - ▶ this can further lead to the reduction in stress required for dislocation motion

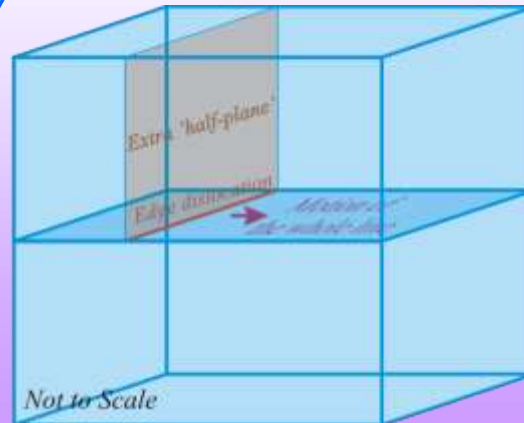


Shearing of a whole part of a crystal with respect to another



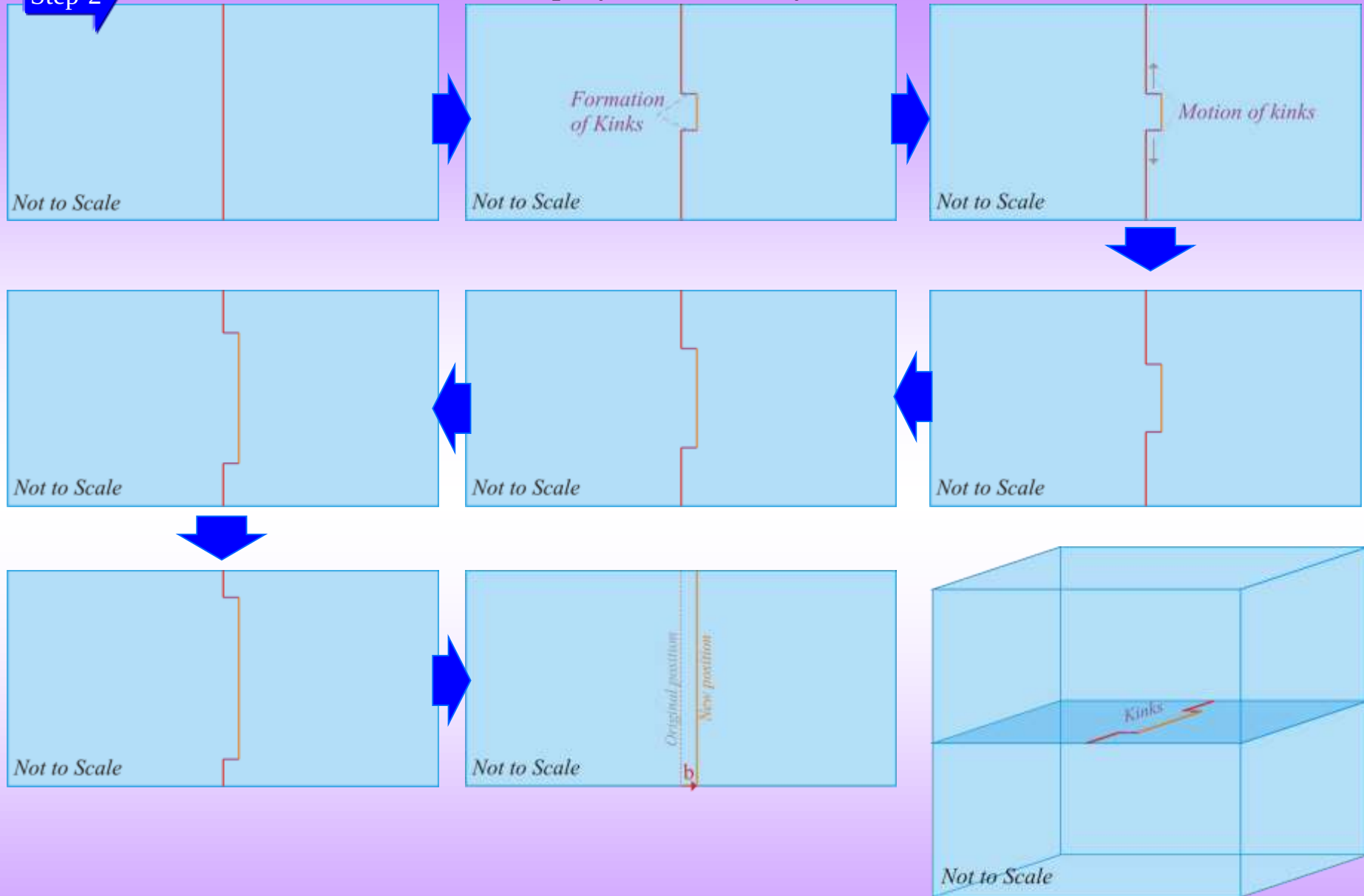
Motion of dislocation line
→ leaving the crystal

Step-1



Step-2

Further break up of the motion of the dislocation

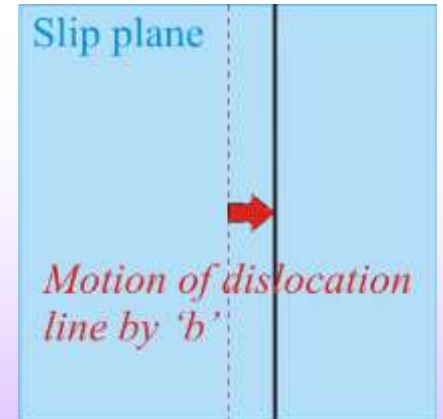
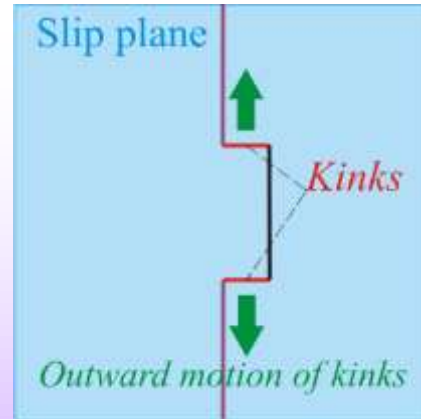
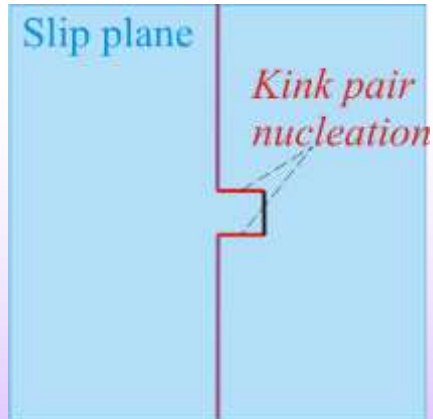
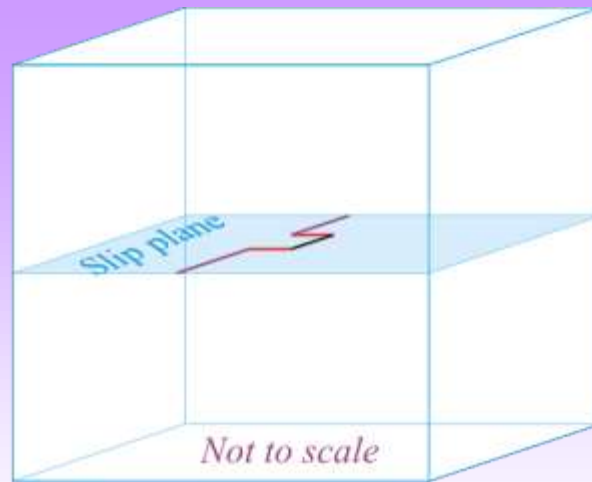


► In BCC metals and Ge thermally assisted formation of kink pairs can cause slip at stresses

$$\tau < \tau_{PN}$$

Continued...

More views



DISLOCATIONS

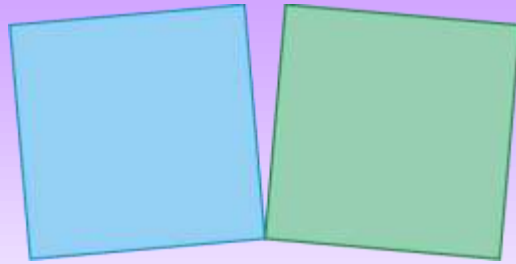
Random

Structural

Distinct from ‘Geometrically Necessary Dislocations’ (GND)

- ❑ As mentioned before: **Structural defects** play a very different role in material behaviour as compared to “**Random Statistical Defects**” (*non-structural*)
- ❑ Structural dislocations may be associated with a boundary and hence their role in plasticity may be very different from random (“statistically stored”) dislocations
- ❑ Often a related term- Geometrically Necessary Dislocations is used in literature (*we have intentionally avoided the use of the term here*)
- ❑ Structural dislocations include:
 - Dislocations at low angle grain boundaries ([will be discussed along with other 2D defects](#))
→ responsible for a tilt or twist
 - Dislocation at semi-coherent interfaces
→ responsible for matching misfit (between adjacent crystals)

Accommodated by an array of edge dislocations



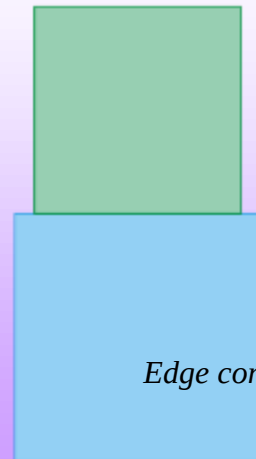
Tilt

Structural dislocations can accommodate

between two crystals

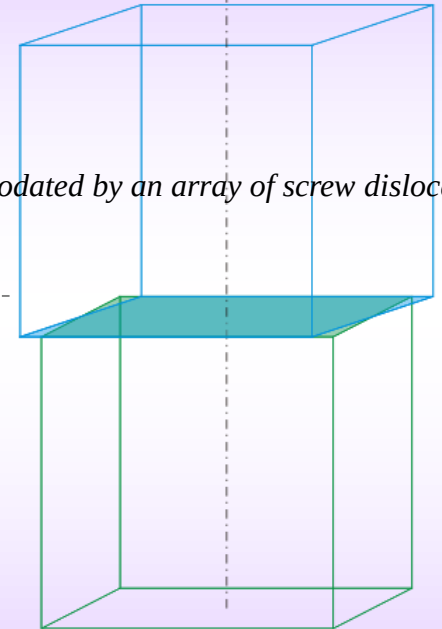
Twist

Misfit



Edge component relieves misfit strain

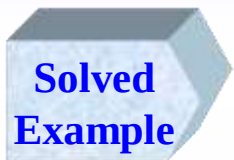
Axis of Twist



Accommodated by an array of screw dislocations

What determines the Burgers vector?

- There are two distinct questions we can ask:
 - Q1** ➤ If you already have a dislocation how to determine the Burgers vector?
 - Q2** ➤ What determines the Burgers vector?
- The answer to Q1 is by constructing a **Burgers circuit**
- The answer to Q2 is: **Crystallography** → the Burgers vector is the shortest lattice translation vector (for a perfect/full dislocation)



How to find the character of a dislocation? (*Edge or Screw*).

$$\vec{b} = \frac{1}{2}[110] \quad \vec{t} = [112] \quad \cos\theta = \frac{\vec{b} \cdot \vec{t}}{|\vec{b}| |\vec{t}|} \quad \cos\theta = \left(\frac{1+1+0}{\sqrt{2}\sqrt{6}} \right) = \frac{2}{\sqrt{2}\sqrt{6}} = \frac{1}{\sqrt{3}}$$

Drop the $\frac{1}{2}$ for the angle calculation

$$\theta = 54.74^\circ$$

- If the angle is 90° , then the dislocation has pure edge character and if it is 0° , then it has pure screw character. An angle between 90° and 0° implies a mixed character.
- Closer to 90° implies a more edge like character.

Solved Example

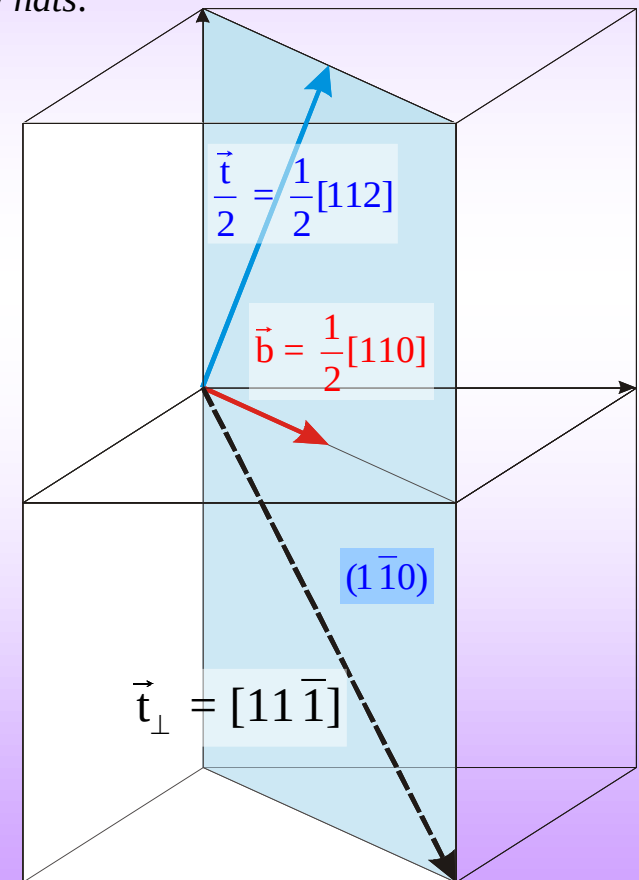
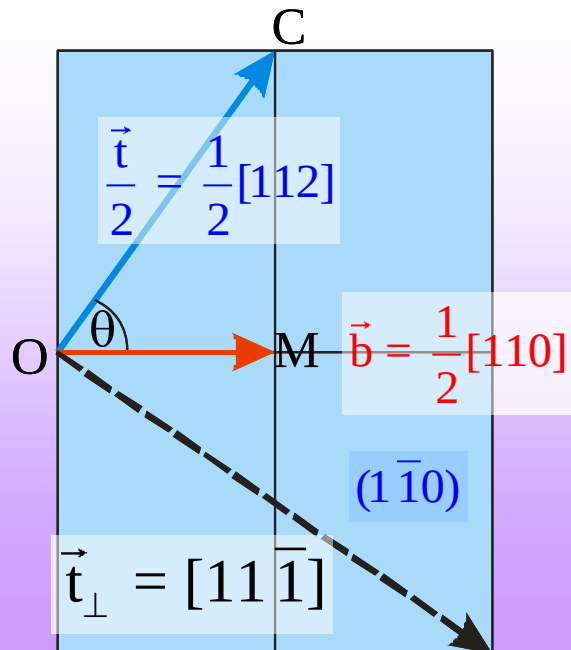
In a cubic crystal a dislocation line of mixed character lies along the $[112]$ direction. Burgers vector = $\frac{1}{2}[110]$. What are the edge and screw components of the Burgers vector? Which is the slip plane?

Given: $\vec{b} = \frac{1}{2}[110]$, $\vec{t} = [112]$

$\vec{t}_\perp$ refers to the direction perpendicular to $\vec{t}$ and unit vectors are shown by *hats*.

$$\vec{b} = \frac{1}{2}[110], |\vec{b}| = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}, \hat{b} = \frac{1}{\sqrt{2}}[110]$$

$$\vec{t} = [112], |\vec{t}| = \sqrt{6}, \hat{t} = \frac{1}{\sqrt{6}}[112]. (\vec{t}' = \frac{1}{2}[112], |\vec{t}'| = \frac{\sqrt{6}}{2} = \sqrt{\frac{3}{2}}).$$



Slip plane contains both $\vec{b}$ & $\vec{t}$. Let slip plane (hkl).

Applying Weiss zone law:

(on [110]) On $\vec{b} \rightarrow h + k = 0$

(on [112]) On $\vec{t} \rightarrow h + k + 2l = 0 \Rightarrow l = 0, h = -k \Rightarrow$ the slip plane (s) = $(1\bar{1}0)$

For the screw segment of a dislocation: $\vec{b} \parallel \vec{t}$

For the edge segment of the dislocation: $\vec{b} \perp \vec{t}$

Looking at the figure: $\cos\theta = \frac{1/\sqrt{2}}{\sqrt{6}/2} = \frac{1}{\sqrt{3}}, \sin\theta = \frac{\sqrt{2}}{\sqrt{3}}, \theta = 54.73^\circ$

Let the direction $\perp$ to $\vec{t}$ be $\vec{t}_\perp = [uvw]$. This direction lies on $(1\bar{1}0)$ plane and is $\perp$ to $\vec{t}$.

Applying Weiss zone law for these conditions:

$u - v = 0, u + v + 2w = 0, \Rightarrow u = -w, u = v \Rightarrow \vec{t}_\perp$ is of the form $[uu\bar{u}]$

$\Rightarrow \vec{t}_\perp = [11\bar{1}], \hat{t}_\perp = \frac{1}{\sqrt{3}}[11\bar{1}]$

$|\vec{b}_\parallel| = \vec{b} \cdot \vec{t} = \frac{1}{2}([110] \cdot [112])\cos\theta = \frac{1}{2}\sqrt{2}\frac{1}{\sqrt{3}} = \frac{1}{\sqrt{6}}, \vec{b}_\parallel = |\vec{b}_\parallel|\hat{t} = \frac{1}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}}[112] = \frac{1}{6}[112]$

$|\vec{b}_\perp| = \vec{b} \cdot \vec{t}_\perp = \frac{1}{2}([110] \cdot [11\bar{1}])\cos(90 - \theta) = \frac{1}{2}\sqrt{2}\sqrt{\frac{2}{3}} = \frac{1}{\sqrt{3}}, \vec{b}_\perp = |\vec{b}_\perp|\hat{t}_\perp = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{3}}[11\bar{1}] = \frac{1}{3}[11\bar{1}]$

Solved Example

In a CCP crystal is the dislocation reaction shown below feasible energetically? What is the significance of the vectors on the RHS of the reaction?

$$\frac{1}{2}[110] \rightarrow \frac{1}{6}[\bar{2}1\bar{1}] + \frac{1}{6}[121]$$

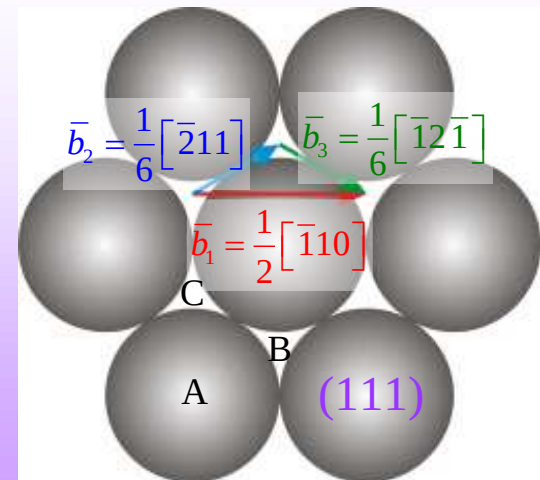
This is of the form $b_1 \rightarrow b_2 + b_3$ The dislocation reaction is feasible if: $b_1^2 > b_2^2 + b_3^2$

As the energy of a dislocation (per unit length of the dislocation line is proportional to b^2

$$|b_1|^2 = \left(\frac{\sqrt{1^2 + 1^2}}{2} \right)^2 = \frac{1}{2} \quad |b_2|^2 = \left(\frac{\sqrt{2^2 + 1^2 + \bar{1}^2}}{6} \right)^2 = \frac{1}{6} \quad |b_3|^2 = \left(\frac{\sqrt{1^2 + 2^2 + 1^2}}{6} \right)^2 = \frac{1}{6}$$

$\frac{1}{2} > \frac{1}{6} + \frac{1}{6} \quad \left(= \frac{1}{3} \right) \Rightarrow$ the dislocation reaction is feasible (i.e. the full dislocation can lower its energy by splitting into partials)

The vectors on the RHS lie on the (111) close packed plane in a CCP crystal and they connect B to C sites and C to B sites respectively. Equivalent vectors (belonging to the same family) are shown in the figure on the right.



Solved Example

What is the image force experienced by an edge dislocation at a distance of $100b$ from the free surface of a semi infinite Al crystal? Is this force sufficient to move the dislocation given that the Peierls Force ($= \text{Peierls Stress} \times b$) $= 2.5 \times 10^{-4} \text{ N/m}$

Data for Al:

- $a_0 = 4.04 \text{ \AA}$, Slip system: $\langle 110 \rangle \{111\}$, $b = \sqrt{2}a_0/2 = 2.86 \text{ \AA}$, $G = 26.18 \text{ GPa}$, $\nu = 0.348$

$$F_{\text{Image}} = \frac{-Gb^2}{4\pi(1-\nu)d} \qquad F_{\text{Image}} = \frac{-Gb^2}{4\pi(1-\nu)100b} \quad \text{--ve sign implies an attraction towards the free surface}$$

$$F_{\text{Image}} = \frac{-(26.18 \times 10^9)(2.86 \times 10^{-10})}{4\pi(1-0.348)(100)} = 9.1 \times 10^{-3} \text{ N/m}$$

As $F_{\text{image}} > F_{\text{Peierls}} \Rightarrow$ that the dislocation will spontaneously move to the surface (*creating a step*) under the action of the image force, **without the application of an externally applied stress.**

**Solved
Example**

Compute the individual number of slip systems for $\langle 111 \rangle \{110\}$.

This is a slip system for the 'BCC crystal'.

For each of the individual directions in the family, we have to locate all the planes such that the dot product between the direction and plane is zero.

Direction	Planes (including -ves of planes)	No.	Including negative of direction	No.
[111]	(1-10), (10-1), (01-1) - ve of above: (-110), (-101), (0-11)	6	$\times 2$ (for [-1-1-1])	12
[-111]	(110), (101), (01-1) - ve of above: (-1-10), (-10-1), (0-11)	6	$\times 2$ (for [1-1-1])	12
[1-11]	(110), (10-1), (011) - ve of above: (-1-10), (-101), (0-1-1)	6	$\times 2$ (for [-11-1])	12
[11-1]	(-110), (101), (011) - ve of above: (1-10), (-10-1), (0-1-1)	6	$\times 2$ (for [-1-11])	12
			Total	48